



US 20250283062A1

(19) **United States**(12) **Patent Application Publication**
DICKINSON et al.(10) **Pub. No.: US 2025/0283062 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **SYSTEMS AND METHODS FOR
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Chicago, IL (US)(21) Appl. No.: **18/861,526**(22) PCT Filed: **May 1, 2023**(86) PCT No.: **PCT/US2023/066438**

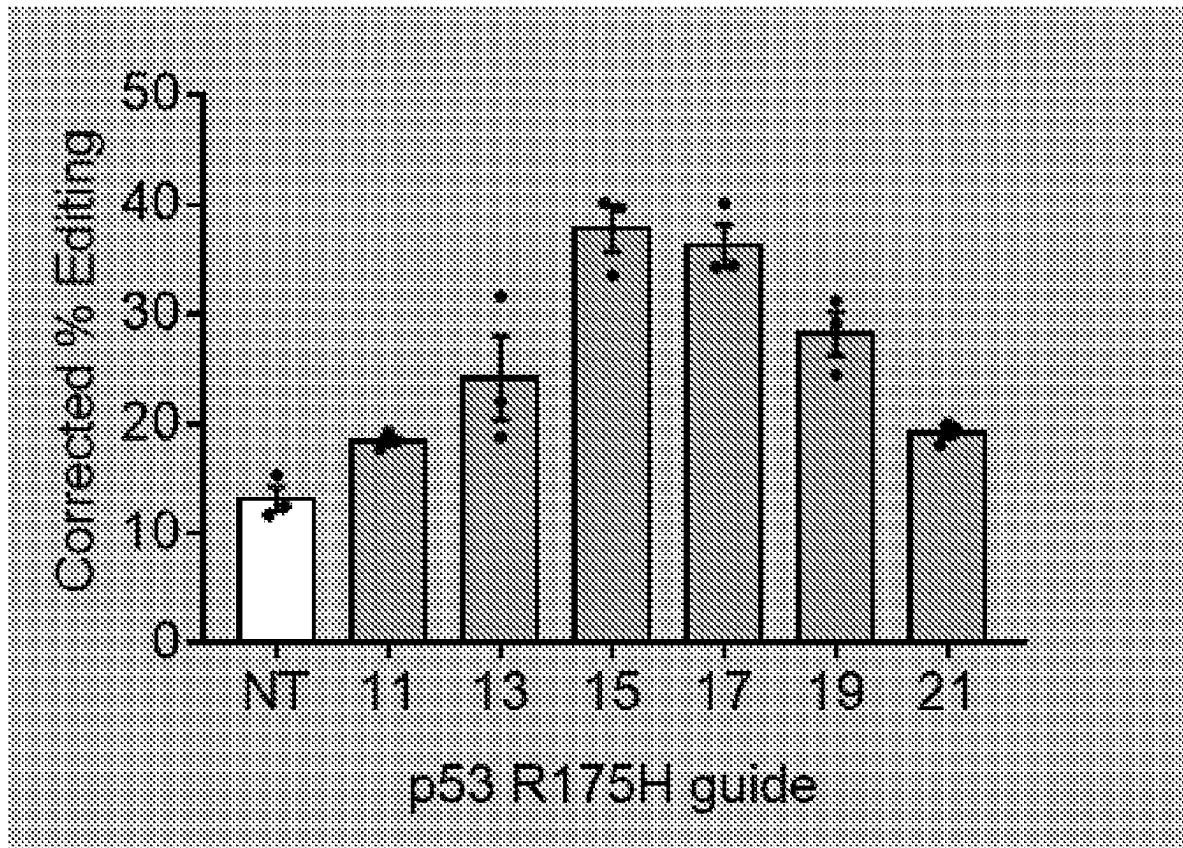
§ 371 (c)(1),

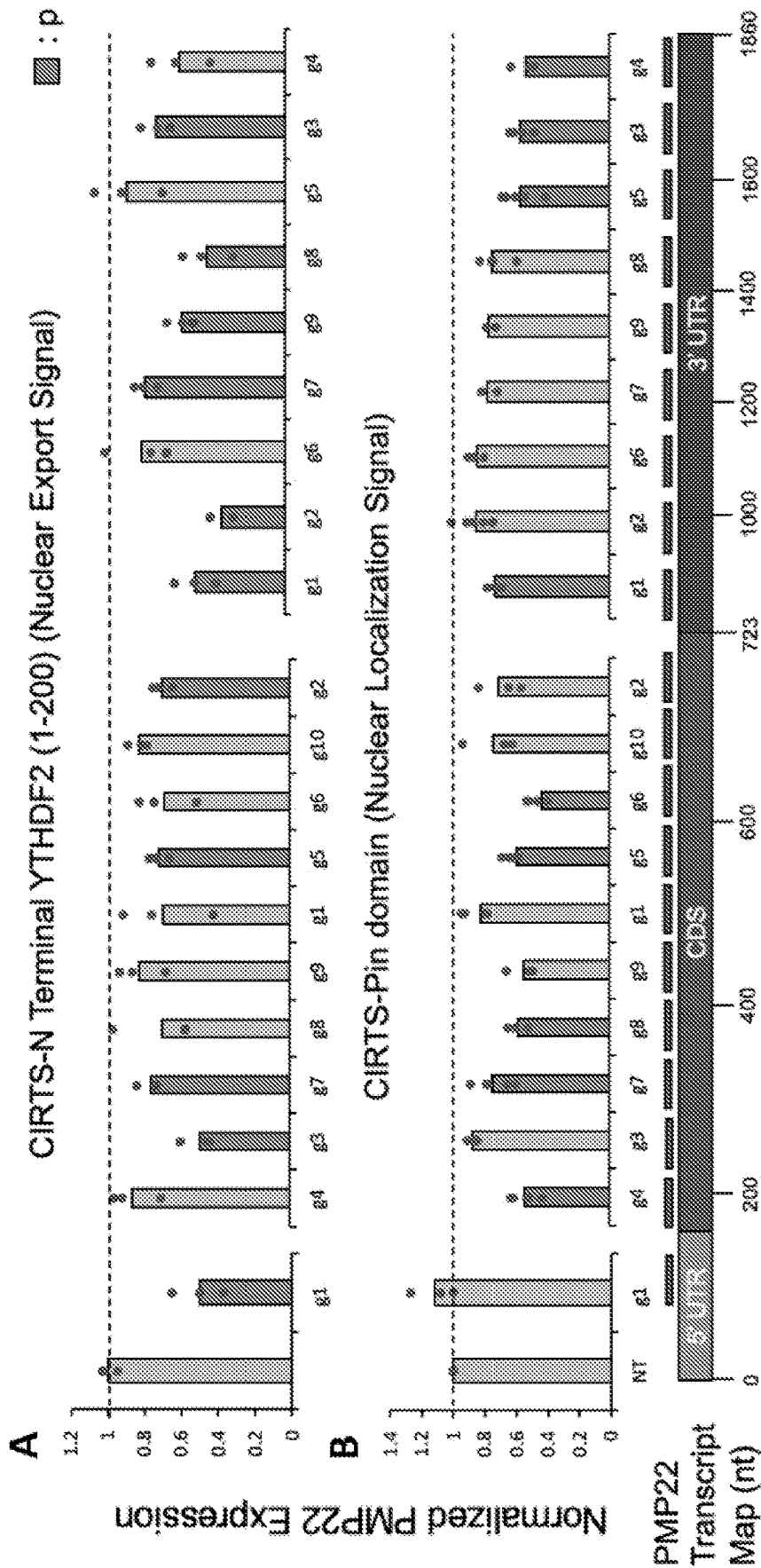
(2) Date: **Oct. 29, 2024****Related U.S. Application Data**(60) Provisional application No. 63/337,481, filed on May
2, 2022.**Publication Classification**(51) **Int. Cl.***C12N 9/22* (2006.01)*C12N 15/88* (2006.01)(52) **U.S. Cl.**CPC *C12N 9/22* (2013.01); *C12N 15/88*
(2013.01)

(57)

ABSTRACT

Aspects of the disclosure relate to a Effector system comprising at least one of each: i) a RNA hairpin binding domain; ii) a RNA targeting molecule comprising a RNA targeting region and at least one hairpin structure, wherein the hairpin structure of the RNA targeting molecule specifically binds to i; and iii) a Effector domain.

Specification includes a Sequence Listing.



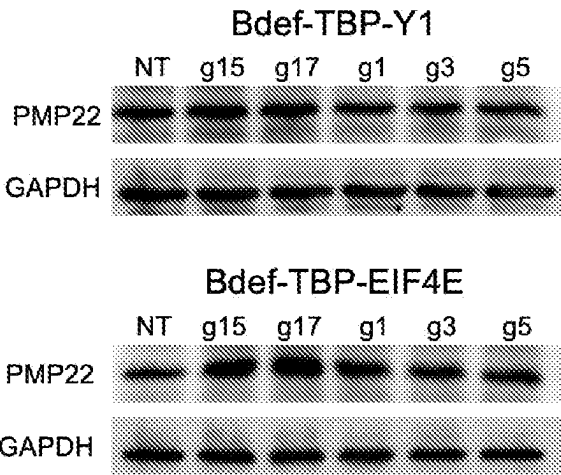


FIG. 2

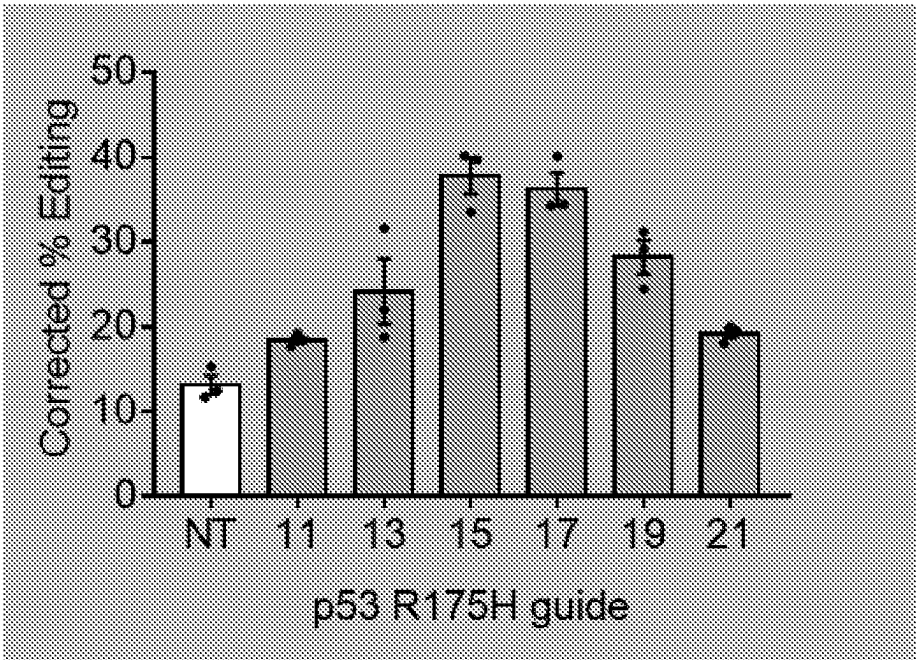


FIG. 3

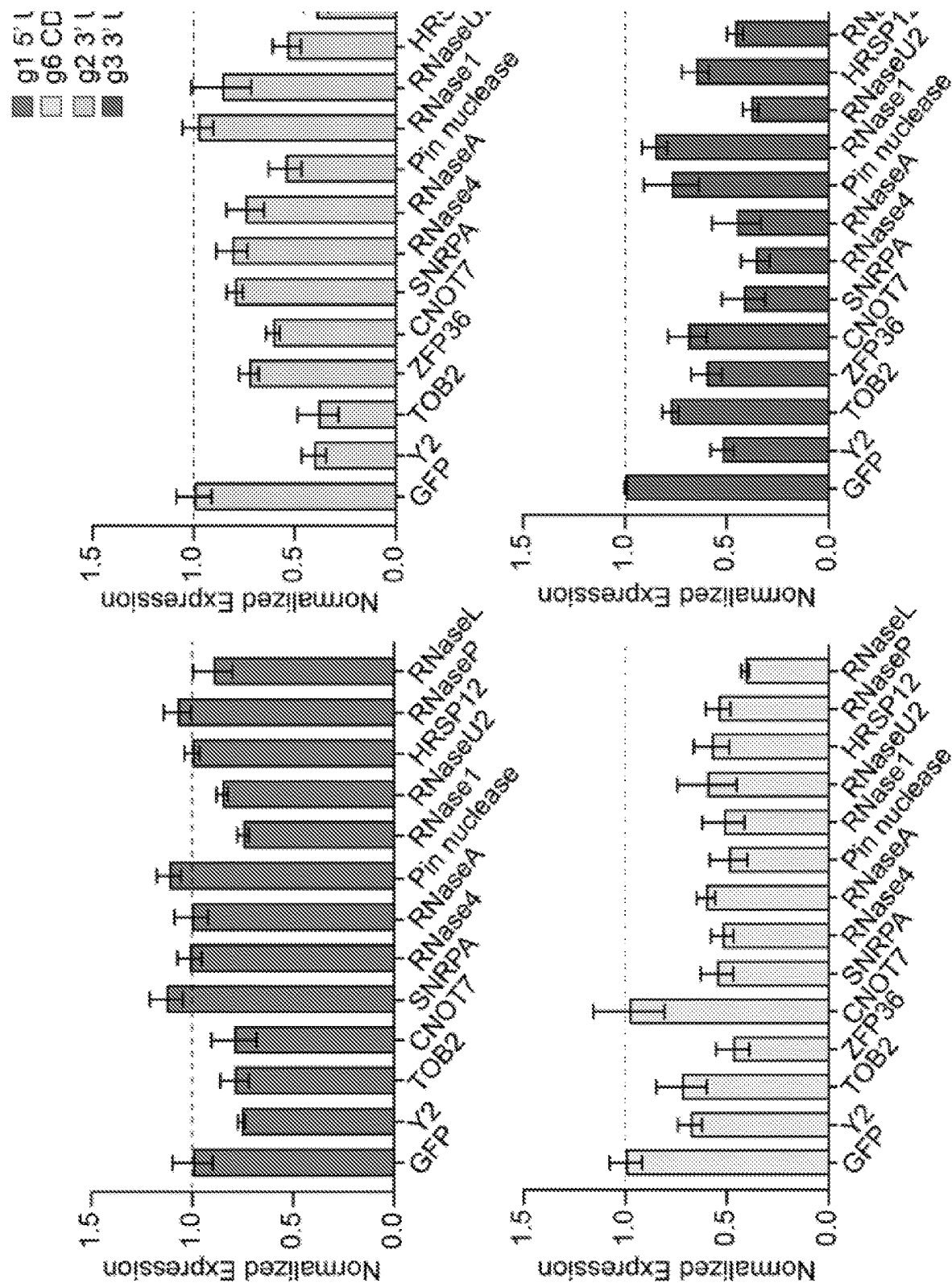
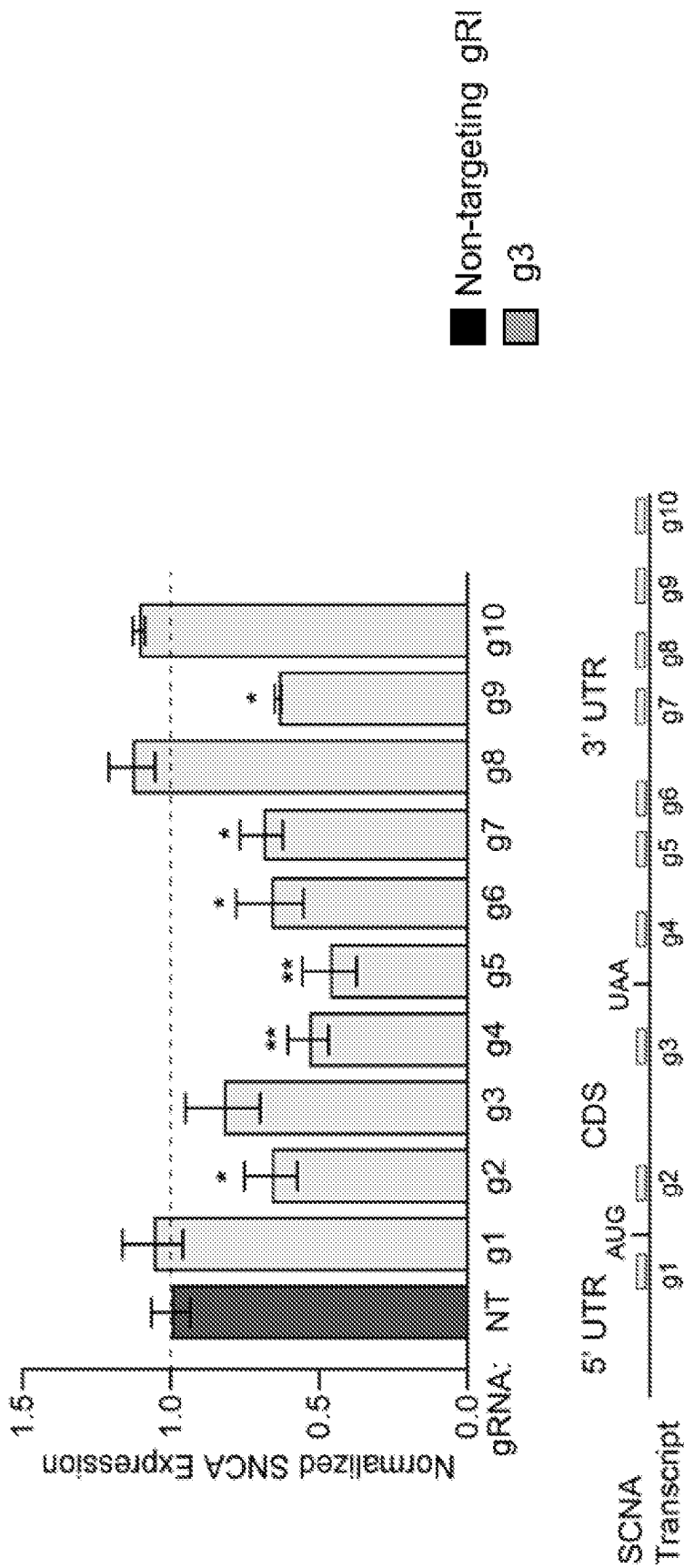


FIG. 4



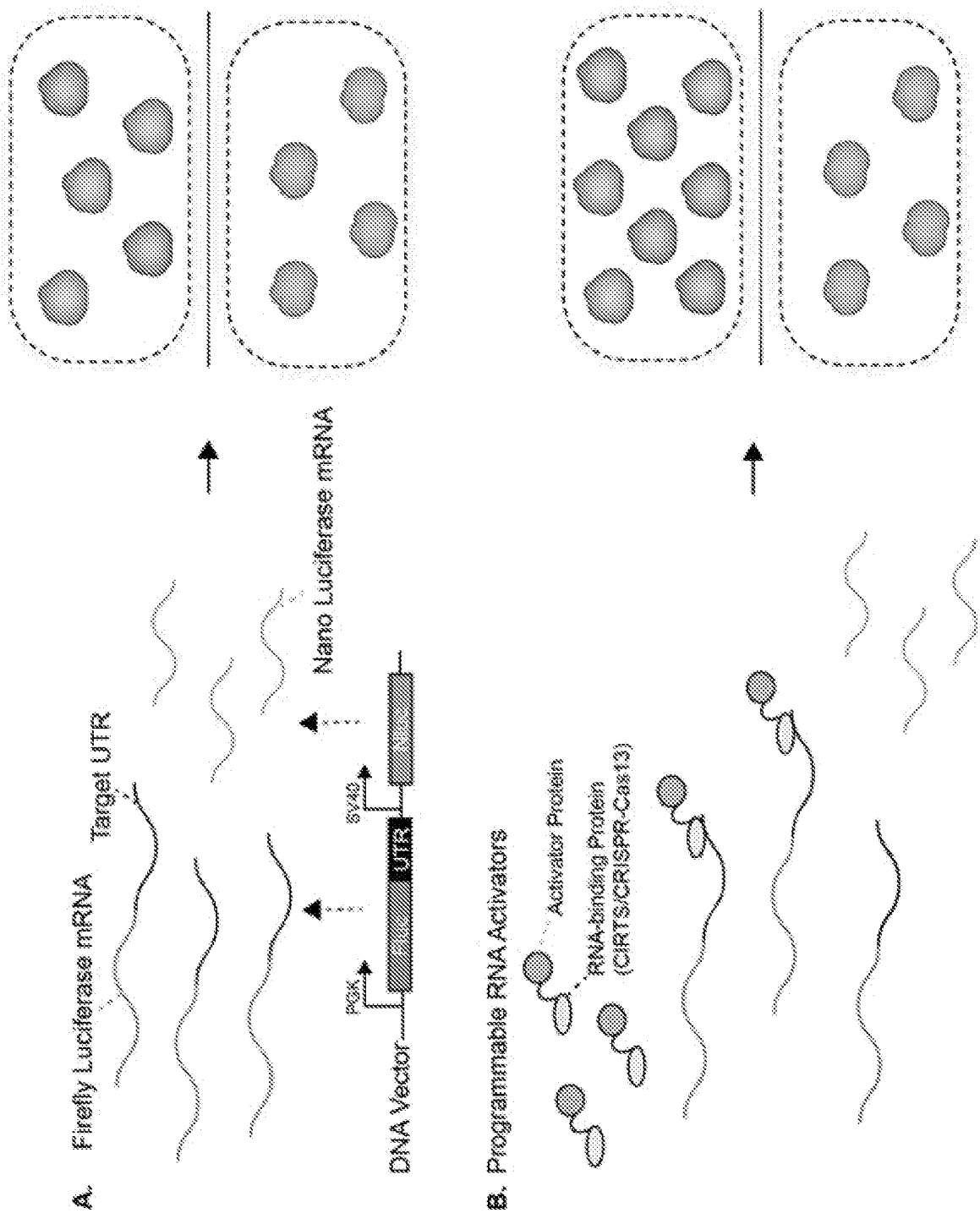
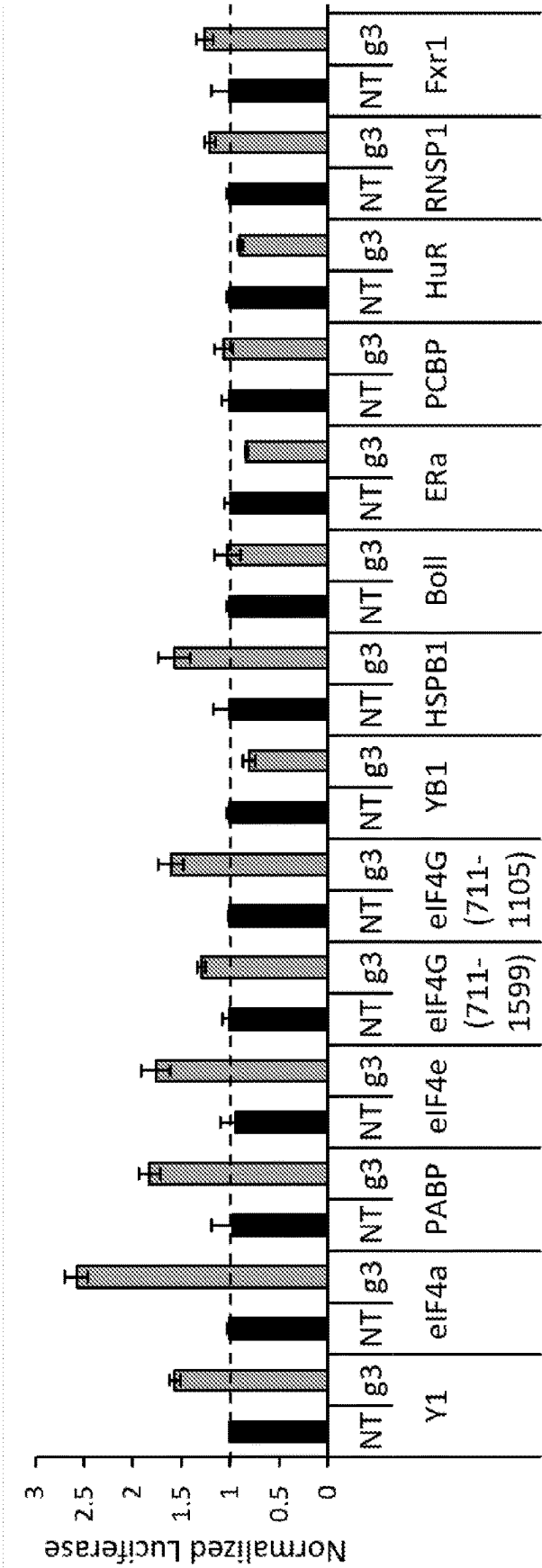


FIG. 6A-B



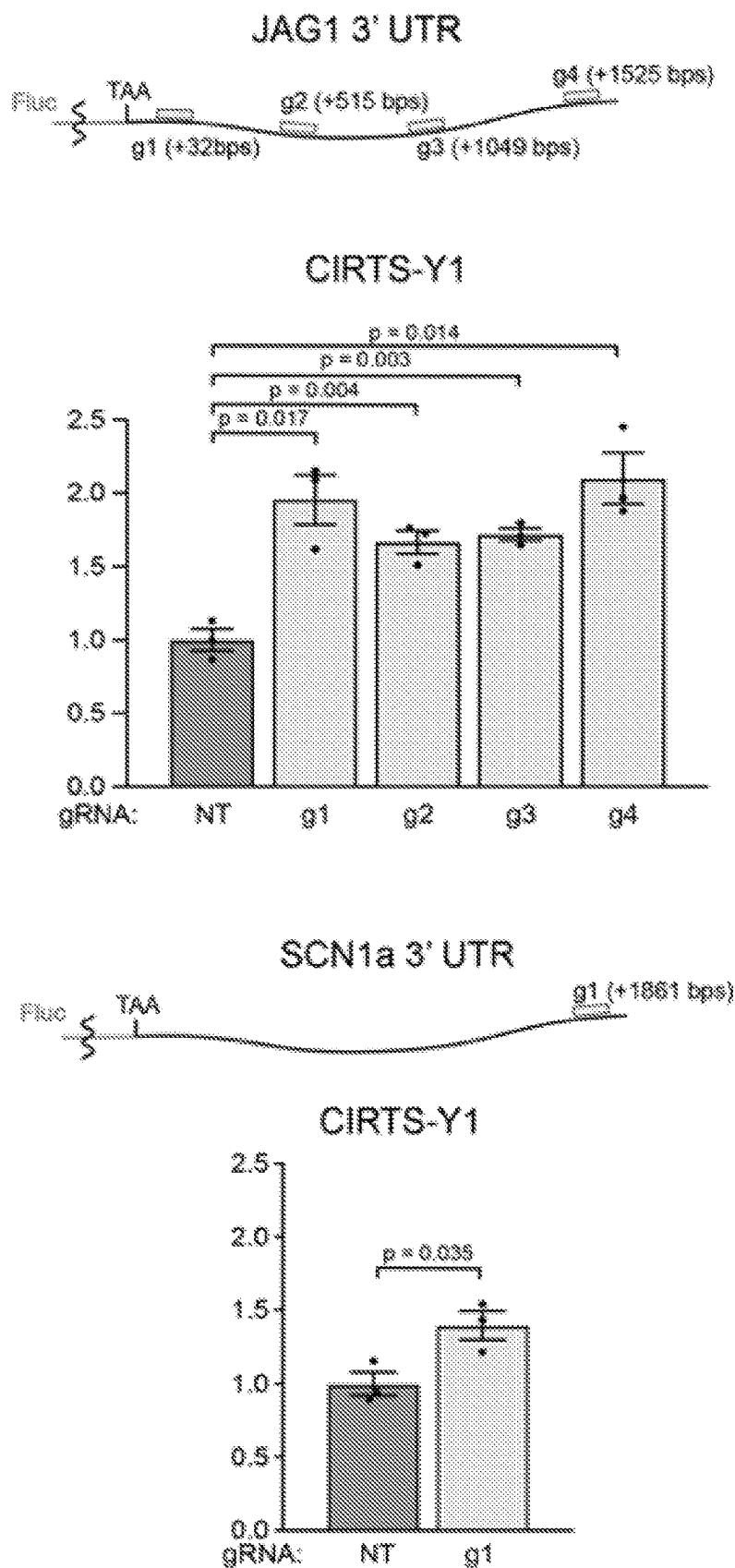


FIG. 8

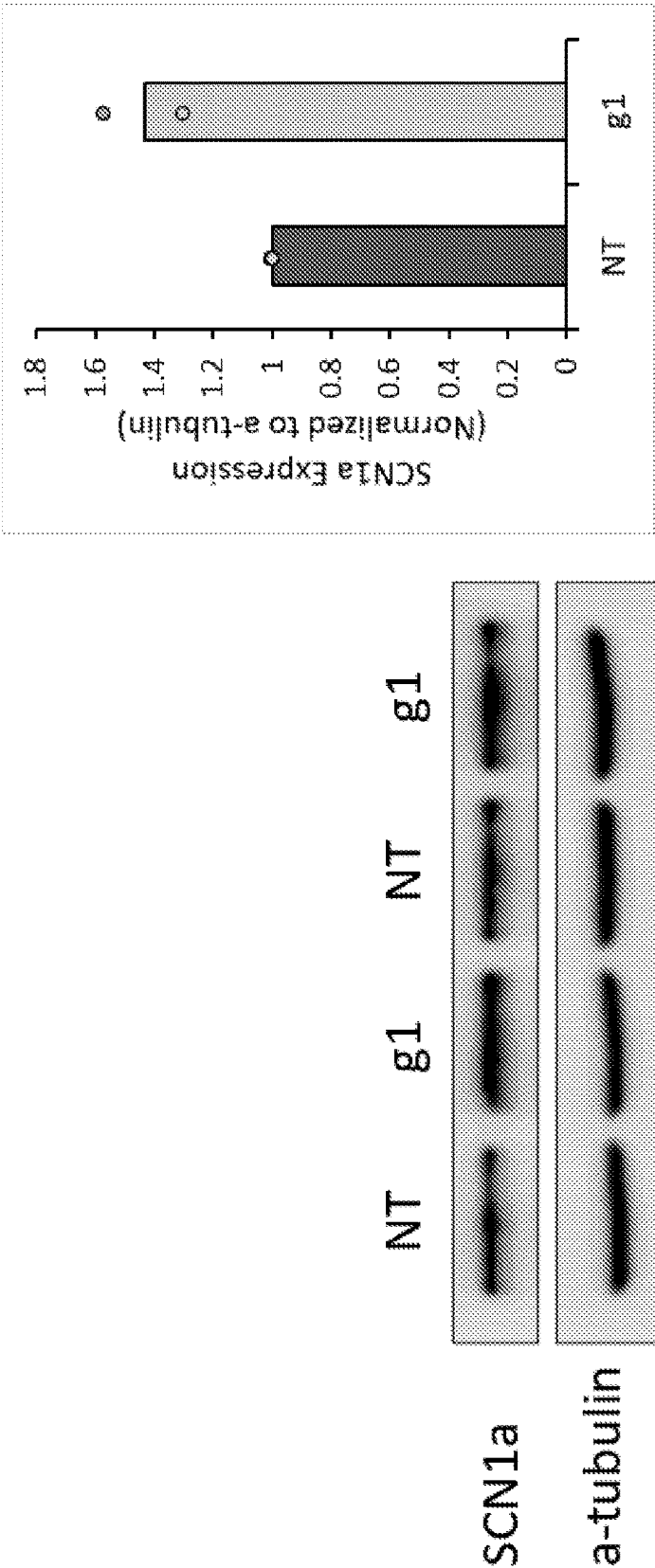


FIG. 9

SYSTEMS AND METHODS FOR MODULATING RNA

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of priority to U.S. Provisional Patent Application Ser. No. 63/337,481, filed May 2, 2022, which is hereby incorporated by reference in its entirety.

STATEMENT OF GOVERNMENT SUPPORT

[0002] This invention was made with government support under grant number GM119840 awarded by the National Institutes of Health. The government has certain rights in the invention.

SEQUENCE LISTING

[0003] The application contains a Sequence Listing prepared in compliance with ST.26 format and is hereby incorporated by reference in its entirety. Said Sequence Listing, created on May 1, 2023 is named ARCDP0734WO.xml and is 194,423 bytes in size.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0004] The present invention relates generally to the field of chemistry and medicine. More particularly, it concerns the use of a system for modulating RNA.

2. Description of Related Art

[0005] Programmable nucleic acid-binding proteins have revolutionized genome studies and editing technologies (Chandrasegaran and Carroll, 2016; Filipovska et al., 2011; Gootenberg et al., 2018; Hilton et al., 2015; Joung and Sander, 2012; Kearns et al., 2015; Strutt et al., 2018) and are opening up new therapeutic opportunities to treat human diseases (Liao et al., 2017; Monteys et al., 2017). In particular, the CRISPR/Cas9 system, which evolved as a bacterial immune defense mechanism, has transformed the ability to study and manipulate cellular DNA site-specifically (Cong et al., 2013; Jiang et al., 2013; O'Connell et al., 2014; Wiedenheft et al., 2012). A key advantage of CRISPR/Cas systems compared to previous methods (Desjarlais and Berg, 1993; Hockemeyer et al., 2011; Joung and Sander, 2012; Schierling et al., 2012) is that they are easily programmable to target virtually any locus of interest. The CRISPR/Cas system is a ribonucleoprotein complex that uses base pair interactions of a displayed guide RNA (gRNA) to interact with a target nucleic acid sequence. The simple nature of base pair-guided targeting opens up the possibility to program systems to interact with a defined nucleic acid sequence by simply changing the nucleic acid sequence on the guiding strand.

[0006] While targeting DNA directly will have profound clinical ramifications, diseases that involve subtle alterations to many genes will be challenging to target using DNA editing technologies (Fuxman Bass et al., 2015). Additionally, the potential side effects or risks of permanent genetic alteration might not be tolerated. For example, the genes one may want to target to activate an enhanced wound healing response are likely targets that could pose a risk for cancer

development, making permanent DNA-based strategy risky. Targeting information flow at the RNA level presents several opportunities for therapeutic intervention, including but not limited to the ability to halt treatment if side effects emerge, the ability to target genes that would be too risky to alter at the DNA level, and the ability to manipulate gene expression without permanent alterations to the host genome. While inhibiting or enhancing transcription at the genome level provides one possibility for controlling gene expression (Du et al., 2017; Fuxman Bass et al., 2015; Qi et al., 2013), recently discovered RNA epitranscriptomic regulatory mechanisms offer a broad range of processes to target, including editing, degradation, transport, and translation of RNA transcripts (Nishikura, 2010; Roundtree et al., 2017; Zhao et al., 2017). Although the mechanisms and consequences of this epitranscriptomic regulatory layer are just beginning to be uncovered, it is apparent that the information flow through RNA is tightly regulated, offering many new opportunities for both basic research discoveries as well as therapeutic development.

[0007] Programmable RNA-targeting tools analogous to the dCas9 DNA-targeting systems hold great promise for studying the mechanisms of epitranscriptomic regulation and for therapeutic applications. The current tools for RNA targeting involve the delivery of large complexes and pose immunogenicity issues. From a basic science perspective, the large size of the delivery vehicle could lead to potential perturbations to the RNA under interrogation, convoluting the study of RNA regulatory mechanisms. From a translational perspective, the large size presents challenges for viral packaging or direct protein delivery. Additionally, while DNA-editing therapies will likely consist of a one-time, irreversible treatment, RNA-targeting therapies will need to be continually administered, making delivery concerns especially important. Moreover, it was recently discovered that 85% of people already have circulating antibodies to CRISPR/Cas proteins (Kim et al., 2018; Wagner et al., 2018), suggesting immunogenicity issues may prove problematic in clinical applications. Therefore, there is a need in the art for improved systems that can target RNA and be delivered efficiently without activating an immune response.

SUMMARY OF THE INVENTION

[0008] To overcome the large size and microbial-derived nature of current RNA-targeting systems, the inventors present a CRISPR/Cas-inspired RNA targeting system (CIRTS), a general method for engineering programmable RNA effector systems. Similar to CRISPR/Cas-based systems, CIRTS is a ribonucleoprotein complex that uses Watson-Crick-Franklin base pair interactions to deliver protein cargo site-selectively in the transcriptome. The inventors show they can easily engineer CIRTS that deliver a range of regulatory proteins to transcripts, including nucleases for degradation, deadenylation regulatory machinery for degradation, or translational activation machinery for enhanced protein production. However, CIRTS complexes are up to 5-fold smaller than the smallest current CRISPR/Cas systems and can be engineered entirely from human parts.

[0009] Aspects of the disclosure relate to a polypeptide comprising an effector domain wherein the effector domain comprises the amino acid sequence of one of SEQ ID NOs: 19-39, 60-62, 69-76, 113-121, or 136-143 or comprises an amino acid sequence with at least 80% sequence identity to one of SEQ ID NOs: 19-39, 60-62, 69-76, 113-121, or

136-143 or a functional portion thereof; and wherein the polypeptide further comprises one or more of a stabilizer domain, linker, RNA hairpin binding domain, nuclear export signal, and molecular tag. In some aspects, the effector domain comprises an amino acid sequence having, having at least, or having at most 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, or 100% sequence identity to one of SEQ ID NOs: 19-39, 60-62, 69-76, 113-121, or 136-143 or a functional portion thereof. Further aspects relate to a polypeptide comprising an effector domain, wherein the effector domain comprises a fusion between at least two polypeptides, wherein each polypeptide is independently selected from a polypeptide comprising an amino acid sequence of one of SEQ ID NOs: 22-24, 28-39, 60-62, 69-76, 113-121, or 136-143 or a polypeptide having an amino acid sequence with at least 80% sequence identity to one of SEQ ID NOs: 22-24, 28-39, 60-62, 69-76, 113-121, or 136-143 or a functional portion thereof. In some aspects, the effector domain comprises a fusion between at least two polypeptides, wherein each polypeptide is independently selected from a polypeptide that has an amino acid sequence that has, has at least, or has at most 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, or 100% sequence identity to one of SEQ ID NOs: 22-24, 28-39, 60-62, 69-76, 113-121, or 136-143 or a functional portion thereof.

[0010] Further aspects of the disclosure relate to a polypeptide comprising an effector domain, wherein the effector domain comprises a fusion between exactly, at least, or at most 2, 3, 4, 5, or 6 polypeptides, wherein each polypeptide is independently selected from a polypeptide comprising an amino acid sequence of one of SEQ ID NOs: 22-24, 28-39, 60-62, 69-76, 113-121, or 136-143 or a polypeptide having an amino acid sequence with at least 80% sequence identity to one of SEQ ID NOs: 22-24, 28-39, 60-62, 69-76, 113-121, or 136-143 or a functional portion thereof. In some aspects, the effector domain comprises a fusion between exactly, at least, or at most 2, 3, 4, 5, or 6 polypeptides, wherein each polypeptide is independently selected from a polypeptide that has an amino acid sequence that has, has at least, or has at most 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, or 100% sequence identity to one of SEQ ID NOs: 22-24, 28-39, 60-62, 69-76, 113-121, or 136-143 or a functional portion thereof.

[0011] Further aspects relate to nucleic acids encoding the polypeptides of the disclosure. Nucleic acids also include a nucleic acid encoding a RNA targeting molecule comprising a RNA targeting region and at least one hairpin structure, wherein the hairpin structure of the RNA targeting molecule specifically binds to the RNA hairpin binding domain of the polypeptide. Also disclosed are expression vectors comprising the nucleic acids of the disclosure and host cells comprising the nucleic acid or expression vector of the disclosure. Yet further aspects relate to a delivery vehicle comprising a polypeptide, nucleic acid, or expression vector of the disclosure. Also described are compositions comprising a polypeptide, nucleic acid, expression vector, host cell, or delivery vehicle of the disclosure. Further aspects relate to a method of making a polypeptide comprising transferring a nucleic acid or expression vector of the disclosure into a cell. The method may further comprise isolating the

expressed polypeptide. Further aspects relate to a method of modulating at least one target RNA comprising contacting the target RNA with a polypeptide, delivery vehicle, or composition of the disclosure. Also described is a method for modulating at least one target RNA in a subject, the method comprising administering the polypeptide, delivery vehicle, or composition of the disclosure to the subject.

[0012] Further aspects relate to a cell or progeny thereof comprising modulated target RNA, wherein the target RNA has been modulated according to the methods of the disclosure. Yet further aspects relate to a multicellular organism comprising one or more cells of the disclosure. Also described is a plant or animal comprising one or more cells or the disclosure. Further aspects relate to a kit comprising a polypeptide, nucleic acid, expression vector, host cell, delivery vehicle, or composition of the disclosure.

[0013] Polypeptides provided and/or utilized in accordance with the present disclosure may comprise or further comprise one or more linkers. The linkers may be between different regions or domains of the polypeptide. In some aspects, the effector domain comprises a linker between the two fused polypeptides. For example, the linker may be between an effector domain and a molecular tag, between an effector domain and a RNA hairpin binding domain, and/or between a stabilizer domain and an RNA hairpin binding domain. In some aspects, the linker comprises or consists of glycine and serine residues. The linker may comprise, may comprise at least, or may comprise at most 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, or 50 amino acids (or any derivable range therein). In some aspects the linker comprises or consists of GS. In some aspects, the linker has an amino acid sequence that comprises or consists of SEQ ID NO:15. Alternative linkers, including glycine-serine linkers, that are employable in methods, compositions, delivery systems, polypeptides, and nucleic acids of the disclosure are known in the art and described herein. The polypeptides may be ones that do not have a linker between different regions or domains of the polypeptide. The polypeptides may exclude having a linker between an effector domain and a molecular tag, between an effector domain and a RNA hairpin binding domain, and/or between a stabilizer domain and an RNA hairpin binding domain.

[0014] In some aspects, the polypeptide has an amino acid sequence that is or comprises one of SEQ ID NOs: 32-39 or 69-76, or a functional portion thereof, or an amino acid sequence that shares at least 80% sequence identity to one of SEQ ID NOs: 32-39 or 69-76, or a functional portion thereof. In some aspects, the polypeptide is independently selected from a polypeptide having, having at least, or having at most 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, or 100% sequence identity to one of SEQ ID NOs: 32-39 or 69-76, or a functional portion thereof.

[0015] In some aspects, the effector domain comprises a fusion between a polypeptide comprising an amino acid sequence with at least 80% sequence identity to SEQ ID NO:32 or a functional portion thereof and a polypeptide comprising an amino acid sequence with at least 80% sequence identity to SEQ ID NO:33 or a functional portion thereof. In some aspects, the effector domain comprises a fusion between a polypeptide comprising an amino acid

[0024] The polypeptides of the disclosure may comprise or further comprise one or more of a stabilizer domain, linker, RNA hairpin binding domain, nuclear export signal, and molecular tag. In some aspects, the polypeptide comprises one or more RNA hairpin binding domain(s). In some aspects, the polypeptide comprises, comprises at least, or comprises at most 1, 2, 3, 4, 5, or 6 RNA hairpin binding domains. The RNA hairpin binding domain may comprise the amino acid sequence of SEQ ID NO:16 or a polypeptide with at least 80% sequence identity to a polypeptide having the amino acid sequence of SEQ ID NO:16 or a functional portion thereof. In some aspects, the RNA hairpin binding domain comprises an amino acid sequence that has, has at least, or has at most 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, or 100% sequence identity to SEQ ID NO:16 or a functional portion thereof. The polypeptides of the disclosure may comprise or further comprise one or more stabilizer domains. The polypeptides may comprise, comprise at least, or comprise at most 1, 2, 3, 4, 5, or 6 RNA stabilizer domains. The stabilizer domain may comprise the amino acid sequence of SEQ ID NO: 14, 58, or 59 or a polypeptide having an amino acid sequence with at least 80% sequence identity to SEQ ID NO:14, 58, or 59 or a functional portion thereof. The stabilizer domain may comprise SEQ ID NO:14. The stabilizer domain may comprise SEQ ID NO:58 or 59 or a polypeptide having an amino acid sequence with at least 80% sequence identity to SEQ ID NO: 58 or 59 or a functional portion thereof. In some aspects, the stabilizer domain comprises an amino acid sequence that has, has at least, or has at most 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, or 100% sequence identity to SEQ ID NO:14 or a functional portion thereof. In some aspects, the stabilizer domain may comprises an amino acid sequence that has, has at least, or has at most 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, or 100% sequence identity to SEQ ID NO:58 or 59 or a functional portion thereof. The polypeptides of the disclosure may comprise or further comprise one or more nuclear export signals (NESs). The polypeptides may comprise, comprise at least, or comprise at most 1, 2, 3, 4, 5, or 6 NESs. The NES may comprise the amino acid sequence of SEQ ID NO:17 or a polypeptide having an amino acid sequence with at least 80% sequence identity to SEQ ID NO:17 or a functional portion thereof. In some aspects, the NES comprises an amino acid sequence that has, has at least, or has at most 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, or 100% sequence identity to SEQ ID NO:17 or a functional portion thereof. The polypeptides of the disclosure may comprise or further comprise one or more molecular tags. The polypeptides may comprise, comprise at least, or comprise at most 1, 2, 3, 4, 5, or 6 molecular tags. The molecular tag may comprise the amino acid sequence of SEQ ID NO:18 or a polypeptide comprising an amino acid sequence with at least 80% sequence identity to SEQ ID NO:18 or a functional portion thereof. In some aspects, the NES comprises an amino acid sequence that has, has at least, or has at most 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87,

88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, or 100% sequence identity to SEQ ID NO:18 or a functional portion thereof. In some aspects, the molecular tag comprises a FLAG tag.

[0025] In aspects of the disclosure, the hairpin binding domain is amino-proximal to the effector domain. In aspects of the disclosure, the stabilizer domain is amino-proximal to the RNA hairpin binding domain and/or effector domain. In aspects of the disclosure, the nuclear export signal is amino-proximal to the effector domain. In aspects of the disclosure, the molecular tag is carboxy-proximal to the effector domain. In aspects of the disclosure, the hairpin binding domain is carboxy-proximal to the effector domain. In aspects of the disclosure, the stabilizer domain is carboxy-proximal to the RNA hairpin binding domain and/or effector domain. In aspects of the disclosure, the nuclear export signal is carboxy-proximal to the effector domain. In aspects of the disclosure, the molecular tag is amino-proximal to the effector domain. In some aspects, the polypeptide comprises, from N-terminus to the C-terminus, a stabilizer domain, a glycine serine linker, a RNA hairpin binding region, a nuclear export signal, an effector domain, and a molecular tag. In some aspects, the polypeptide comprises the amino acid sequence of one of SEQ ID NOS: 1-13, 63-68, 77-82, or 148-169, or an amino acid with at least 80% sequence identity to one of SEQ ID NOS: 1-13, 63-68, 77-82, or 148-169 or a functional portion thereof. The polypeptide may comprise an amino acid sequence that has, has at least, or has at most 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, or 100% sequence identity to one of SEQ ID NOS: 1-13, 63-68, 77-82, or 148-169 or a functional portion thereof. In some aspects, the polypeptide comprises an amino acid sequence that has at least one substitution relative to SEQ ID NOS: 1-13, 63-68, 77-82, or 148-169. In some aspects, the polypeptide has an amino acid sequence that comprises, comprises at least, or comprises at most 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, or 100 substitutions relative one of SEQ ID NOS: 1-13, 63-68, 77-82, or 148-169.

[0026] The delivery vehicles of the disclosure may comprise liposome(s), particle(s), exosome(s), microvesicle(s), a gene-gun or one or more nucleic acid vector(s). In some aspects, the delivery vehicle or composition of the disclosure comprises or further comprises a RNA targeting molecule comprising a RNA targeting region and at least one hairpin structure, wherein the hairpin structure of the RNA targeting molecule specifically binds to the RNA hairpin binding domain of the polypeptide. In some aspects, the RNA targeting molecule comprises a TAR hairpin scaffold or a human SLBP hairpin scaffold. In some aspects, the RNA targeting molecule comprises SEQ ID NO:40 or 41. The RNA targeting region may hybridizes to the 5'UTR of a RNA. The RNA targeting region may hybridize to the 3'UTR of a RNA. The RNA targeting region may hybridize to the 5'UTR or 3'UTR of a mammalian mRNA. In some aspects, the RNA targeting region comprises a nucleic acid sequence of one of SEQ ID NOS: 55, 83-86, 88-112, 122-131, or 144-147 or a nucleic acid sequence having at

least 80% sequence identity to one of SEQ ID NOs: 55, 83-86, 88-112, 122-131, or 144-147. The RNA targeting region may comprise a nucleic acid sequence having or having at least 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, or 100% sequence identity (or any derivable range therein) to one of SEQ ID NOs: 55, 83-86, 88-112, 122-131, or 144-147. The RNA targeting molecule may comprise at least one of SEQ ID NO:40-54, 56, 57, or the sequence UCCCGA. In some aspects, the RNA targeting molecule comprises, comprises at least, or comprises at most 1, 2, 3, 4, 5, 6, 7, or 8 hairpin structures (or any derivable range therein), wherein each hairpin structure is independently chosen from one of one of SEQ ID NO:40-54, 56, 57, or the sequence UCCCGA.

[0027] Modulating at least one target RNA may comprise cleaving, demethylating, methylating, activating translation, repressing translation, promoting degradation, or binding to the RNA. Modulating at least one target RNA may exclude cleaving, demethylating, methylating, activating translation, repressing translation, promoting degradation, or binding to the RNA. In some aspects, at least two target RNAs are modulated. In some aspects, at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, or 25 (or any derivable range therein) target RNAs are modulated. The multiple RNAs may be modulated by the same effector domain or by an effector domain with the same activity. In some aspects, the different target RNAs are modulated with a different activity, such as by cleaving, demethylating, methylating, activating translation, repressing translation, promoting degradation, or binding to the RNA. The effector domain may be one that does not bind to RNA. In some aspects, the effector domain comprises a polypeptide that does not have RNA binding activity. In some aspects, the effector domain does not bind to modified RNA. In some aspects, the effector domain does not bind to m6A modified RNA.

[0028] A stabilizer domain, a glycine serine linker, a RNA hairpin binding region, a nuclear export signal, an effector domain, a molecular tag, a hairpin structure, and/or RNA targeting region may exclude one or more of SEQ ID NOs: 1-169.

[0029] The term “RNA hairpin” refers to a RNA molecule with stem-loop intramolecular base pairing. A hairpin can occur when two regions of the same strand, usually complementary in nucleotide sequence when read in opposite directions, base-pair to form a double helix that ends in an unpaired loop. The disclosure relates to engineered RNA targeting molecules comprising a RNA targeting region and one or more hairpins. Accordingly, the engineered RNA molecules of the disclosure are chimeric molecules that are non-naturally occurring.

[0030] The term “RNA targeting region” refers to a region of the RNA that is capable of hybridizing to a target RNA. The target RNA may be a disease associated RNA or one that is a modulation target according to the current systems and methods.

[0031] The “effector domain” refers to a peptide or polypeptide that has activity directed to RNA. Examples of activity include methylation activity, RNA-binding activity, nuclease activity, and translational activation or repression activity.

[0032] In some aspects, the RNA hairpin binding domain and the effector domain are operably linked. The term “operably linked” refers to two proteins that are linked

through either covalent or non-covalent interactions. For example, the two proteins may be covalently linked through a peptide bond. In some aspects, the proteins are non-covalently linked. One or more proteins of the disclosure may be operably linked to another protein through linkage to a pair of accessory proteins that have a strong affinity for each other. Such accessory proteins are known in the art. For example, the SunTag is one such system that includes an antibody with a strong affinity for a peptide. One protein, polypeptide, or domain of the disclosure may be linked to a SunTag peptide and another protein, polypeptide, or domain of the disclosure may be linked to an antibody to allow operable linkage of the two proteins, polypeptides, or domains through the interaction of the SunTag peptide and antibody. Further examples include biotin and avidin/streptavidin and spytag and spycatcher.

[0033] In some aspects, the system is inducible by providing the effector domain and the hairpin binding domain as two unlinked polypeptides that become linked upon the presence of a stimulant. The induction may be, for example, by light induction or by chemical induction. Such inducibility allows for activation of the RNA regulation at a desired moment in time. In some aspects, the effector domain is covalently linked to a first dimerization domain and the RNA hairpin binding domain is covalently linked to a second dimerization domain and wherein the first and second dimerization domain are capable of dimerizing to form a non-covalent or covalent linkage. In some aspects, the dimerization is inducible. In some aspects, the dimerization is induced through binding of the dimerization domains to a ligand. The term inducible refers to dimerization that is formed in response to a stimulus, such as a ligand, a chemical, a temperature change, or light, for example.

[0034] Light inducibility is for instance achieved by designing a fusion complex wherein the first and second dimerization domains comprise CRY2 PHR and CIBN. This system is particularly useful for light induction of protein interactions in living cells and is further described in Konermann S, et al. *Nature*. 2013; 500:472-476, which is herein incorporated by reference.

[0035] Suitable dimerization domains and corresponding ligands are known in the art. For example, Liang, F. S., Ho, W. Q., and Crabtree, G. R. (2011). Engineering the ABA plant stress pathway for regulation of induced proximity. *Sci. Signal.* 4, rs2, which is incorporated by reference, describes suitable dimerization/ligand systems that are useful in aspects of the disclosure. In some aspects, one of the first or second dimerization domain comprises PYR/PYL1-like (PYL1), the other of the first or second domain comprises ABA insensitive 1 (ABI1), and the ligand comprises abscisic acid (ABA) or derivatives or fragments thereof. The dimerization domain may be a fragment or portion of the whole protein and may be a substituted or modified. In some aspects, the first and/or second dimerization domain comprises FKBP12 and the ligand comprises FK1012 or derivatives or fragments thereof. In some aspects, one of the first or second dimerization domain comprises FK506 binding protein (FKBP), the other of the first or second domain comprises FKBP-Rap binding domain of mammalian target of Rap mTOR (Frb), and the ligand comprises rapamycin (Rap) or derivatives or fragments thereof.

[0036] Derivatives refer to modified ligands and domains that retain binding or have enhanced binding to their dimerization domain or ligand, respectively. Fragments refer

to contiguous portions of the dimerization domains that retain binding to the ligand. In some aspects, the dimerization domain may be a modified fragment.

[0037] In some aspects, the domains of the disclosure are human or are human-derived. In some aspects, the polypeptide is non-immunogenic. A human protein, polypeptide, domain, or nucleic acid refers to a protein, polypeptide, domain, or nucleic acid that is from the human genome, although it may be produced recombinantly in non-human systems. The term “human-derived” refers to a protein, polypeptide, domain, or nucleic acid that is a variant or fragment of a protein, polypeptide, domain, or nucleic acid from the human genome, although it may be produced recombinantly in non-human systems. In some aspects, the fusion protein, conjugate, system, or parts thereof, such as parts i, ii, and/or iii are non-immunogenic and/or non-toxic when expressed in or administered to humans.

[0038] In some aspects, the nucleic acids or polypeptides of the disclosure are synthetic, are non-natural, and/or do not occur naturally in nature. The polypeptides and methods of the disclosure may exclude a Cas9 enzyme or a polypeptide that comprises or comprises at least 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, 200, 205, 210, 215, 220, 225, 230, 235, 240, 245, 250, 255, 260, 265, 270, 275, 280, 285, 290, 295, 300, 305, 310, 315, 320, 325, 330, 335, 340, 345, 350, 355, 360, 365, 370, 375, 380, 385, 390, 395, 400, 405, 410, 415, 420, 425, 430, 435, 440, 445, 450, 455, 460, 465, 470, 475, 480, 485, 490, 495, or 500 amino acids that have or have at least 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, or 100% sequence identity to a Cas9 protein. Nucleic acids and methods of the disclosure exclude nucleic acids that bind to a Cas9 protein.

[0039] In some aspects, the RNA targeting molecule comprises exactly one hairpin. In some aspects, the RNA targeting molecule comprises at least one hairpin. In some aspects, the RNA targeting molecule comprises exactly two hairpins. In some aspects, the RNA targeting molecule comprises at least two hairpins. In some aspects, the RNA targeting molecule comprises exactly three hairpins. In some aspects, the RNA targeting molecule comprises at least three hairpins. In some aspects, the RNA targeting molecule comprises exactly four hairpins. In some aspects, the RNA targeting molecule comprises at least four hairpins. In some aspects, the RNA targeting molecule comprises exactly five hairpins. In some aspects, the RNA targeting molecule comprises at least five hairpins. In some aspects, the RNA targeting molecule comprises 1–4 hairpins. In some aspects, the RNA targeting molecule comprises 1–3 hairpins. In some aspects, the RNA targeting molecule comprises 1–2 hairpins. In some aspects, the RNA targeting molecule comprises 2–4 hairpins. In some aspects, the RNA targeting molecule comprises 2–3 hairpins. In some aspects, the RNA targeting molecule comprises at least, at most, or exactly 1, 2, 3, 4, 5, or 6 hairpins (or any range derivable therein). In some aspects, the RNA targeting molecule comprises at least one hairpin that does not bind to the RNA hairpin binding protein and at least one hairpin that binds to the RNA hairpin binding protein. In some aspects, the RNA targeting molecule binds to more than one RNA binding protein. In some aspects, the RNA targeting molecule comprises two, three,

or four hairpin structures and binds to at least two RNA binding proteins. In some aspects, the effector system comprises at least two regulatory domains, wherein each regulatory domain binds to a different RNA binding molecule.

[0040] In some aspects, the RNA targeting molecule comprises one or more modified nucleotides. In some aspects, the modified nucleotides comprise a modification such as a phosphorothioate, locked nucleotides, ethylene bridged nucleotides, peptide nucleic acids, 5'E-VP, or is modified to a morpholino. In some aspects, the modification includes one described herein.

[0041] In some aspects, the effector domain comprises a nuclease, methylase, demethylase, translational activator, translational repressor, single-stranded RNA cleavage activity, double-stranded RNA cleavage activity, or RNA binding activity. In some aspects, the effector domain comprises an activity described herein.

[0042] In some aspects, the effector domain increases translation of a target RNA (e.g., by recruiting the translational machinery). In some aspects, the effector domain increases degradation of a target RNA. In some aspects, the effector domain modifies the localization of a target RNA. In some aspects, the effector domain modifies the processing of the target RNA (e.g., by active nuclease activity on the target or by triggering endogenous epitranscriptomic regulatory pathways).

[0043] In some aspects, the vectors of the disclosure may comprise or further comprise a regulatory element operably linked to the nucleotide of the disclosure. Regulatory elements, in addition to a NLS and NES, as previously described, also include promoters, polyadenylation signals, enhancers, etc. Other regulatory elements are known in the art and described herein and may be used in the aspects of the disclosure. In some aspects, the one or more nucleic acid vectors are optimized for expression in an eukaryotic cell. In some aspects, the expression of the domains, RNA, or polypeptides in the cell or from a vector is constitutive. In some aspects, the expression of the domains, RNA, or polypeptides in the cell or from a vector is conditional. In some aspects, the expression vector further comprises a nucleic acid encoding a RNA targeting molecule. In some aspects, one or more of the vectors are viral vectors. In some aspects, the one or more vectors comprise one or more retroviral, lentiviral, adenoviral, adeno-associated or herpes simplex viral vectors. In some aspects, one or more of the vectors are non-viral vectors. In some aspects, the system or composition is non-viral, which denotes that it does not contain any viral components.

[0044] In some aspects, there is a system or kit comprising one or more of the following components: a polypeptide comprising an effector domain, a polypeptide comprising a RNA binding domain, a polypeptide comprising a stabilizer, a nucleic acid encoding for a effector domain, a nucleic acid encoding for a RNA binding domain, a nucleic acid encoding a stabilizer, a nucleic acid encoding a RNA targeting molecule comprising a RNA targeting region and at least one hairpin structure; a conjugate of the disclosure; a vector of the disclosure, a fusion protein of the disclosure, a recombinant host cell, an expression construct, an engineered viral vector, or an engineered attenuated virus. In certain aspects, a polypeptide of the disclosure is under the control of a heterologous promoter. It is specifically contemplated that any protein or polypeptide function that are used in aspects, may be used a nucleic acid encoding that protein or poly-

peptide function. Also, any and all polypeptides, proteins, nucleic acid molecules may be contained within a cell or other living organism, such as a virus (for instance, a phage).

[0045] A kit may include one or more components that are separate or together in a suitable container means, such as a sterile, non-reactive container. In some aspects, cells or viruses are provided that contain one or more nucleic acid constructs that encode the polypeptides of the disclosure. The term “promoter” is used according to its ordinary meaning to those in the field of molecular biology; it generally refers to a site on a nucleic acid in which a polymerase can bind to initiate transcription. In specific aspects, the promoter is recognized by a T7 RNA polymerase.

[0046] The compositions, vectors, systems, methods, and proteins of the disclosure are useful for a variety of clinical and research-related applications. The aspects of the disclosure may be useful for the treatment of a disease or condition, such as cancer or autoimmunity. In some aspects, the methods and compositions are for the acute treatment of a disease or condition. In some aspects, the methods and compositions are useful for the temporary modulation of RNA. In some aspects exclude permanent modification of gene activity. In some aspects, the methods and compositions are safer due to the acute modulation of RNA and/or due to the ability to control the expression of the system in vivo.

[0047] Throughout this application, the term “about” is used according to its plain and ordinary meaning in the area of cell and molecular biology to indicate that a value includes the standard deviation of error for the device or method being employed to determine the value.

[0048] The use of the word “a” or “an” when used in conjunction with the term “comprising” may mean “one,” but it is also consistent with the meaning of “one or more,” “at least one,” and “one or more than one.”

[0049] As used herein, the terms “or” and “and/or” are utilized to describe multiple components in combination or exclusive of one another. For example, “x, y, and/or z” can refer to “x” alone, “y” alone, “z” alone, “x, y, and z,” “(x and y) or z,” “x or (y and z),” or “x or y or z.” It is specifically contemplated that x, y, or z may be specifically excluded from an embodiment or aspect.

[0050] The words “comprising” (and any form of comprising, such as “comprise” and “comprises”), “having” (and any form of having, such as “have” and “has”), “including” (and any form of including, such as “includes” and “include”), “characterized by” (and any form of including, such as “characterized as”), or “containing” (and any form of containing, such as “contains” and “contain”) are inclusive or open-ended and do not exclude additional, unrecited elements or method steps.

[0051] The compositions and methods for their use can “comprise,” “consist essentially of,” or “consist of” any of the ingredients or steps disclosed throughout the specification. The phrase “consisting of” excludes any element, step, or ingredient not specified. The phrase “consisting essentially of” limits the scope of described subject matter to the specified materials or steps and those that do not materially affect its basic and novel characteristics. It is contemplated that embodiments and aspects described in the context of the term “comprising” may also be implemented in the context of the term “consisting of” or “consisting essentially of.”

[0052] Any method in the context of a therapeutic, diagnostic, or physiologic purpose or effect may also be described in “use” claim language such as “Use of” any compound, composition, or agent discussed herein for achieving or implementing a described therapeutic, diagnostic, or physiologic purpose or effect.

[0053] Use of the one or more sequences or compositions may be employed based on any of the methods described herein. Other embodiments and aspects are discussed throughout this application. Any embodiment or aspect discussed with respect to one aspect of the disclosure applies to other aspects of the disclosure as well and vice versa.

[0054] It is specifically contemplated that any limitation discussed with respect to one embodiment or aspect of the invention may apply to any other embodiment or aspect of the invention. Furthermore, any composition of the invention may be used in any method of the invention, and any method of the invention may be used to produce or to utilize any composition of the invention. Aspects of an embodiment set forth in the Examples are also embodiments that may be implemented in the context of embodiments discussed elsewhere in a different Example or elsewhere in the application, such as in the Summary of Invention, Detailed Description of the Embodiments, Claims, and description of Figure Legends.

[0055] Other objects, features and advantages of the present invention will become apparent from the following detailed description. It should be understood, however, that the detailed description and the specific examples, while indicating specific embodiments and aspects of the invention, are given by way of illustration only, since various changes and modifications within the spirit and scope of the invention will become apparent to those skilled in the art from this detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

[0056] The following drawings form part of the present specification and are included to further demonstrate certain aspects of the present invention. The invention may be better understood by reference to one or more of these drawings in combination with the detailed description of specific embodiments presented herein.

[0057] FIG. 1A-B: gRNA screen across mPMP22 measuring gRNA-dependent RNA decay by RT-qPCR in NIH/3T3 cells. Significant changes in degradation were observed depending on where the transcript is targeted as well as the effectors and CIRT5 localization tag used. (A) CIRT5-YTHDF2 (1-200aa) with nuclear export tag (NES). (B) CIRT5-Pin nuclease with nuclear localization tag (NLS). All guides represent 3 or more biological replicates.

[0058] FIG. 2. PMP22 activation. Targeting PMP22 with eIF4E and Y1-based CIRT5 activators with 5 gRNAs. g15 (targets 5'UTR) and g17 (targets 3'UTR) activate PMP22 expression, especially with an eIF4E-based CIRT5 system.

[0059] FIG. 3. p53 RNA editing. FIG. 3 demonstrates that a CIRT5-ADAR editor delivered to cells along with gRNAs can edit p53 mismatch sites and the highest editing is observed when the mismatch occurs 15 nucleotides from the 5' end of the gRNA sequence. CIRT5-ADAR editor delivered to cells along with a p53 reporter vector. Guide tiling along the mismatch site and monitoring editing efficiency by Sanger sequencing RT-qPCR products. HEK293T cells with

CIRTS-ADAR and 2TAR guides with lipofectamine 2000 (P53 R175H reporter (12 ng)+CIRTS-ADAR (125 ng)+guide (100 ng)).

[0060] FIG. 4: CIRTS degrader panel targeting the PMP22 transcript in NIH-3T3 cells measured by RT-qPCR. Each CIRTS degrader is normalized to a gRNA-matched, CIRTS-GFP negative control. Data are mean $\pm$ SEM; n=3 or more biological replicates.

[0061] FIG. 5: CIRTS Y2-based degrader can decrease SNCA transcript levels. Values shown as mean $\pm$ SEM from n=3 biological replicates. Student's t-test: * p<0.05 and ** p<0.01

[0062] FIG. 6A-B: Dual luciferase reporter system for assessing translational activators targeting native UTR contexts. A) Dual luciferase reporter expresses two separate luciferase transcripts. Only one (Firefly luciferase, Fluc) will carry the target UTR sequence. B) CIRTS activators directed by gRNA complementary to the target UTR are expected to specifically increase Fluc translation, and not that of Nano luciferase (Nluc).

[0063] FIG. 7: CIRTS activators that boost Fluc expression when targeted to the PMP22 3' UTR and normalized to an effector matched NT gRNA control. Data are mean $\pm$ SEM; n=3 or more biological replicates.

[0064] FIG. 8: CIRTS-Y1 activators can boost protein levels in JAG1 and SNC1a reporter systems. Data are mean $\pm$ SEM; n=3 biological replicates. Significance determined by Student's t-test.

[0065] FIG. 9: CIRTS-Y1 increases SCN1a proteins levels in Neuro 2a cells when directed by a gRNA complimentary to the SCN1a transcript 3' UTR (g1) compared to a non-

targeting gRNA (NT). Band intensity quantification completed in EmpiriaStudio. n=2 biological replicates.

DETAILED DESCRIPTION

[0066] Epitranscriptomic regulation controls information flow through the central dogma and provides unique opportunities for manipulating cell state at the RNA level. However, both fundamental mechanistic studies and potential translational applications are impeded by a lack of effective methods to target specific RNAs with effector proteins. Here, the inventors present the design and validation of a CRISPR/Cas-inspired RNA targeting system (CIRTS), a new strategy for constructing programmable systems. The inventors show that CIRTS is a simple and generalizable approach to deliver a range of effector proteins, including nucleases, degradation machinery, and translational activators, to target transcripts. CIRTS complexes are not only smaller than naturally-occurring CRISPR/Cas programmable RNA binding systems, but can be built entirely from human parts. The small size and human-derived nature of CIRTS provides a less perturbative method for fundamental RNA regulatory studies as well as a potential strategy to avoid immune issues when applied to epitranscriptome-modulating therapies.

I. RNA Effector Domains

[0067] The RNA effector domain may be a polypeptide or functional portion thereof of any one of SEQ ID NOs: 19-39, 60-62, 69-76, 113-121, 136-143 and combinations and fusion proteins thereof. RNA effector domains are described herein. The Table below exemplifies functions of the RNA effector domain.

Description	Exemplary Function	Seq ID
ZFP36	ZFP36 (ZFP36 Ring Finger Protein) is a Zinc-finger RNA-binding protein that destabilizes several cytoplasmic AU-rich element (ARE)-containing mRNA transcripts by promoting their poly(A) tail removal or deadenylation, and hence providing a mechanism for attenuating protein synthesis	32
YTHDF2	YTH N6-Methyladenosine RNA Binding Protein 2 (YTHDF2) specifically recognizes and binds N6-methyladenosine	33
CNOT7	CCR4-NOT Transcription Complex Subunit 7 (CNOT7) has 3'-5' poly(A) exoribonuclease activity for poly(A) RNA substrate.	34
GCN2	General control nonderepressible 2 (GCN2) phosphorylates eIF2 α causing it to function as a competitive inhibitor of eIF2B and therefore decreasing translation.	22
4EBP1	Eukaryotic translation initiation factor 4E-binding protein 1 (4EBP1) is a repressor of translation initiation that regulates EIF4E activity by preventing its assembly into the eIF4F complex: hypophosphorylated form competes with EIF4G1/EIF4G3 and strongly binds to EIF4E, leading to repress translation.	23
NANOS3	NANOS3 is a zinc-finger RNA-binding protein that interacts with the ubiquitously expressed protein Pumilio. The NANOS3-Pumilio complex binds to the 3' UTR of target mRNA and represses translation.	24
EIF4E	Eukaryotic translation initiation factor 4E (eIF4E) recognizes and binds the 7-methylguanosine-containing mRNA cap during an early step in the initiation of protein synthesis and facilitates ribosome binding by inducing the unwinding of the mRNAs secondary structures	35
EIF4A	Eukaryotic initiation factor 4A-I (eIF4A-I) is an ATP-dependent RNA helicase which is a subunit of the eIF4F complex involved in cap recognition and is required for mRNA binding to ribosome.	36
PAIP	Polyadenylate-binding protein-interacting protein 1 (PAIP) acts as a coactivator in the regulation of translation initiation of poly(A)-containing mRNAs. Its stimulatory activity on translation can be mediated via its action on PABPC1.	37

-continued

Description	Exemplary Function	Seq ID
YTHDF1	Specifically recognizes and binds N6-methyladenosine (m6A)-containing mRNAs, and promotes mRNA translation efficiency.	38
YTHDF3	Specifically recognizes and binds N6-methyladenosine (m6A)-containing RNAs and promotes RNA translation efficiency.	39
EIF2S2	Functions in the early steps of protein synthesis by forming a ternary complex with GTP and initiator tRNA. This complex binds to a 40S ribosomal subunit, followed by mRNA binding to form a 43S preinitiation complex. Junction of the 60S ribosomal subunit to form the 80S initiation complex is preceded by hydrolysis of the GTP bound to eIF-2 and release of an eIF-2-GDP binary complex.	28
HSPB1	Heat shock protein B1 has been demonstrated to bind eIF4E and repress translation, but it has also been identified in screens to positively regulation mRNA stability and translation.	29
YWHAE	YWHAE is a member of the 14-3-3 protein family that has been implicated in various regulatory processes and has been identified in screens to positively regulation mRNA stability and translation.	30
SF3B3	Involved in pre-mRNA splicing as a component of the splicing factor SF3B complex, a constituent of the spliceosome. SF3B complex is required for "A" complex assembly formed by the stable binding of U2 snRNP to the branchpoint sequence (BPS) in pre-mRNA.	31
SNRPA	Component of the spliceosomal U1 snRNP, which is essential for recognition of the pre-mRNA 5' splice-site and the subsequent assembly of the spliceosome. U1 snRNP is the first snRNP to interact with pre-mRNA.	60
TNRC6B	Trinucleotide repeat-containing gene 6B (TNRC6B) plays a role in RNA-mediated transcription degradation and can recruit deadenylase complexes to destabilize target mRNA.	61
DAZ4	Deleted in azoospermia protein 4 (DAZ4) is an RNA-binding protein that binds to the 3' UTR of target mRNA and increases their translation by interacting with PABP.	62
YB-1	YB-1 is a DNA- and RNA-binding protein involved in various processes, such as translational repression, RNA stabilization, mRNA splicing, DNA repair and transcription regulation. YB-1 can act as a RNA-binding protein: binds preferentially to the 5'-[CU]CUGCG-3' RNA motif and can specifically recognizes mRNA transcripts modified by C5-methylcytosine (m5C). YB-1 can promote mRNA stabilization: acts by binding to m5C-containing mRNAs and recruiting the mRNA stability maintainer ELAVL1, thereby preventing mRNA decay. YB-1 can contribute to the regulation of translation by modulating the interaction between the mRNA and eukaryotic initiation factors (By similarity). YB-1 can play a role in RNA composition of extracellular exosomes by defining the sorting of small non-coding RNAs, such as tRNAs, Y RNAs, Vault RNAs and miRNAs. YB-1 may sort RNAs in exosomes by recognizing and binding C5-methylcytosine (m5C)-containing RNAs. YB-1 is also involved in pre-mRNA alternative splicing regulation: binds to splice sites in pre-mRNA and regulates splice site selection.	69
ADAR2 (E488Q)	ADAR2 (E488Q) features a point mutation compared to ADAR2 that increases its base editing catalytic rate.	70
ADAR2 (wt)	Catalyzes the hydrolytic deamination of adenosine to inosine in double-stranded RNA (dsRNA) referred to as A-to-I RNA editing.	71
FXR1	Fragile X-related-1 (FXR1) binds to AU-rich elements and stabilizes the expression of target mRNA. FXR1 has also been demonstrated to interact with eIF4A and eIF4E.	74
APPLE	APPLE is a micropeptide that is encoded by a non-coding RNA transcript that promotes translation by facilitating the PABP-eIF4G interaction.	75
eIF4E2	Similar to eIF4E, eIF4E2 also binds the 7-methylguanosine-containing mRNA cap, but unlike eIF4E, cannot recruit eIF4G. Therefore, eIF4E2 inhibits translation by preventing eIF4F complex formation and translation initiation.	76

II. RNA Hairpin Binding Domains and Hairpin Structures

[0068] Various RNA hairpin binding domains and hairpin structures that they bind are known in the art and can be used in the systems, compositions, fusion proteins, kits, vectors,

and methods of the disclosure. For example, aspects include a RNA hairpin binding domain and hairpin structure according to the following table (Table 1), which lists proteins comprising RNA hairpin binding domains and the hairpin structure to which they specifically bind"

TABLE 1

RNA hairpin binding domain	
RNA hairpin binding domain	Hairpin structure
MS2 coat protein (MCP) (bacteriophage)	AAACAUGAGGAUUACCCAUGU (SEQ ID NO: 42) AAACAUGAGGAUCACCCAUGU (SEQ ID NO: 43)
λ N (bacteriophage)	GCCUGAAGAAGGC (SEQ ID NO: 44); GCCUGAAAAAGGC (SEQ ID NO: 45)
PP7 (<i>Pseudomonas aeruginosa</i>)	UAAGGAGUUUAUUGGAAACCCUUA (SEQ ID NO: 46)
Qβ	AAACAUGAGGAUUACCCAUGU (SEQ ID NO: 47); AAACAUGAGGAUCACCCAUGU (SEQ ID NO: 48); AUGCAUGUCUAAGACAGCAU (SEQ ID NO: 49)
GA	AAACAUGAGGAUUACCCAUGU (SEQ ID NO: 50); AAACAUGAGGAUCACCCAUGU (SEQ ID NO: 51); AAAACAUAAAGGAAAACCUAUGUU (SEQ ID NO: 52)
Bovine immunodeficiency virus (BIV) transactivator of transcription (tat)	GGCUCGUGUAGCUCAUUAAGCUCGAGCC (SEQ ID NO: 53)
KU70 (HUMAN)	GGCGUCCCUCCGAAGCUGCGCGCUCGGUCGAACAGGACG ACC (SEQ ID NO: 54)
NUCLEOLIN (HUMAN)	UCCCGA; GGCCGAAAUCCCGAAUGAGGCC (SEQ ID NO: 56); GGAUGCCUCCCGAGUGCAUCC (SEQ ID NO: 57).

[0069] It is contemplated that multiple RNA hairpin binding domains and/or the Effector domain may be used in a multiplexed fashion by using RNA hairpin binding domains that bind to different hairpin structures to target multiple different RNAs in the same cell. Different targeted RNAs may be modulated in the same or in different ways. For example, one RNA may be modulation with translational activation, while a second RNA may be modulated with translational repression in the same cell. Therefore, the systems of the disclosure can be used in a multiplexed fashion for the modulation of at least 2, 3, 4, 5, 6, 7, 8, 9, 10 or more RNAs in one cell, tissue, or organisms.

III. Stabilizer Domain

[0070] Exemplary stabilizer domains include those in the table below as well as variants of those represented by the amino acid sequence below.

Description	Amino Acid Sequence
ORF5	MDDPSFLTGRSTYAKRRRARMNVCKCGAILHNNKDCRSSTISGHKL DRLRFVKEGRVALEGETPVYRTWVKWVETEHINILETSDDDEE (SEQ ID NO: 58)
HBEGF	MRVTLSKPKQALATPNKEEHGKRKKKGKGLGKGRDPCLRKYKDFCI HGECKYVKELRAPSCICHPPGYHGERCHGLS (SEQ ID NO: 59)
β-defensin	MGIINTLQKYYCRVRGGRCAVLSCLPKEEQIGKCKSTRGRKCCRKK (SEQ ID NO: 14)

[0071] Stabilizer proteins are proteins that can bind to the gRNA sequence non-specifically, in some embodiments via charge-based interactions, and can be used to stabilize and protect the guiding RNA prior to target engagement.

IV. Nucleic Acids

[0072] In certain aspects, there are recombinant nucleic acids encoding the proteins, polypeptides, regulatory domains, or RNA targeting molecules described herein.
[0073] As used in this application, the term “polynucleotide” refers to a nucleic acid molecule that either is recombinant or has been isolated free of total genomic nucleic acid. Included within the term “polynucleotide” are oligonucleotides (nucleic acids 100 residues or fewer in length), recombinant vectors, including, for example, plasmids, cosmids, phage, viruses, and the like. Polynucleotides include, in certain aspects, regulatory sequences, isolated substan-

tially away from their naturally occurring genes or protein encoding sequences. Polynucleotides may be single-stranded (coding or antisense) or double-stranded, and may be RNA, DNA (genomic, cDNA or synthetic), analogs thereof, or a combination thereof. Additional coding or non-coding sequences may, but need not, be present within a polynucleotide.

[0074] In this respect, the term “gene,” “polynucleotide,” or “nucleic acid” is used to refer to a nucleic acid that encodes a protein, polypeptide, or peptide (including any sequences required for proper transcription, post-translational modification, or localization). As will be understood by those in the art, this term encompasses genomic sequences, expression cassettes, cDNA sequences, and smaller engineered nucleic acid segments that express, or may be adapted to express, proteins, polypeptides, domains, peptides, fusion proteins, and mutants. A nucleic acid encoding all or part of a polypeptide may contain a contiguous nucleic acid sequence encoding all or a portion of such a polypeptide. It also is contemplated that a particular polypeptide may be encoded by nucleic acids containing variations having slightly different nucleic acid sequences but, nonetheless, encode the same or substantially similar protein (see above).

[0075] In particular aspects, there are isolated nucleic acid segments and recombinant vectors incorporating nucleic acid sequences that encode a polypeptides (e.g., a polymerase, RNA polymerase, one or more truncated polymerase domains or interaction components that are polypeptides) that drive gene transcription dependent on polymerase activity from the polymerase domains when the interaction components interact. The term “recombinant” may be used in conjunction with a polypeptide or the name of a specific polypeptide, and this generally refers to a polypeptide produced from a nucleic acid molecule that has been manipulated in vitro or that is a replication product of such a molecule . . .

[0076] The nucleic acid segments, regardless of the length of the coding sequence itself, may be combined with other nucleic acid sequences, such as promoters, polyadenylation signals, additional restriction enzyme sites, multiple cloning sites, other coding segments, and the like, such that their overall length may vary considerably. It is therefore contemplated that a nucleic acid fragment of almost any length may be employed, with the total length preferably being limited by the ease of preparation and use in the intended recombinant nucleic acid protocol. In some cases, a nucleic acid sequence may encode a polypeptide sequence with additional heterologous coding sequences, for example to allow for purification of the polypeptide, transport, secretion, post-translational modification, or for therapeutic benefits such as targeting or efficacy. As discussed above, a tag or other heterologous polypeptide may be added to the modified polypeptide-encoding sequence, wherein “heterologous” refers to a polypeptide that is not the same as the modified polypeptide.

[0077] In certain aspects, there are polynucleotide variants having substantial identity to the sequences disclosed herein; those comprising at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% or higher sequence identity, including all values and ranges there between, compared to a polynucleotide sequence provided herein using the methods described herein (e.g., BLAST analysis using standard parameters). In certain aspects, the isolated polynucleotide

will comprise a nucleotide sequence encoding a polypeptide that has at least 90%, preferably 95% and above, identity to an amino acid sequence described herein, over the entire length of the sequence; or a nucleotide sequence complementary to said isolated polynucleotide.

A. Vectors

[0078] Polypeptides may be encoded by a nucleic acid molecule. The nucleic acid molecule can be in the form of a nucleic acid vector. The term “vector” is used to refer to a carrier nucleic acid molecule into which a heterologous nucleic acid sequence can be inserted for introduction into a cell where it can be replicated and expressed. A nucleic acid sequence can be “heterologous,” which means that it is in a context foreign to the cell in which the vector is being introduced or to the nucleic acid in which is incorporated, which includes a sequence homologous to a sequence in the cell or nucleic acid but in a position within the host cell or nucleic acid where it is ordinarily not found. Vectors include DNAs, RNAs, plasmids, cosmids, viruses (bacteriophage, animal viruses, and plant viruses), and artificial chromosomes (e.g., YACs). One of skill in the art would be well equipped to construct a vector through standard recombinant techniques (for example Sambrook et al., 2001; Ausubel et al., 1996, both incorporated herein by reference). Vectors may be used in a host cell to produce a polymerase, RNA polymerase, one or more truncated polymerase domains or interaction components that are fused, attached or linked to the one or more truncated RNA polymerase domains.

[0079] The term “expression vector” refers to a vector containing a nucleic acid sequence coding for at least part of a gene product capable of being transcribed. In some cases, RNA molecules are then translated into a protein, polypeptide, or peptide. Expression vectors can contain a variety of “control sequences,” which refer to nucleic acid sequences necessary for the transcription and possibly translation of an operably linked coding sequence in a particular host organism. In addition to control sequences that govern transcription and translation, vectors and expression vectors may contain nucleic acid sequences that serve other functions as well and are described herein.

B. Cells

[0080] The disclosure provides methods for modifying a target RNA of interest, in particular in prokaryotic cells, eukaryotic cells, tissues, organs, or organisms, more in particular in mammalian cells, tissues, organs, or organisms. The target RNA may be comprised in a nucleic acid molecule within a cell. In some aspects, the target RNA is in a eukaryotic cell, such as a mammalian cell or a plant cell. The mammalian cell may be a human, non-human primate, bovine, porcine, rodent or mouse cell. The cell may be a non-mammalian eukaryotic cell such as poultry, fish or shrimp. The plant cell may be of a crop plant such as cassava, corn, sorghum, wheat, or rice. The plant cell may also be of an algae, tree or vegetable. The modulation of the RNA induced in the cell by the methods, systems, and compositions of the disclosure may be such that the cell and progeny of the cell are altered for improved production of biologic products such as an antibody, starch, alcohol or other desired cellular output. The modulation of the RNA induced in the cell may be such that the cell and progeny of the cell include an alteration that changes the biologic product produced.

[0081] The mammalian cell may be a human or non-human mammal, e.g., primate, bovine, ovine, porcine, canine, rodent, Leporidae such as monkey, cow, sheep, pig, dog, rabbit, rat or mouse cell. The cell may be a non-mammalian eukaryotic cell such as poultry bird (e.g., chicken), vertebrate fish (e.g., salmon) or shellfish (e.g., oyster, clam, lobster, shrimp) cell. The cell may also be a plant cell. The plant cell may be of a monocot or dicot or of a crop or grain plant such as cassava, corn, sorghum, soybean, wheat, oat or rice. The plant cell may also be of an algae, tree or production plant, fruit or vegetable (e.g., trees such as citrus trees, e.g., orange, grapefruit or lemon trees; peach or nectarine trees; apple or pear trees; nut trees such as almond or walnut or pistachio trees; nightshade plants; plants of the genus *Brassica*; plants of the genus *Lactuca*; plants of the genus *Spinacia*; plants of the genus *Capsicum*; cotton, tobacco, asparagus, carrot, cabbage, broccoli, cauliflower, tomato, eggplant, pepper, lettuce, spinach, strawberry, blueberry, raspberry, blackberry, grape, coffee, cocoa, etc.).

[0082] As used herein, the terms “cell,” “cell line,” and “cell culture” may be used interchangeably. All of these terms also include their progeny, which is any and all subsequent generations. It is understood that all progeny may not be identical due to deliberate or inadvertent mutations. In the context of expressing a heterologous nucleic acid sequence, “host cell” refers to a prokaryotic or eukaryotic cell, and it includes any transformable organism that is capable of replicating a vector or expressing a heterologous gene encoded by a vector. A host cell can, and has been, used as a recipient for vectors or viruses. A host cell may be “transfected” or “transformed,” which refers to a process by which exogenous nucleic acid, such as a recombinant protein-encoding sequence, is transferred or introduced into the host cell. A transformed cell includes the primary subject cell and its progeny.

[0083] Some vectors may employ control sequences that allow it to be replicated and/or expressed in both prokaryotic and eukaryotic cells. One of skill in the art would further understand the conditions under which to incubate all of the above described host cells to maintain them and to permit replication of a vector. Also understood and known are techniques and conditions that would allow large-scale production of vectors, as well as production of the nucleic acids encoded by vectors and their cognate polypeptides, proteins, or peptides.

C. Expression Systems

[0084] Numerous expression systems exist that comprise at least a part or all of the compositions discussed above. Prokaryote- and/or eukaryote-based systems can be employed for use with an aspect to produce nucleic acid sequences, or their cognate polypeptides, proteins and peptides. For example, the vectors, fusion proteins, RNA hairpin binding proteins, RNA targeting molecules, Effector domain, and accessory proteins of the disclosure may utilize an expression system, such as an inducible or constitutive expression system. Many such systems are commercially and widely available.

[0085] The insect cell/baculovirus system can produce a high level of protein expression of a heterologous nucleic acid segment, such as described in U.S. Pat. Nos. 5,871,986, 4,879,236, both herein incorporated by reference, and which can be bought, for example, under the name MAXBAC®

2.0 from INVITROGEN® and BACPACK™ BACULOVIRUS EXPRESSION SYSTEM FROM CLONTECH®.

[0086] In addition to the disclosed expression systems, other examples of expression systems include STRATAGENER's COMPLETE CONTROL Inducible Mammalian Expression System, which involves a synthetic ecdysone-inducible receptor, or its pET Expression System, an *E. coli* expression system. Another example of an inducible expression system is available from INVITROGEN®, which carries the T-REX™ (tetracycline-regulated expression) System, an inducible mammalian expression system that uses the full-length CMV promoter. INVITROGEN® also provides a yeast expression system called the *Pichia methanolica* Expression System, which is designed for high-level production of recombinant proteins in the methylotrophic yeast *Pichia methanolica*. One of skill in the art would know how to express a vector, such as an expression construct, to produce a nucleic acid sequence or its cognate polypeptide, protein, or peptide.

D. Conjugation of Nucleic Acids and Polypeptides

[0087] Aspects of the disclosure relate to the conjugation of nucleic acids to polypeptides. Methods of conjugation of nucleic acids to polypeptides are known in the art and include those described below. Aspects of the disclosure relate to methods of making nucleic acid-polypeptide molecules and the molecules themselves wherein the nucleic acid has been conjugated to the polypeptide by way of a method described herein. One such example includes click chemistry. The “click reaction”, also known as “click chemistry” is a name often used to describe a stepwise variant of the Huisgen 1,3-dipolar cycloaddition of azides and alkynes to yield 1,2,3-triazole. This reaction is carried out under ambient conditions, or under mild microwave irradiation, typically in the presence of a Cu(I) catalyst, and with exclusive regioselectivity for the 1,4-disubstituted triazole product when mediated by catalytic amounts of Cu(I) salts [V. Rostovtsev, L. G. Green, V. V. Fokin, K. B. Sharpless, *Angew. Chem. Int. Ed.* 2002, 41, 2596; H. C. Kolb, M. Finn, K. B. Sharpless, *Angew. Chem., Int. Ed.* 2001, 40, 2004].

[0088] In other conjugation methods, a mutant form of the human DNA repair protein 06-alkylguanine-DNA alkyl-transferase reacts rapidly and specifically with 06-benzylguanine (BG) and also with derivatives that carry a large moiety linked to the benzyl group. With guanine as the leaving group, the benzyl moiety becomes covalently attached to a cysteine in the active site of the enzyme. The enzyme has also been mutagenized to become specific for 06-benzylcytosine (BC) in a similar manner. These enzyme domains (about 20 kDa) are commercially available as SNAP and CLIP tags, respectively.

[0089] A further conjugation method utilizes the Halo tag. The Halo tag makes use of a chemical reaction orthogonal to eukaryotes, i.e. the dehalogenation of haloalkane ligands, thus, leading to highly specific covalent labelling of the tag, and therefore protein, in both live and fixed cells.

E. Nucleic Acid Modifications

[0090] The oligonucleotides of the disclosure, such as the RNA targeting molecules and other nucleic acids described herein may have modifications, for example that increase the stability of the nucleic acid. In some aspects, the RNA targeting molecule is an oligonucleotide analog. The term

“oligonucleotide analog” refers to compounds which function like oligonucleotides but which have non-naturally occurring portions (e.g., that include one or more non-natural residues or linkages). Oligonucleotide analogs can have altered sugar moieties, altered base moieties or altered inter-sugar linkages. The term “oligomers” is intended to encompass oligonucleotides, oligonucleotide analogs or oligonucleosides. Thus, in speaking of “oligomers” reference is made to a series of nucleosides or nucleoside analogs that are joined via either natural phosphodiester bonds or other linkages, including the four atom linkers. Although the linkage generally is from the 3' carbon of one nucleoside to the 5' carbon of a second nucleoside, the term “oligomer” can also include other linkages such as 2'-5' linkages.

[0091] Oligonucleotide analogs also can include other modifications, particularly modifications that increase nuclease resistance, improve binding affinity, and/or improve binding specificity. For example, when the sugar portion of a nucleoside or nucleotide is replaced by a carbocyclic moiety, it is no longer a sugar. Moreover, when other substitutions, such a substitution for the inter-sugar phosphodiester linkage are made, the resulting material is no longer a true nucleic acid species. All such compounds are considered to be analogs. Throughout this specification, reference to the sugar portion of a nucleic acid species shall be understood to refer to either a true sugar or to a species taking the structural place of the sugar of wild type nucleic acids. Moreover, reference to inter-sugar linkages shall be taken to include moieties serving to join the sugar or sugar analog portions in the fashion of wild type nucleic acids.

[0092] The present disclosure concerns modified oligonucleotides, i.e., oligonucleotide analogs or oligonucleosides, and methods for effecting the modifications. These modified oligonucleotides and oligonucleotide analogs may exhibit increased chemical and/or enzymatic stability relative to their naturally occurring counterparts. Extracellular and intracellular nucleases generally do not recognize and therefore do not bind to the backbone-modified compounds. When present as the protonated acid form, the lack of a negatively charged backbone may facilitate cellular penetration.

[0093] The modified internucleoside linkages are intended to replace naturally-occurring phosphodiester-5'-methylene linkages with four atom linking groups to confer nuclease resistance and enhanced cellular uptake to the resulting compound. Preferred linkages have structure CH₂--RA--NR₁ CH₂, CH₂--NR₁--RA--CH₂, RA--NR₁--CH₂--CH₂, CH₂--CH₂--NR₁--RA, CH₂--CH₂--RA--NR₁, or NR₁--RA--CH₂--CH₂ where RA is O or NR₂.

[0094] Modifications to the nucleic acid molecules of the disclosure may be achieved using solid supports which may be manually manipulated or used in conjunction with a DNA synthesizer using methodology commonly known to those skilled in DNA synthesizer art. Generally, the procedure involves functionalizing the sugar moieties of two nucleosides which will be adjacent to one another in the selected sequence. In a 5' to 3' sense, an “upstream” synthon is modified at its terminal 3' site, while a “downstream” synthon is modified at its terminal 5' site.

[0095] Oligonucleosides linked by hydrazines, hydroxylamines, and other linking groups can be protected by a dimethoxytrityl group at the 5'-hydroxyl and activated for coupling at the 3'-hydroxyl with cyanoethyl-diisopropylphosphite moieties. These compounds can be inserted into

any desired sequence by standard, solid phase, automated DNA synthesis techniques. One of the most popular processes is the phosphoramidite technique. Oligonucleotides containing a uniform backbone linkage can be synthesized by use of CPG-solid support and standard nucleic acid synthesizing machines such as Applied Biosystems Inc. 380B and 394 and Milligen/Bioscience 7500 and 8800s. The initial nucleotide (number 1 at the 3'-terminus) is attached to a solid support such as controlled pore glass. In sequence specific order, each new nucleotide is attached either by manual manipulation or by the automated synthesizer system.

[0096] Free amino groups can be alkylated with, for example, acetone and sodium cyanoborohydride in acetic acid. The alkylation step can be used to introduce other, useful, functional molecules on the macromolecule. Such useful functional molecules include but are not limited to reporter molecules, RNA cleaving groups, groups for improving the pharmacokinetic properties of an oligonucleotide, and groups for improving the pharmacodynamic properties of an oligonucleotide. Such molecules can be attached to or conjugated to the macromolecule via attachment to the nitrogen atom in the backbone linkage. Alternatively, such molecules can be attached to pendent groups extending from a hydroxyl group of the sugar moiety of one or more of the nucleotides. Examples of such other useful functional groups are provided by WO1993007883, which is herein incorporated by reference, and in other of the above-referenced patent applications.

[0097] Solid supports may include any of those known in the art for polynucleotide synthesis, including controlled pore glass (CPG), oxalyl controlled pore glass [53], Tenta-Gel Support—an aminopolyethyleneglycol derivatized support or Poros—a copolymer of polystyrene/divinylbenzene. Attachment and cleavage of nucleotides and oligonucleotides can be effected via standard procedures [55]. As used herein, the term solid support further includes any linkers (e.g., long chain alkyl amines and succinyl residues) used to bind a growing oligonucleoside to a stationary phase such as CPG.

1. Locked Nucleotides

[0098] In some aspects, the nucleic acid of the disclosure, such as the RNA targeting molecule comprises a locked nucleic acid. A locked nucleic acid (LNA or L_n), also referred to as inaccessible RNA, is a modified RNA nucleotide. The ribose moiety of an LNA nucleotide is modified with an extra bridge connecting the 2' oxygen and 4' carbon. The bridge “locks” the ribose in the 3'-endo (North) conformation, which is often found in the A-form duplexes. LNA nucleotides can be mixed with DNA or RNA residues in the oligonucleotide whenever desired and hybridize with DNA or RNA according to Watson-Crick base-pairing rules. Such oligomers are synthesized chemically and are commercially available. The locked ribose conformation enhances base stacking and backbone pre-organization. This significantly increases the hybridization properties (melting temperature) of oligonucleotides.

2. Ethylene Bridged Nucleotides

[0099] In some aspects, the nucleic acid of the disclosure, such as the RNA targeting molecule comprises one or more ethylene bridged nucleotides. Ethylene-bridged nucleic

acids (ENA or En) are modified nucleotides with a 2'-O, 4'C ethylene linkage. Like locked nucleotides, these nucleotides also restrict the sugar puckering to the N-conformation of RNA.

3. Peptide Nucleic Acids

[0100] In some aspects, the nucleic acid of the disclosure, such as the RNA targeting molecule comprises one or more peptide nucleic acids. Peptide nucleic acids (PNA or Pn) mimic the behavior of DNA and binds complementary nucleic acid strands. The term, "peptide," when used herein may also refer to a peptide nucleic acid. PNA is an artificially synthesized polymer similar to DNA or RNA. DNA and RNA have a deoxyribose and ribose sugar backbone, respectively, whereas PNA's backbone is composed of repeating N-(2-aminoethyl)-glycine units linked by peptide bonds. The various purine and pyrimidine bases are linked to the backbone by a methylene bridge ($-\text{CH}_2-$) and a carbonyl group ($-(\text{C}=\text{O})-$). PNAs are depicted like peptides, with the N-terminus at the first (left) position and the C-terminus at the last (right) position.

[0101] Since the backbone of PNAs contains no charged phosphate groups, the binding between PNA/DNA strands is stronger than between DNA/DNA strands due to the lack of electrostatic repulsion. PNAs are not easily recognized by either nucleases or proteases, making them resistant to degradation by enzymes. PNAs are also stable over a wide pH range. In some aspects, the PNAs described herein have improved cytosolic delivery over other PNAs.

4. 5' (E)-Vinyl-Phosphonate (VP) Modification

[0102] In some aspects, the nucleic acid of the disclosure, such as the RNA targeting molecule comprises one or more 5' (E)-vinyl-phosphonate (VP) modifications. 5'-Vinyl-phosphonate modifications (metabolically stable phosphate mimics) have been reported to enhance the metabolic stability and the potency of oligonucleotides.

5. Morpholinos

[0103] In some aspects, the nucleic acid of the disclosure, such as the RNA targeting molecule comprises a morpholino. Morpholinos are synthetic molecules that are the product of a redesign of natural nucleic acid structure. Usually 25 bases in length, they bind to complementary sequences of RNA or single-stranded DNA by standard nucleic acid base-pairing. In terms of structure, the difference between Morpholinos and DNA is that, while Morpholinos have standard nucleic acid bases, those bases are bound to methylenemorpholine rings linked through phosphorodiamidate groups instead of phosphates. The figure compares the structures of the two strands depicted there, one of RNA and the other of a Morpholino. Replacement of anionic phosphates with the uncharged phosphorodiamidate groups eliminates ionization in the usual physiological pH range, so Morpholinos in organisms or cells are uncharged molecules. The entire backbone of a Morpholino is made from these modified subunits.

V. Delivery Vehicles

[0104] The current disclosure contemplates several delivery systems compatible with nucleic acids that provide for roughly uniform distribution and have controllable rates of release. A variety of different media are described below that

are useful in creating nucleic acid delivery systems. It is not intended that any one medium or carrier is limiting to the present invention. Note that any medium or carrier may be combined with another medium or carrier; for example, in one aspect a polymer microparticle carrier attached to a compound may be combined with a gel medium.

[0105] Carriers or mediums contemplated by this disclosure comprise a material selected from the group comprising gelatin, collagen, cellulose esters, dextran sulfate, pentosan polysulfate, chitin, saccharides, albumin, fibrin sealants, synthetic polyvinyl pyrrolidone, polyethylene oxide, polypropylene oxide, block polymers of polyethylene oxide and polypropylene oxide, polyethylene glycol, acrylates, acrylamides, methacrylates including, but not limited to, 2-hydroxyethyl methacrylate, poly (ortho esters), cyanoacrylates, gelatin-resorcinol-aldehyde type bioadhesives, polyacrylic acid and copolymers and block copolymers thereof.

a. Microparticles

[0106] Some aspects of the present disclosure contemplate a delivery system comprising a microparticle. Preferably, microparticles comprise liposomes, nanoparticles, microspheres, nanospheres, microcapsules, and nanocapsules. Preferably, some microparticles contemplated by the present invention comprise poly (lactide-co-glycolide), aliphatic polyesters including, but not limited to, poly-glycolic acid and poly-lactic acid, hyaluronic acid, modified polysaccharides, chitosan, cellulose, dextran, polyurethanes, polyacrylic acids, pseudo-poly (amino acids), polyhydroxybutyrate-related copolymers, polyanhydrides, polymethylmethacrylate, poly (ethylene oxide), lecithin and phospholipids.

B. Liposomes

[0107] One aspect of the disclosure contemplates liposomes capable of attaching and releasing nucleic acids conjugates, polypeptides, and fusion proteins as described herein. Liposomes are microscopic spherical lipid bilayers surrounding an aqueous core that are made from amphiphilic molecules such as phospholipids. For example, a liposome may trap a nucleic acid between the hydrophobic tails of the phospholipid micelle. Water soluble agents can be entrapped in the core and lipid-soluble agents can be dissolved in the shell-like bilayer. Liposomes have a special characteristic in that they enable water soluble and water insoluble chemicals to be used together in a medium without the use of surfactants or other emulsifiers. Liposomes can form spontaneously by forcefully mixing phospholipids in aqueous media. Water soluble compounds are dissolved in an aqueous solution capable of hydrating phospholipids. Upon formation of the liposomes, therefore, these compounds are trapped within the aqueous liposomal center. The liposome wall, being a phospholipid membrane, holds fat soluble materials such as oils. Liposomes provide controlled release of incorporated compounds. In addition, liposomes can be coated with water soluble polymers, such as polyethylene glycol to increase the pharmacokinetic half-life. One aspect of the present invention contemplates an ultra high-shear technology to refine liposome production, resulting in stable, unilamellar (single layer) liposomes having specifically designed structural characteristics. These unique properties of liposomes allow the simultaneous storage of normally immiscible compounds and the capability of their controlled release.

[0108] In some aspects, the disclosure contemplates cationic and anionic liposomes, as well as liposomes having neutral lipids. Preferably, cationic liposomes comprise negatively-charged materials by mixing the materials and fatty acid liposomal components and allowing them to charge-associate. Clearly, the choice of a cationic or anionic liposome depends upon the desired pH of the final liposome mixture. Examples of cationic liposomes include lipofectin, lipofectamine, and lipofectace.

[0109] One aspect of the present disclosure contemplates a delivery system comprising liposomes that provides controlled release of at least one molecule described herein. Preferably, liposomes that are capable of controlled release: i) are biodegradable and non-toxic; ii) carry both water and oil soluble compounds; iii) solubilize recalcitrant compounds; iv) prevent compound oxidation; v) promote protein stabilization; vi) control hydration; vii) control compound release by variations in bilayer composition such as, but not limited to, fatty acid chain length, fatty acid lipid composition, relative amounts of saturated and unsaturated fatty acids, and physical configuration; viii) have solvent dependency; iv) have pH-dependency and v) have temperature dependency.

[0110] The compositions of liposomes are broadly categorized into two classifications. Conventional liposomes are generally mixtures of stabilized natural lecithin (PC) that may comprise synthetic identical-chain phospholipids that may or may not contain glycolipids. Special liposomes may comprise: i) bipolar fatty acids; ii) the ability to attach antibodies for tissue-targeted therapies; iii) coated with materials such as, but not limited to lipoprotein and carbohydrate; iv) multiple encapsulation and v) emulsion compatibility.

[0111] Liposomes may be easily made in the laboratory by methods such as, but not limited to, sonication and vibration. Alternatively, compound-delivery liposomes are commercially available. For example, Collaborative Laboratories, Inc. are known to manufacture custom designed liposomes for specific delivery requirements.

C. Microspheres, Microparticles and Microcapsules

[0112] Microspheres and microcapsules are useful due to their ability to maintain a generally uniform distribution, provide stable controlled compound release and are economical to produce and dispense. Preferably, an associated delivery gel or the compound-impregnated gel is clear or, alternatively, said gel is colored for easy visualization by medical personnel.

[0113] Microspheres are obtainable commercially (Prolease™, Alkermes: Cambridge, Mass.). For example, a freeze dried medium comprising at least one therapeutic agent is homogenized in a suitable solvent and sprayed to manufacture microspheres in the range of 20 to 90 μm. Techniques are then followed that maintain sustained release integrity during phases of purification, encapsulation and storage. Scott et al., Improving Protein Therapeutics With Sustained Release Formulations, *Nature Biotechnology*, Volume 16:153-157 (1998). Modification of the microsphere composition by the use of biodegradable polymers can provide an ability to control the rate of nucleic acid release. Miller et al., Degradation Rates of Oral Resorbable Implants {Polylactates and Polyglycolates: Rate Modification and Changes in PLA/PGA Copolymer Ratios, *J. Biomed. Mater. Res.*, Vol. 11:711-719 (1977).

[0114] Alternatively, a sustained or controlled release microsphere preparation is prepared using an in-water drying method, where an organic solvent solution of a biodegradable polymer metal salt is first prepared. Subsequently, a dissolved or dispersed medium of a nucleic acid is added to the biodegradable polymer metal salt solution. The weight ratio of a nucleic acid to the biodegradable polymer metal salt may for example be about 1:100000 to about 1:1, preferably about 1:20000 to about 1:500 and more preferably about 1:10000 to about 1:500. Next, the organic solvent solution containing the biodegradable polymer metal salt and nucleic acid is poured into an aqueous phase to prepare an oil/water emulsion. The solvent in the oil phase is then evaporated off to provide microspheres. Finally, these microspheres are then recovered, washed and lyophilized. Thereafter, the microspheres may be heated under reduced pressure to remove the residual water and organic solvent.

[0115] Other methods useful in producing microspheres that are compatible with a biodegradable polymer metal salt and nucleic acid mixture are: i) phase separation during a gradual addition of a coacervating agent; ii) an in-water drying method or phase separation method, where an anti-flocculant is added to prevent particle agglomeration and iii) by a spray-drying method.

[0116] In one aspect, the present invention contemplates a medium comprising a microsphere or microcapsule capable of delivering a controlled release of a nucleic acid for a duration of approximately between 1 day and 6 months. In one aspect, the microsphere or microparticle may be colored to allow the medical practitioner the ability to see the medium clearly as it is dispensed. In another aspect, the microsphere or microcapsule may be clear. In another aspect, the microsphere or microparticle is impregnated with a radio-opaque fluoroscopic dye.

[0117] Controlled release microcapsules may be produced by using known encapsulation techniques such as centrifugal extrusion, pan coating and air suspension. Such microspheres and/or microcapsules can be engineered to achieve desired release rates. For example, Oliosphere™ (Macromed) is a controlled release microsphere system. These particular microsphere's are available in uniform sizes ranging between 5-500 μm and composed of biocompatible and biodegradable polymers. Specific polymer compositions of a microsphere can control the nucleic acid release rate such that custom-designed microspheres are possible, including effective management of the burst effect. ProMaxx™ (Epic Therapeutics, Inc.) is a protein-matrix delivery system. The system is aqueous in nature and is adaptable to standard pharmaceutical delivery models. In particular, ProMaxx™ are bioerodible protein microspheres that deliver both small and macromolecular drugs, and may be customized regarding both microsphere size and desired release characteristics.

[0118] In one aspect, a microsphere or microparticle comprises a pH sensitive encapsulation material that is stable at a pH less than the pH of the internal mesentery. The typical range in the internal mesentery is pH 7.6 to pH 7.2. Consequently, the microcapsules should be maintained at a pH of less than 7. However, if pH variability is expected, the pH sensitive material can be selected based on the different pH criteria needed for the dissolution of the microcapsules. The encapsulated nucleic acid, therefore, will be selected for the pH environment in which dissolution is desired and stored in a pH preselected to maintain stability. Examples of

pH sensitive material useful as encapsulants are Eudragit™ L-100 or S-100 (Rohm GMBH), hydroxypropyl methylcellulose phthalate, hydroxypropyl methylcellulose acetate succinate, polyvinyl acetate phthalate, cellulose acetate phthalate, and cellulose acetate trimellitate. In one aspect, lipids comprise the inner coating of the microcapsules. In these compositions, these lipids may be, but are not limited to, partial esters of fatty acids and hexitol anhydrides, and edible fats such as triglycerides. Lew C. W., Controlled-Release pH Sensitive Capsule And Adhesive System And Method. U.S. Pat. No. 5,364,634 (herein incorporated by reference).

[0119] In one aspect, the present invention contemplates a microparticle comprising a gelatin, or other polymeric cation having a similar charge density to gelatin (i.e., poly-L-lysine) and is used as a complex to form a primary microparticle. A primary microparticle is produced as a mixture of the following composition: i) Gelatin (60 bloom, type A from porcine skin), ii) chondroitin 4-sulfate (0.005%-0.1%), iii) glutaraldehyde (25%, grade 1), and iv) 1-ethyl-3-(3-dimethylaminopropyl)-carbodiimide hydrochloride (EDC hydrochloride), and ultra-pure sucrose (Sigma Chemical Co., St. Louis, Mo.). The source of gelatin is not thought to be critical; it can be from bovine, porcine, human, or other animal source. Typically, the polymeric cation is between 19,000-30,000 daltons. Chondroitin sulfate is then added to the complex with sodium sulfate, or ethanol as a coacervation agent.

[0120] Following the formation of a microparticle, a nucleic acid is directly bound to the surface of the microparticle or is indirectly attached using a “bridge” or “spacer”. The amino groups of the gelatin lysine groups are easily derivatized to provide sites for direct coupling of a compound. Alternatively, spacers (i.e., linking molecules and derivatizing moieties on targeting ligands) such as avidin-biotin are also useful to indirectly couple targeting ligands to the microparticles. Stability of the microparticle is controlled by the amount of glutaraldehyde-spacer crosslinking induced by the EDC hydrochloride. A controlled release medium is also empirically determined by the final density of glutaraldehyde-spacer crosslinks.

[0121] In one aspect, the present invention contemplates microparticles formed by spray-drying a composition comprising fibrinogen or thrombin with a nucleic acid. Preferably, these microparticles are soluble and the selected protein (i.e., fibrinogen or thrombin) creates the walls of the microparticles. Consequently, the nucleic acids are incorporated within, and between, the protein walls of the microparticle. Heath et al., Microparticles And Their Use In Wound Therapy. U.S. Pat. No. 6,113,948 (herein incorporated by reference). Following the application of the microparticles to living tissue, the subsequent reaction between the fibrinogen and thrombin creates a tissue sealant thereby releasing the incorporated compound into the immediate surrounding area.

[0122] One having skill in the art will understand that the shape of the microspheres need not be exactly spherical; only as very small particles capable of being sprayed or spread into or onto a surgical site (i.e., either open or closed). In one aspect, microparticles are comprised of a biocompatible and/or biodegradable material selected from the group consisting of polylactide, polyglycolide and copolymers of lactide/glycolide (PLGA), hyaluronic acid, modified polysaccharides and any other well known material.

VI. Proteinaceous and Nucleic Acid Compositions

[0123] The polypeptides or polynucleotides of the disclosure, such as the CIRT fusion proteins, stabilizer polypeptide, linker, RNA hairpin binding domain, NES, effector domain, tag, NLS, RNA targeting molecule, hairpin region of the RNA targeting molecule, helical region, or targeting region of the RNA targeting molecule may include 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, or 50 or more variant amino acids or nucleic acid substitutions or be at least 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% similar, identical, or homologous with at least, or at most 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 300, 400, 500, 550, 1000 or more contiguous amino acids or nucleic acids, or any range derivable therein, of SEQ ID NOs: 1-169.

[0124] The polypeptides or polynucleotides of the disclosure, such as the CIRT fusion proteins, stabilizer polypeptide, linker, RNA hairpin binding domain, NES, effector domain, tag, NLS, RNA targeting molecule, hairpin region of the RNA targeting molecule, helical region, or targeting region of the RNA targeting molecule may include 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 300, 400, 500, 550, 1000 or more contiguous amino acids, or any range derivable therein, of SEQ ID NOs: 1-169.

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[0128] In other aspects, the stabilizer polypeptide may comprise amino acids 1 to 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, or 90, or any derivable range therein of any one of SEQ ID NOs: 14, 58, or 59.

[0129] In some aspects, the stabilizer polypeptide may comprise 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, or 90 (or any derivable range therein) contiguous amino acids of any one of SEQ ID NOs: 14, 58, or 59.

[0130] In some aspects, the stabilizer polypeptide may comprise 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, or 90 (or any derivable range therein) contiguous amino acids of any one of SEQ ID NOs: 14, 58, or 59 that are at least, at most, or exactly 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%,

91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% similar, identical, or homologous with one of any one of SEQ ID NOs: 14, 58, or 59.

[0131] In some aspects, the RNA hairpin binding domain may comprise amino acids 1 to 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, or 102, (or any derivable range therein) of SEQ ID NO:16.

[0132] In some aspects, the RNA hairpin binding domain may comprise 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, or 102 (or any derivable range therein) contiguous amino acids of SEQ ID NO:16.

[0133] In some aspects, the RNA hairpin binding domain may comprise 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, or 102 (or any derivable range therein) contiguous amino acids of SEQ ID NO:16 that are at least, at most, or exactly 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% similar, identical, or homologous with one of SEQ ID NO:16.

[0134] In some aspects, the effector domain may comprise amino acids 1 to 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348,

[0136] In some aspects there is a nucleic acid molecule or polypeptide starting at position 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357.

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610, 611, 612, 613, 614, or 615 of any of SEQ ID NOS:
1-169 and comprising 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14,
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[0137] In some aspects, the effector domain may comprise 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, 359, 360, 361, 362, 363, or 364 (or any derivable range therein) contiguous amino acids of SEQ ID NOs: 19-39, 60-62, 69-76, 113-121, or 136-143 that are at least, at most, or exactly 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% similar, identical, or homologous with one of SEQ ID NOs: 19-39, 60-62, 69-76, 113-121, or 136-143.

[0138] In some aspects, the hairpin structure, such as the stem, loop, or both stem and loop, of the RNA targeting molecule may comprise nucleic acids 1 to 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, or 43 (or any derivable range therein) of SEQ ID NOs: 40-57.

[0139] In some aspects, the hairpin structure, such as the stem, loop, or both stem and loop, of the RNA targeting molecule may comprise 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, or 43 (or any derivable range therein) contiguous nucleic acids of SEQ ID NOs: 40-57.

[0140] In some aspects, the hairpin structure, such as the stem, loop, or both stem and loop, of the RNA targeting molecule may comprise 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,

13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, or 43 (or any derivable range therein) contiguous nucleic acids of SEQ ID NOs: 40-57 that are at least, at most, or exactly 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100% similar or identical with one of SEQ ID NOs: 40-57.

[0141] In some aspects, the stabilizer polypeptide may have one or more substitutions that reduce or eliminate binding to endogenous proteins. In some aspects, the stabilizer polypeptide may have one or more substitutions that reduce or eliminate an activity directed to an endogenous protein.

[0142] The polypeptides and nucleic acids of the disclosure, such as the CIRT proteins, stabilizer domain, linker, RNA hairpin binding domain, NES, effector domain, molecular tag, RNA hairpin region of the RNA targeting molecule, may comprise, comprise at least, or comprise at most 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, 359, 360, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370, 371, 372, 373, 374, 375, 376, 377, 378, 379, 380, 381, 382, 383, 384, 385, 386, 387, 388, 389, 390, 391, 392, 393, 394, 395, 396, 397, 398, 399, 400, 401, 402, 403, 404, 405, 406, 407, 408, 409, 410, 411, 412, 413, 414, 415, 416, 417, 418, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 430, 431, 432, 433, 434, 435, 436, 437, 438, 439, 440, 441, 442, 443, 444, 445, 446, 447, 448, 449, 450, 451, 452, 453, 454, 455, 456, 457, 458, 459, 460, 461, 462, 463, 464, 465, 466, 467, 468, 469, 470, 471, 472, 473, 474, 475, 476, 477, 478, 479, 480, 481, 482, 483, 484, 485, 486, 487, 488, 489, 490, 491, 492, 493, 494, 495, 496, 497, 498, 499, 500, 501, 502, 503, 504, 505, 506, 507, 508, 509, 510, 511, 512, 513, 514, 515, 516, 517, 518, 519, 520, 521, 522, 523, 524, 525, 526, 527, 528, 529, 530, 531, 532, 533, 534, 535, 536, 537, 538, 539, 540, 541, 542, 543, 544, 545, 546, 547, 548, 549, 550, 551, 552, 553,

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[0143] The substitution may be at amino acid position or nucleic acid position 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, 359, 360, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370, 371, 372, 373, 374, 375, 376, 377, 378, 379, 380, 381, 382, 383, 384, 385, 386, 387, 388, 389, 390, 391, 392, 393, 394, 395, 396, 397, 398, 399, 400, 401, 402, 403, 404, 405, 406, 407, 408, 409, 410, 411, 412, 413, 414, 415, 416, 417, 418, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 430, 431, 432, 433, 434, 435, 436, 437, 438, 439, 440, 441, 442, 443, 444, 445, 446, 447, 448, 449, 450, 451, 452, 453, 454, 455, 456, 457, 458, 459, 460, 461, 462, 463, 464, 465, 466, 467, 468, 469, 470, 471, 472, 473, 474, 475, 476, 477, 478, 479, 480, 481, 482, 483, 484, 485, 486, 487, 488, 489, 490, 491, 492, 493, 494, 495, 496, 497, 498, 499, 500, 501, 502, 503, 504, 505, 506, 507, 508, 509, 510, 511, 512, 513, 514, 515, 516, 517, 518, 519, 520, 521, 522, 523, 524, 525, 526, 527, 528, 529, 530, 531, 532, 533, 534, 535, 536, 537, 538, 539, 540, 541, 542, 543, 544, 545, 546, 547, 548, 549, 550, 551, 552, 553, 554, 555, 556, 557, 558, 559, 560, 561, 562, 563, 564, 565, 566, 567, 568, 569, 570, 571, 572, 573, 574, 575, 576, 577, 578, 579, 580, 581, 582, 583, 584, 585, 586, 587, 588, 589, 590, 591, 592, 593, 594, 595, 596, 597, 598, 599, 600, 601, 602, 603, 604, 605, 606, 607, 608, 609, 610, 611, 612, 613, 614, or 615 of one of SEQ ID NO: 1-169.

[0144] The substitution at amino acid position 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101,

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[0145] Substitutional variants typically contain the exchange of one amino acid for another at one or more sites within the protein, and may be designed to modulate one or more properties of the polypeptide, with or without the loss of other functions or properties. Substitutions may be conservative, that is, one amino acid is replaced with one of similar shape and charge. Conservative substitutions are well known in the art and include, for example, the changes of: alanine to serine; arginine to lysine; asparagine to glutamine or histidine; aspartate to glutamate; cysteine to serine; glutamine to asparagine; glutamate to aspartate; glycine to proline; histidine to asparagine or glutamine; isoleucine to leucine or valine; leucine to valine or isoleucine; lysine to arginine; methionine to leucine or isoleucine; phenylalanine to tyrosine, leucine or methionine; serine to threonine; threonine to serine; tryptophan to tyrosine; tyro-

sine to tryptophan or phenylalanine; and valine to isoleucine or leucine. Alternatively, substitutions may be non-conservative such that a function or activity of the polypeptide is affected. Non-conservative changes typically involve substituting a residue with one that is chemically dissimilar, such as a polar or charged amino acid for a nonpolar or uncharged amino acid, and vice versa.

[0146] Proteins may be recombinant or synthesized *in vitro*. Alternatively, a non-recombinant or recombinant protein may be isolated from bacteria. It is also contemplated that bacteria containing such a variant may be implemented in compositions and methods. Consequently, a protein need not be isolated.

[0147] The term “functionally equivalent codon” is used herein to refer to codons that encode the same amino acid, such as the six codons for arginine or serine, and also refers to codons that encode biologically equivalent amino acids.

[0148] It also will be understood that amino acid and nucleic acid sequences may include additional residues, such as additional N- or C-terminal amino acids, or 5' or 3' sequences, respectively, and yet still be essentially as set forth in one of the sequences disclosed herein, so long as the sequence meets the criteria set forth above, including the maintenance of biological protein activity where protein expression is concerned. The addition of terminal sequences particularly applies to nucleic acid sequences that may, for example, include various non-coding sequences flanking either the 5' or 3' portions of the coding region.

[0149] The following is a discussion based upon changing of the amino acids of a protein to create an equivalent, or even an improved, second-generation molecule. For example, certain amino acids may be substituted for other amino acids in a protein structure without appreciable loss of interactive binding capacity. Structures such as, for example, an enzymatic catalytic domain or interaction components may have amino acid substituted to maintain such function. Since it is the interactive capacity and nature of a protein that defines that protein's biological functional activity, certain amino acid substitutions can be made in a protein sequence, and in its underlying DNA coding sequence, and nevertheless produce a protein with like properties. It is thus contemplated by the inventors that various changes may be made in the DNA sequences of genes without appreciable loss of their biological utility or activity.

[0150] In other aspects, alteration of the function of a polypeptide is intended by introducing one or more substitutions. For example, certain amino acids may be substituted for other amino acids in a protein structure with the intent to modify the interactive binding capacity of interaction components. Structures such as, for example, protein interaction domains, nucleic acid interaction domains, and catalytic sites may have amino acids substituted to alter such function. Since it is the interactive capacity and nature of a protein that defines that protein's biological functional activity, certain amino acid substitutions can be made in a protein sequence, and in its underlying DNA coding sequence, and

nevertheless produce a protein with different properties. It is thus contemplated by the inventors that various changes may be made in the DNA sequences of genes with appreciable alteration of their biological utility or activity.

[0151] In making such changes, the hydropathic index of amino acids may be considered. The importance of the hydropathic amino acid index in conferring interactive biologic function on a protein is generally understood in the art (Kyte and Doolittle, 1982). It is accepted that the relative hydropathic character of the amino acid contributes to the secondary structure of the resultant protein, which in turn defines the interaction of the protein with other molecules, for example, enzymes, substrates, receptors, DNA, antibodies, antigens, and the like.

[0152] It also is understood in the art that the substitution of like amino acids can be made effectively on the basis of hydrophilicity. U.S. Pat. No. 4,554,101, incorporated herein by reference, states that the greatest local average hydrophilicity of a protein, as governed by the hydrophilicity of its adjacent amino acids, correlates with a biological property of the protein. It is understood that an amino acid can be substituted for another having a similar hydrophilicity value and still produce a biologically equivalent and immunologically equivalent protein.

[0153] As outlined above, amino acid substitutions generally are based on the relative similarity of the amino acid side-chain substituents, for example, their hydrophobicity, hydrophilicity, charge, size, and the like. Exemplary substitutions that take into consideration the various foregoing characteristics are well known and include: arginine and lysine; glutamate and aspartate; serine and threonine; glutamine and asparagine; and valine, leucine and isoleucine.

[0154] In specific aspects, all or part of proteins described herein can also be synthesized in solution or on a solid support in accordance with conventional techniques. Various automatic synthesizers are commercially available and can be used in accordance with known protocols. See, for example, Stewart and Young, (1984); Tam et al., (1983); Merrifield, (1986); and Barany and Merrifield (1979), each incorporated herein by reference. Alternatively, recombinant DNA technology may be employed wherein a nucleotide sequence that encodes a peptide or polypeptide is inserted into an expression vector, transformed or transfected into an appropriate host cell and cultivated under conditions suitable for expression.

[0155] One aspect includes the use of gene transfer to cells, including microorganisms, for the production and/or presentation of proteins. The gene for the protein of interest may be transferred into appropriate host cells followed by culture of cells under the appropriate conditions. A nucleic acid encoding virtually any polypeptide may be employed. The generation of recombinant expression vectors, and the elements included therein, are discussed herein. Alternatively, the protein to be produced may be an endogenous protein normally synthesized by the cell used for protein production.

VII. Sequences

RNA targeting molecule-hairpin domain	Sequence	SEQ ID NO:
TAR hairpin Scaffold	ggccagaucugagccugggagcucucuggcc	40
Human SLBP hairpin scaffold	ccaaaggcucucucagagccacca	41

Description	Sequence	Seq ID
B-defensin	MGIINTLQKYYCRVGRGRCVLSCLPKEEQIGKCSTRGRKCCR RKK	14
Linker	GGSGGSGGSGGSGGSGGS	15
TBP6.7	MAVPETRPNHTIYINNLSNKIKDELKKSLEYAIFSQQGILDLVLP RQRTPRGQAFVI FKEVSSATNALRSMQGPFPPFDKPMRIQYAKT DKRIPAKMKGTFFV	16
linker	GS	
HIV NES	LQLPPLERLTL	17
FLAG	DYKDDDDK	18
CIRT-1	MGIINTLQKYYCRVGRGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGSGGSGGSGGSGGSGGSMAPVETRPNHTIYINNLSNKIK KDELKKSLEYAIFSQQGILDLVLPQRTPRGQAFVI FKEVSSATN ALRSMQGPFPPYDKPMRIQYAKTDKRIPAKMKGTFFVGSGLQLPPL ERLTLGSGGSGGSGGSGGSGGSMDLTAIYESLLSLSPDVPVPS DHGGTESSPGWGSSGPWSLSPSDSSPSGVTSRLPGRSTSLVEGR SCGWVPPPPGFAPLAPRLGPELSPSPTSTATSTTPSRKYTELCSR TFESGRCRYGAKCQFAHGLGELRQANRHPKYKTELCHKFYLG GRCPYGSRCRFIHNPSEDLAAPGHPPVLRQSISFSGLPSGRRTSP PPGLAGPSLSSSSFSPSSSPPPPGDLPLSPSAFSAAPGTPLARRDPT PVCCPSRRATPI SVWGPLGGLVRTPSVQSLGSDPDEYASSGSS LGGSDSPVFEGAVFAPPQPVAAPRRLPINFNRISVSEGSGGSGGS GGSGGSGGDAMFGQPGALGSTPFLGQHGFNFPPSGIDFSAWG NNSSQGSTQSSGYSSNYAPSSLGGAMIDGQSAPANETLNK APGMNTIDQMAAGSDYKDDDDK*	1
ZFP36	MDLTAIYESLLSLSPDVPVPSDHGGTESSPGWGSSGPWSLSPSD SSPSGVTSRLPGRSTSLVEGRSCGWVPPPPGFAPLAPRLGPELSP SPTSPTATSTTPSRKYTEL CRTTFESGRCRYGAKCQFAHGLGEL RQANRHPKYKTELCHKFYLGRCPCPYGSRCRFIHNPSEDLAAPG HPVLRQSISFSGLPSGRRTSPPPGLAGPSLSSSSFSPSSSPPPPG DLPLSPSAFSAAPGTPLARRDPTPVCCPSRRATPI SVWGPLGGL VRTPSVQSLGSDPDEYASSGSS LGGSDSPVFEGAVFAPPQPVA PRRLPIFNIRISVSE	32
YTHDF2	DAMFGQPGALGSTPFLGQHGFNFPPSGIDFSAWGNNSSQGST QSSGYSSNYAPSSLGGAMIDGQSAPANETLNKAPGMNTIDQ GMAA	33
ZFP36 - YTHDF2	MDLTAIYESLLSLSPDVPVPSDHGGTESSPGWGSSGPWSLSPSD SSPSGVTSRLPGRSTSLVEGRSCGWVPPPPGFAPLAPRLGPELSP SPTSPTATSTTPSRKYTEL CRTTFESGRCRYGAKCQFAHGLGEL RQANRHPKYKTELCHKFYLGRCPCPYGSRCRFIHNPSEDLAAPG HPVLRQSISFSGLPSGRRTSPPPGLAGPSLSSSSFSPSSSPPPPG DLPLSPSAFSAAPGTPLARRDPTPVCCPSRRATPI SVWGPLGGL VRTPSVQSLGSDPDEYASSGSS LGGSDSPVFEGAVFAPPQPVA PRRLPIFNIRISVSEGSGGSGGSGGSGGSDAMFGQPGALGS TPFLGQHGFNFPPSGIDFSAWGNNSSQGSTQSSGYSSNYAP SSLGGAMIDGQSAPANETLNKAPGMNTIDQGMAA	19
CIRT-2	MGIINTLQKYYCRVGRGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGSGGSGGSGGSGGSGGSMAPVETRPNHTIYINNLSNKIK KDELKKSLEYAIFSQQGILDLVLPQRTPRGQAFVI FKEVSSATN ALRSMQGPFPPYDKPMRIQYAKTDKRIPAKMKGTFFVGSGLQLPPL ERLTLGSGGSGGSGGSGGSGGSMPEATVDHSQRICEVWCANL DEEMKKIRQVKINYVAMDTEFPGVVARP IGEFRSNADYQYQ LLRCNVLLKIIQLGLTFMNEOGEYPPGTS TWOFNFKNLTEDM	2

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Description	Sequence	Seq ID
	YAQDSIELLTSGIQFKKHEEEGIETQYFAELLMTSGVVLCCEGV KWLSPHSGYDFGYLIKILTNSNLPPEELDFEILRLFFPVIYDVKY LMKSCKNLKGGLQEVAEQLELERIGPQHAGSDSLTGMMAFFK MREMFEDHIDDAKYCGHLYGLSGSSSYVQNGTGNAYEEAN KQSGSGSGSGSGSGSGSDAMFGQPGALGSTPFLGQHGF NFFPSGIDFSAWGNSSQGGSTQSSGYSSNYAYAPSSLGGAMID GQSAFANETLNKAPGMNTIDQMAAGSDYKDDDDK*	
CNOT7	MPAATVDHSQRICEVWACNLDEEMKKIRQVIRKYNVAMDE FPGVVARPIGEFRSNADYQYQLLRNVDLLKI IQLGLTFMNEQG EYPPGTSTWQFNFKFNLTEDMYAQDSIELLTSGIQFKKHEEEGI ETQYFAELLMTSGVVLCCEGVKWLSPHSGYDFGYLIKILTNSNLP EEELDFEILRLFFPVIYDVKYLKMSCKNLKGGLQEVAEQLELE RIGPQHAGSDSLTGMMAFFKMRMFFEDHIDDAKYCGHLYGL SGSSSYVQNGTGNAYEEANQKS	34
YTHDF2	DAMFGQPGALGSTPFLGQHGFNFFPSGIDFSAWGNSSQGGST QSSGYSSNYAYAPSSLGGAMIDGQSAFANETLNKAPGMNTIDQ GMAA	33
CNOT7- YTHDF2	MPAATVDHSQRICEVWACNLDEEMKKIRQVIRKYNVAMDE FPGVVARPIGEFRSNADYQYQLLRNVDLLKI IQLGLTFMNEQG EYPPGTSTWQFNFKFNLTEDMYAQDSIELLTSGIQFKKHEEEGI ETQYFAELLMTSGVVLCCEGVKWLSPHSGYDFGYLIKILTNSNLP EEELDFEILRLFFPVIYDVKYLKMSCKNLKGGLQEVAEQLELE RIGPQHAGSDSLTGMMAFFKMRMFFEDHIDDAKYCGHLYGL SGSSSYVQNGTGNAYEEANQKS GSGSGSGSGSGSGSGSD AMFGQPGALGSTPFLGQHGFNFFPSGIDFSAWGNSSQGGSTQS SGYSSNYAYAPSSLGGAMIDGQSAFANETLNKAPGMNTIDQ MAA	20
CIRT-3	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGGSGSGSGSGSGSGSGSMAPETRPNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTFFVGSLLQPLPL ERLTGSGSGSGSGSGSGSGSMPAATVDHSQRICEVWACNL DEEMKKIRQVIRKYNVAMDETFPGVVARPIGEFRSNADYQYQ LLRCNVDLLKI IQLGLTFMNEQGEYPPGTSTWQFNFKFNLTEDM YAQDSIELLTSGIQFKKHEEEGIETQYFAELLMTSGVVLCCEGV KWLSPHSGYDFGYLIKILTNSNLPPEELDFEILRLFFPVIYDVKY LMKSCKNLKGGLQEVAEQLELERIGPQHAGSDSLTGMMAFFK MREMFEDHIDDAKYCGHLYGLSGSSSYVQNGTGNAYEEAN KQSGSGSGSGSGSGSGSGSMDLTAIYESLLSLSPDVPVPSDH GTESSPGWGSSGFWSLSPDSSPSGVTSLRPLGRSTSLVEGRSCG WVPPPPGFAPLAPRLGPELSPSPSTPTATSTPSRYKTELCRTFSE SGRCRYGAKCQFAHGLGELRQANRHPKYKTELCCHKFYLGRC PYGSRCHF IHNPSEDLAAPGHPPVLRQSI SFGSLPSGRRTSPPPPG LAGPSLSSSFSPSSPPPPGDLPLSPSAFSAAPGTPLARRDPTPVC CPSRRATPISVWGPLGGLVRTPSVQSLGSDPDEYASSGSSLGG SDSPVFEAGVFAPPQVAPRRLP I FNRI SVS VSGSDYKDDDDK*	3
CNOT7	MPAATVDHSQRICEVWACNLDEEMKKIRQVIRKYNVAMDE FPGVVARPIGEFRSNADYQYQLLRNVDLLKI IQLGLTFMNEQG EYPPGTSTWQFNFKFNLTEDMYAQDSIELLTSGIQFKKHEEEGI ETQYFAELLMTSGVVLCCEGVKWLSPHSGYDFGYLIKILTNSNLP EEELDFEILRLFFPVIYDVKYLKMSCKNLKGGLQEVAEQLELE RIGPQHAGSDSLTGMMAFFKMRMFFEDHIDDAKYCGHLYGL SGSSSYVQNGTGNAYEEANQKS	34
ZFP36	MDLTAIYESLLSLSPDVPVPSDHGGTESSPGWGSSGFWSLSPD SSPSGVTSLRPLGRSTSLVEGRSCGWVPPPPGFAPLAPRLGPELSP SPTSPTATSTPSRYKTELCRTFSESGRCRYGAKCQFAHGLGEL RQANRHPKYKTELCCHKFYLGRCPYGSRCHF IHNPSEDLAAPG HPVLRQSI SFGSLPSGRRTSPPPPGLAGPSLSSSFSPSSPPPPG DLPLSPSAFSAAPGTPLARRDPTPVCCPSRRATPISVWGPLGGL VRTPSVQSLGSDPDEYASSGSSLGSDSPVFEAGVFAPPQVAA PRRLPIFNRI SVSE	32
CNOT7- ZFP36	MPAATVDHSQRICEVWACNLDEEMKKIRQVIRKYNVAMDE FPGVVARPIGEFRSNADYQYQLLRNVDLLKI IQLGLTFMNEQG EYPPGTSTWQFNFKFNLTEDMYAQDSIELLTSGIQFKKHEEEGI ETQYFAELLMTSGVVLCCEGVKWLSPHSGYDFGYLIKILTNSNLP EEELDFEILRLFFPVIYDVKYLKMSCKNLKGGLQEVAEQLELE RIGPQHAGSDSLTGMMAFFKMRMFFEDHIDDAKYCGHLYGL	21

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Description	Sequence	Seq ID
	GSGSSYVQNGTGNAYEEEEANKQSGGGSGGSGGGSGGSGGSM DLTAIYESLLSLSPDVPVPSDHGGTESSPGWSSGPWSLSPSDSS PSGVTSRLPGRSTSLVEGRSCGWVPPPGFAPLAPRLGPELSPSP TSPATATSTTPSRYTELCRTFSESGRCRYGAKCQFAHGLGELRQ ANRHPKYKTELCHKFYLGRCPCYGRCHFIHNPSEDLAAPGHP PVLRSISFSGPLSGRRTSPPPGLAGPSLSSSSFSPPSSPPPPGDLPL LSPSAFSAAPGTPLARRDPTPVCCPSCRRATPISVWGPLGGLVRT PSVQSLGSDPDEYASSGSSLGSDSPVFEAGVFAPPQVVAAPRR LPIFNRI SVSE	
CIRT-4	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGGSGGSGGSGGSGGSGGSMVETRPNTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTfVGSLLPPL ERLTLGGSGGSGGSGGSGGSGGSTRSRYASDFEETAVLGQGAFG QVVKARNALDSRYAIAKKIRHTEKSLTILSEVMLLASLNHQYV VRYAAWLEEDSMDENVFESTDEESDLESSESDFEENDLLDQSS IFKNRTNHDLDNSNWD FISGSGYPDIVFENSRRDENEDLDHDT SSTSSSESQDDTDKESKSIQNVPRRRNFVKPMTAVKKKSTLFIQ MEYCENGTLVDLIHSENLNQORDEYWRLEFRQILEALSYIHSQGI HRDLKPMNILIDESRNVKIGDFGLAKNVHRSGLDILKLDSONLPG SSDNLTSAIGTAMYVATEVLDGTGHYNEKIDMYSLGIIFFEMIY PFSTGMERNVILKKLRVSIEFFPPD FDDNKMKEKKIIRLLIDHD PNKRPGARTLLNSGWL PVKHQDEVIKEALKSLSNPSSPWQQQV REGSDYKDDDDK*	4
GCN2	TRSRYASDFEETAVLGQGAFGQVVKARNALDSRYAIAKKIRHTE EKLSTILSEVMLLASLNHQYVRYAAWLEEDSMDENVFESTD EESDLESSESDFEENDLLDQSSI FKNRTNHDLDNSNWD FISGSGY PDIVFENSRRDENEDLDHDT SSTSSSESQDDTDKESKSIQNVPR RRRNFVKPMTAVKKKSTLFIQMEYCENGTLVDLIHSENLNQOR EYWRLEFRQILEALSYIHSQGIHRDLKPMNILIDESRNVKIGDFGL AKNVHRSGLDILKLDSONLPGSSDNLTSAIGTAMYVATEVLDGT GHYNEKIDMYSLGIIFFEMIYPFSTGMERNVILKKLRVSIEFFPPD FDDNKMKEKKIIRLLIDHDPNKRPGARTLLNSGWL PVKHQDE VIKEALKSLSNPSSPWQQQVRE	22
CIRT-5	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGGSGGSGGSGGSGGSGGSMVETRPNTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTfVGSLLPPL ERLTLGGSGGSGGSGGSGGSGGSMGSSCSQTPSRAIPTRRVA LGDGVQLPPGDYSTAPGGTLFSTAPGGTRIIYDRKFLMECRNAP VAKAPPKDLPTIPGVTSPTSDEPPMQASQSHLHSSPEDKRAGGE ESQFEMDIGSDYKDDDDK*	5
4EBP1	MSGSSCSQTPSRAIPTRRVALGDGVQLPPGDYSTAPGGTLFST APGGTRIIYDRKFLMECRNAPVAKAPPKDLPTIPGVTSPTSDEPP MQASQSHLHSSPEDKRAGGEESQFEMDI	23
CIRT-6	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGGSGGSGGSGGSGGSGGSMVETRPNTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTfVGSLLPPL ERLTLGGSGGSGGSGGSGGSGGSMGTFDLWTDYLGHLVRA LSGKEGPETRLSPQPEPEPMLEPVSALEPMPAPESVPVPGPKDQ KRSLESSPAPERLCSFCKHNGESRAIYQSHVLKDEAGRVLCPILR DYVCPQCGATRERAHTRRFCLTGQGYTSVYSHTTRNSAGKKL VRPDKAKTQDTGHRGGGGAGFRGAGKSEPSPCSPSMSTGS DYKDDDDK*	6
NANOS3	MGTFDLWTDYLGHLVRLSGKEGPETRLSPQPEPEPMLEPV SALEPMPAPESVPVPGPKDQKRSLESSPAPERLCSFCKHNGESR AIYQSHVLKDEAGRVLCPILRDYVCPQCGATRERAHTRRFCLT GQGYTSVYSHTTRNSAGKKLVRPDKAKTQDTGHRGGGGGA GFRGAGKSEPSPCSPSMST	24
CIRT-7	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGGSGGSGGSGGSGGSGGSMVETRPNTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTfVGSLLPPL ERLTLGGSGGSGGSGGSGGSGGSMATVEPETTPNPPTTEEEK TESNQEVANPEHYIKHPLQNRWALWFFKNDKSKTWQANLRLIS KFDTVEDFWALYNIQLSSNLMPCDYSLFKDGIEPMWEDEKN	7

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Description	Sequence	Seq ID
	KRGGRWLI TLNKQRRSDLDRFWLETLLCLIGESFDDYSDDVC GAVVNVRAKGDKIAIWTTECENREAVTHIGRVYKERLGLPPKI VIGYQSHADTATKSGSTTKNRFVVGSGSGSGSGSGSGSGSM SASQDSRSRDNGPDGMEPEGVIESNWNEIVDSFDDMNLSESLR GIYAYGFEKPSAIQQRAILPCIKGYDVIAQAQSGTGKTATFAISIL QQIELDLKATQALVLAPTRELAQQIQKVVMALGDYMGASCHA CIGGTNVRAEVQKLQMEAPHI IVGTPGRVFDMLNRRYLSPKYIK MFVLDEADEMLSRGFKDQIYDIFQKLNSNTQVLLSATMPSDV LEVTKKFMRDPPIRILVKKEELTLEGIRQFYINVEREEWKLDTLCL DLYETLTI TQAVIFINTRRKVDWLT EKM HARDFTVSAMHGDM QKRDVIMREFRSGSSRVLITD L LARGIDVQQVSLVINYLPTN RENYIHRIGRGRFGRKGVA INMVTEEDKRTL RDIETFYNTSIEE MPLNVADLIGSDYKDDDK*	
EIF4E	MATVEPETTPTPNPPTTEEEKTESNQEVANPEHYIKHPLQNRWA LWFFKNDKSKTWQANLRLISKFDTVEDFWALYNHILQSSNLMP GCDYSLFKDGI EPWDEKKNRGRWLI TLNKQRRSDLDRF WLETLLCLIGESFDDYSDDVC GAVVNVRAKGDKIAIWTTECEN REAVTHIGRVYKERLGLPPKIVIGYQSHADTATKSGSTTKNRFV V	35
EIF4A	MSASQDSRSRDNGPDGMEPEGVIESNWNEIVDSFDDMNLSESL LRGIYAYGFEKPSAIQQRAILPCIKGYDVIAQAQSGTGKTATFAI SILQQIELDLKATQALVLAPTRELAQQIQKVVMALGDYMGASC HACIGGTNVRAEVQKLQMEAPHI IVGTPGRVFDMLNRRYLSPK YIKMFVLDEADEMLSRGFKDQIYDIFQKLNSNTQVLLSATMPS DVLEVTKKFMRDPPIRILVKKEELTLEGIRQFYINVEREEWKLD LCDLYETLTI TQAVIFINTRRKVDWLT EKM HARDFTVSAMHG MDQKRDVIMREFRSGSSRVLITD L LARGIDVQQVSLVINYL PTNRENYIHRIGRGRFGRKGVA INMVTEEDKRTL RDIETFYNT SIEEMPLNVADLI	36
EIF4E- EIF4A	MATVEPETTPTPNPPTTEEEKTESNQEVANPEHYIKHPLQNRWA LWFFKNDKSKTWQANLRLISKFDTVEDFWALYNHILQSSNLMP GCDYSLFKDGI EPWDEKKNRGRWLI TLNKQRRSDLDRF WLETLLCLIGESFDDYSDDVC GAVVNVRAKGDKIAIWTTECEN REAVTHIGRVYKERLGLPPKIVIGYQSHADTATKSGSTTKNRFV VGSGSGSGSGSGSGSGSGSMASQDSRSRDNGPDGMEPEGVIE SNWNEIVDSFDDMNLSESLLRGIYAYGFEKPSAIQQRAILPCIKG YDVIAQAQSGTGKTATFAISILQQIELDLKATQALVLAPTRELA QQIQKVVMALGDYMGASCHACIGGTNVRAEVQKLQMEAPHI I VGTPGRVFDMLNRRYLSPKYIKMFVLDEADEMLSRGFKDQIYD IFQKLNSNTQVLLSATMPSDVLEVTKKFMRDPPIRILVKKEELT LEGIRQFYINVEREEWKLDTLCDLYETLTI TQAVIFINTRRKVDW LTEKM HARDFTVSAMHGMDQKRDVIMREFRSGSSRVLITD LLARGIDVQQVSLVINYLPTNRENYIHRIGRGRFGRKGVA IN MVTEEDKRTL RDIETFYNTSIEEMPLNVADLI	25
CIRT-8	MGIINTLQKYICRVRGRCVLSCLPKEEQIGKCS TRGRKCCR RKKGGSGSGSGSGSGSGSGSMAPETRPNTIYINNLSKIK KDELKKS LYAIFSQQGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPYDKPMRIQYAKTDKRIPAKMKGT FVGS LQLPPL ERLTGGSGSGSGSGSGSGSGSMASQDSRSRDNGPDGMEPE GVIESNWNEIVDSFDDMNLSESLLRGIYAYGFEKPSAIQQRAILP CIKG YDVIAQAQSGTGKTATFAISILQQIELDLKATQALVLAPTR ELAQQIQKVVMALGDYMGASCHACIGGTNVRAEVQKLQMEA PHIIVGTPGRVFDMLNRRYLSPKYIKMFVLDEADEMLSRGFKD QIYDIFQKLNSNTQVLLSATMPSDVLEVTKKFMRDPPIRILVKK EELTLEGIRQFYINVEREEWKLDTLCDLYETLTI TQAVIFINTRR KVDWLT EKM HARDFTVSAMHGMDQKRDVIMREFRSGSSR VLITD L LARGIDVQQVSLVINYLPTNRENYIHRIGRGRFGRK GVAINMVTEEDKRTL RDIETFYNTSIEEMPLNVADLIGSGSGSG SGSGSGSGSGMSDGFDRAPGAGRGRSRLGRGGGGPEGGGF PNGAGPAERARHQPPQPKAPGFLQPPPLRQPRTPPPGAQCEVP ASPQRPSRPGALPEQTRPLRAPPSQDKIPQQNESAMAKPQVV VAPVLSKLSVNAPEFYPSGYSSSYTESYEDGCEDYPTLSEYVQ DFLNHLTEQPGSFETIEQFAETLNGCVTTDDALQELVELIYQQ ATSI PNFSYMGARLCNYLSHHLTISPQSGNFRQLLQRCRTEYE VKDQAAKGDEVTRKRHFVFLGELYLNLEIKGTNGQVTRA DILQVGLRELLNALFSNPMDDNLICAVKLLKLTGSLVEDAWKE KGKMDMEEIIQRIENVVLDANCSRDVKQMLLKLVELRSSNWG RVHATSTYREATPENDPNYFMNEPTFYTSDGVPF TAADPDYQE KYQELLEREDFFPDYEENGTDLSGAGDPYLDIDDEM DPEIEEA YEKFCLESERKRKQGS DYKDDDK*	8

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Description	Sequence	Seq ID
EIF4A	MSASQDSRSRDNGPDGMEPEGVIESNWNEIVDSFDDMNLSL LRGIYAYGFKEPSAIQQRAILPCIKGYDVIAQAQSGTGKTATFAI SILQQIELDLKATQALVLAPTRELAQQIKVVMALGDYMGASC HACIGGTNVRAEVQKLQMEAPHIIVGTPGRVEDMLNRRYLSPK YIKMFVLDEADEMLSRGFKDQIYDIFQKLNSNTQVVLSATMPS DVLEVTKKFMRDPILVKKKEELTEGIRQFYINVEREEWKLD LCDLYETLTTIQAVIDINRRKVDWLTEKMHARDFTVSAMHGD MDQKERVIMREFRSGSSRVLITDLDLARGIDVQQVSLVINYL PTNRENYIHRIGRGRFGRKGVAINMVTEDKRTLRLDIETFYNT SIEEMPLNVADLI	36
PAIP	MSDGFDRAPGAGRGRSLGRGGGPEGGGFPNGAGPAERAR HQPPQPKAPGFLQPPPLRQPRTPPPGAQCEVPASQRPSPRGAL PEQTRPLRAPSSQDKIPQONSESAMAKPQVVVAPVLMKLSV NAPEFYPSGYSSSYTESYEDGCEDYPTLSEYVQDFLNHLTEQPG SFETEIEQFAETLNGCVTTDDALQELVELIYQQATSIPNFSYMG RLCNYLSHHLTISPGSGNFRQLLQRCRTEYEVKDQAAKGDEV TRKRFHAFVFLGELYLNLEIKGTNGQVTRADILQVGLRELLNA LFSNPMDNLCVAKLLKLTGSVLEDAWKEKGKMDMEEIIQRI ENVVLDANCSRDVKQMLLKLVELRSSNWGRVHATSTYREATP ENDPNYFMNEPTFYTSDGVPTAADPDYQEKYQELLEREDFFP DYEENGTDLSGAGDPYLDIDDEMDPEIEEAYEKFCLESERKRK Q	37
EIF4A- PAIP	MSASQDSRSRDNGPDGMEPEGVIESNWNEIVDSFDDMNLSL LRGIYAYGFKEPSAIQQRAILPCIKGYDVIAQAQSGTGKTATFAI SILQQIELDLKATQALVLAPTRELAQQIKVVMALGDYMGASC HACIGGTNVRAEVQKLQMEAPHIIVGTPGRVFDMLNRRYLSPK YIKMFVLDEADEMLSRGFKDQIYDIFQKLNSNTQVVLSATMPS DVLEVTKKFMRDPILVKKKEELTEGIRQFYINVEREEWKLD LCDLYETLTTIQAVIDINRRKVDWLTEKMHARDFTVSAMHGD MDQKERVIMREFRSGSSRVLITDLDLARGIDVQQVSLVINYL PTNRENYIHRIGRGRFGRKGVAINMVTEDKRTLRLDIETFYNT SIEEMPLNVADLIGSGSGSGSGSGSGSGSMSDGFDRAPGAG RGRSLGRGGGPEGGGFPNGAGPAERARHQPPQPKAPGFLQ PPPLRQPRTPPPGAQCEVPASQRPSPRGALPEQTRPLRAPSS QDKIPQONSESAMAKPQVVVAPVLMKLSVNAPEFYPSGYSS YTESYEDGCEDYPTLSEYVQDFLNHLTEQPGSFETEIEQFAETLN GCVTTDDALQELVELIYQQATSIPNFSYMGARLCNLSHHLTIS PQSGNFRQLLQRCRTEYEVKDQAAKGDEVTRKRFHAFVFLG ELYLNLEIKGTNGQVTRADILQVGLRELLNALFSNPMDNLCV VKLLKLTGSVLEDAWKEKGKMDMEEIIQRIENVVLDANCSRDV KQMLLKLVELRSSNWGRVHATSTYREATPENDPNYFMNEPTF YTSBGVPTAADPDYQEKYQELLEREDFFPDYEENGTDLSGAG DPYLDIDDEMDPEIEEAYEKFCLESERKRKQ	26
CIRT-9	MGIINTLQKYYCRVRGRCVLSCLPKEEIGKICSTRGRKCCR RKKGGSGSGSGSGSGSGSGSMVPETRPNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPYDKPMRIQYAKTDKRIPAKMKGTFVGSLLPPL ERLTGSGSGSGSGSGSGSGSGSMSATSVDTQRTKGQDNKVQ NGSLHQKDTVHDNDFEPYLTGQSNQSNYPSMSDPYLSYYPP SIGFPYSLNEAPWSTAGDPPIPYLTITYGQLSNGDHHFMDAVFG QPGGLGNNIYQHRFNFFPENAFSAWGTSGSQGQQTQSSAYGS SYTYPPSSLGGTVVDGQPGFHSDTLSKAPGMNSLEQGMVGLKI GDVSSSAVKTVGSVVSSVALTGVLSGNGGTNNMMPVSKPTSW AAIASKPAKPQPKMKTSGPVMGGGLPPPIKHNMDIGTWDNK GPVPKAPVPQQAQSPQAAPQPPQVQPLPAQPPALAQPPQVQSP QQPPQTRVAPRNRAAFQSGGAGSDSNSPGNVQPNAPSV ESGSGSGSGSGSGSGSGSGSMSATSVDQRPKGQGNKVSQNG SIHQKDAVNDDDFEPYLSQTNQSNYPPMSDPYMPSSYAPSIG FPYSLGEAAWSTAGDQPMPLYLTITYGQMSNGEHYIPDGVFSQ GALGNTPPFLGQHGFNFFPGNADFSTWGTSGSQGSTQSSAYSS SYGYPPSSLGRAITDQAGFGNDTLKVPGISSIEQGMTGLKIGG DLTAAVTKTVGTALSSGMTSIATNSVPPVSSAAPKPTSWAAIA RKPAKPQPKLKPKNVIGGSAVPPPIKHNMDIGTWEKGSV VKAPPTQVLPPTIQQPQPLIQQPPLVQSQLPQQPQPPQPPQ QQGPQQAQPHQVQPPQQQLQNRWVAPRNRGAGFNQNNAG SENFGLGVVPSASPSSVEVHPVLEKLKAINNYPKDFDNLK NGSDYKDDDDK*	9

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Description	Sequence	Seq ID
YTHDF1	MSATSVDTQRTKGQDNKVQNGSLHQKDTVHDNDFEPYLTGQS NQSNSYPSMSDPYLSSYYPPSIGFPYSLNEAPWSTAGDPPIPYLT TYGQLSNGDHHFMHDAVFGQPGGLGNNIYQHRFNFFPENPAFS AWGTSGSQGQQTQSSAYGSSYTYPPSSLGGTVVDGQPGFHS LTKAPGMNSLEQGMVGLKIGDVSSSAVKTVGSVVSSVALTGVL SGNGGTNVNMPVSKPTSWAAIASKPAKPQPKMKTSGPVMGG GLPPPIKHNMDIGTWDNKGVPKAPVPQQAPSPQAAPQPQQV AQPLPAQPPALAQFPQYQSPQQPPQTRWVAPRNRNAAFGQSGG AGSDSNSPGNVQPNAPSVE	38
YTHDF3	MSATSVDRPKGQGNKVSQNGSIHQKDAVNDDDFEPYLSQ TNQSNYPMSDPYMPSSYYAPSIGFPYSLGEAAWSTAGDQPM YLTTYGQMSNGEHYIPDGVSQPGALGNTPPFLGQHGFNFF GNADFSWTGSGSQGSTQSSAYSSSYGPPSSLGRAITDQAG FGNDTLKVPGISSEIQQMTGLKIGDLTAAVTKTVGTALSSSG MTSIATNSVPPVSSAAPKPTSWAAIARKPAKPQPKLPGNVGI GSAVPPPIKHNMDIGTWEKGSVVKAPPTQPVLPQTIIQQP QPLIQPPPLVQSQLPQQQPQPQQQQGPQQAQPHQVQPPQ QQLQNRWVAPRNRGAGFNQNGAGSENFGLGVVPSASPSSV EVHPVLEKKAINNYPKDFDWNLKN	39
YTHDF1- YTHDF3	MSATSVDTQRTKGQDNKVQNGSLHQKDTVHDNDFEPYLTGQS NQSNSYPSMSDPYLSSYYPPSIGFPYSLNEAPWSTAGDPPIPYLT TYGQLSNGDHHFMHDAVFGQPGGLGNNIYQHRFNFFPENPAFS AWGTSGSQGQQTQSSAYGSSYTYPPSSLGGTVVDGQPGFHS LTKAPGMNSLEQGMVGLKIGDVSSSAVKTVGSVVSSVALTGVL SGNGGTNVNMPVSKPTSWAAIASKPAKPQPKMKTSGPVMGG GLPPPIKHNMDIGTWDNKGVPKAPVPQQAPSPQAAPQPQQV AQPLPAQPPALAQFPQYQSPQQPPQTRWVAPRNRNAAFGQSGG AGSDSNSPGNVQPNAPSVESSSSSSSSSSSSSSSSSSSSSS SVDQRPKGQGNKVSQNGSIHQKDAVNDDDFEPYLSQTNQ NSYPPMSDPYMPSSYYAPSIGFPYSLGEAAWSTAGDQPMPLYT YGQMSNGEHYIPDGVSQPGALGNTPPFLGQHGFNFFPGNAD FSTWGTSGSQGSTQSSAYSSSYGPPSSLGRAITDQAGFGND TLKVPGISSEIQQMTGLKIGDLTAAVTKTVGTALSSSGMTSIA TNSVPPVSSAAPKPTSWAAIARKPAKPQPKLPGNVGIGGSAV PPPIKHNMDIGTWEKGSVVKAPPTQPVLPQTIIQQPQPLIQPP PLVQSQLPQQQPQPQQQQGPQQAQPHQVQPPQQLQNR WVAPRNRGAGFNQNGAGSENFGLGVVPSASPSSVEVHPVL EKKAINNYPKDFDWNLKN	27
CIRT-10	MGIINTLQKYICRVRGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGGSGSGSGSGSGSGSGSMVAPETRPNHTIYINNLSKIK KDELKSLYAFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPYDKPMRIQYAKTDKRIPAKMKGTFVGSLLPPL ERLTGGSGSGSGSGSGSGSGMSGDEMI F DPTMSKKKKK KKPFMLDEEGDTQTEETQPSETKEVEPEPTEDKLEADEEDTRK KASDDLDLNFNQKKKKKTKKIFIDEAEEGVKDLKIESD VQEPTEPEDDLIMLGNNKKKKKNVFPDEDEILEKDEALEDE DNKKDDGISFSNQTGPWAGSERDYTYEELNRFVFNIMREKNP DMVAGEKRKFVMPKPPQVVRVGTKTSFVNFTDICKLLHRQPK HLLAFLAELGTSGSIDGNNQLVIKGRFQQKIENVLRRYIKEY VTCHTCRSPDTILQKDTRLVFLQCETCHSRCSVASIKTGFQAVT GKRAQLRAKANGSDYKDDDDK*	10
EIF2S2	MSGDEMI F DPTMSKKKKKKKPFMLDEEGDTQTEETQPSETKE VEPEPTEDKLEADEEDTRKKDASDDLDLNFNQKKKKKT KKIFDIDEAEEGVKDLKIESDVQEPTEPEDDLIMLGNNKKKKK NVKFPDEDEILEKDEALEEDNKKDDGISFSNQTGPWAGSER	28

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Description	Sequence	Seq ID
	DYTYEELLNRVFNIMREKNPDMVAGEKRKFVMKPPQVVRVGT KKTsfvNFTDICKLLHRQPKHLLAFLAELGTSGSIDGNNQLVIK GRFQQKQIENVLRRYIKEYVTCHTCRSPDTILQKDTRLFLQCE TCHSRCSVASIKTGFQAVTGKRAQLRAKAN	
CIRT-11	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGGSGSGSGSGSGSGSGSMVPETRPNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTFVGSLLQPLPL ERLTGSGSGSGSGSGSGSGSMTERRVPFSLLRGPSWDPPFR DWYPHSRLFDQAFGLPRLPEEWSQWLGGSSWPGYVRPLPPAAI ESPAVAAPAYSRLSRQLSSGVSEIRHTADRWRVSLDVNHFAP DELTVKTKDGVVEITGKHEERQDEHGYISRCFTRKYTLPPGVDP TQVSSSLSPGTLTVEAMPKLATQSNIEITIPVTFESRAQLGGPE AAKSDETAALKGSDYKDDDDK*	11
HSPB1	MTERRVPFSLLRGPSWDPPFRDWYPHSRLFDQAFGLPRLPEEWS QWLGGSSWPGYVRPLPPAAIESPAVAAPAYSRLSRQLSSGVSE IRHTADRWRVSLDVNHFAPDELTVKTKDGVVEITGKHEERQDE HGYISRCFTRKYTLPPGVDPTQVSSSLSPGTLTVEAMPKLAT QSNIEITIPVTFESRAQLGGPEAAKSDETAALK	29
CIRT-12	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGGSGSGSGSGSGSGSGSMVPETRPNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTFVGSLLQPLPL ERLTGSGSGSGSGSGSGSGSMDDREDLVYQAKLAEQAER YDEMVESMKKVAGMDVELTVEERNLISVAYKNVIGARRASW RIISSIEQKEENKGGEDKLMIREYRQMVELTELKLIICDILDVLD KHLIPAANTGESKVFFYKMGDYHRYLAEFATGNDRKEAEN SLVAYKAASDIAMTELPPTHPIRLGLALNFSVFYYEILNSPDRAC RLAKAAFDDAIAELDTLSEESYKDSLIMQLLRDNLTLWTSDM QGDGEEQNKEALQDVEDENQGSYKDDDDK*	12
YWHAE	MDDREDLVYQAKLAEQAERYDEMVESMKKVAGMDVELTVEE RNLLSVAYKNVIGARRASWRIISSIEQKEENKGGEDKLMIREY RQMVELTELKLIICDILDVLDKHLIPAANTGESKVFFYKMGDY HRYLAEFATGNDRKEAENSLVAYKAASDIAMTELPPTHPIRL GLALNFSVFYYEILNSPDRACRLAKAAFDDAIAELDTLSEESYK DSLIMQLLRDNLTLWTSDMQGDGEEQNKEALQDVEDENQ	30
CIRT-13	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGGSGSGSGSGSGSGSGSMVPETRPNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTFVGSLLQPLPL ERLTGSGSGSGSGSGSGSGSMAEMVEAAGEDERELAAE MAAFLNENLPESIFGAPKAGNGQWASVIRVMNPIQGNLTDLV QLEQNEAAFSVAVCRFSNIGEDWYVLVGAKDLILNPRSVAGG FVYTYKLVNNGEKLEPLHKTPVEEVPAAIAPPQGRVLIQVGLKLL RVYDLGKKLLRKCENKHIANYISGIQTIHGRVIVSDVQESFIW VRYKRNEQLIIFADDTYPRWVTASLLDYDTVAGADKFGNIC VVRLPPNTNDEVEDPTGNKALWDRGLNGASQKAEVIMNYH VGETVLSLQKTTLIPGGESLVYTTLSGGIGILVPFTSHEDHDFQ HVMHLRSEHPPLCGRDHLSFRSYFFPVKNVIDGDLCEQFNSM EPNKQKNVSEELDRTPPEVSKKLEDIRTRYAFGSDYKDDDDK*	13
SF3B3	MAEMVEAAGEDERELAAEMAAFLNENLPESIFGAPKAGNG QWASVIRVMNPIQGNLTDLVQLEQNEAAFSVAVCRFSNIGEDW YVLVGAKDLILNPRSVAGGFVYTYKLVNNGEKLEPLHKTPVE EVPAAIAPPQGRVLIQVGLKLLRVYDLGKKLLRKCENKHIANYI SGIQTIHGRVIVSDVQESFIWVRYKRNEQLIIFADDTYPRWVT ASLLDYDTVAGADKFGNICVVRLPPNTNDEVEDPTGNKALW DRGLNGASQKAEVIMNYHVGETVLSLQKTTLIPGGESLVYTT LSGGIGILVPFTSHEDHDFQHVEMHLRSEHPPLCGRDHLSFRSY YFPVKNVIDGDLCEQFNSMEPNKQKNVSEELDRTPPEVSKKLE DIRTRYAF	31
CIRT-14	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGGSGSGSGSGSGSGSGSMVPETRPNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTFVGSLLQPLPL ERLTGSGSGSGSGSGSGSGSMVPETRPNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVSRSLKMRGQAFVIFKEVSSAT NALRSMQGFPPFYDKPMRIQYAKTDSIIAKMKGTFVERDRKRE	63

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Description	Sequence	Seq ID
	KRKPKSQETPATKKAVQGGGATPVVGAVQGPVPGMPPMTQAP RIMHHMPGQPPYMPPPGMI PPPGLAPGQIPPGAMPPQQLMPGQ MPPAQPLSENPPNHILFLTNLPEETNELMLSMLEFNQFPGFKEVR LVPGRHDIAFVEFDNEVQAGAARDALQGFKITQNNAMKISFAK KGS DYKDDDDK	
SNRPA	MAVPETRPNHITIYINNLSNEKIKKDELKKSLEYAIFSQFGQILDILVS RSLKMRGQAFVIFKEVSSATNALRSMQGFPPYDKPMRIQYAKT DSDIIAKMKGTFVERDRKREKRKPKSQETPATKKAVQGGGATP VVGAVQGPVPGMPPMTQAPRIMHHMPGQPPYMPPPGMI PPPGL APGQIPPGAMPPQQLMPGQMPPAQPLSENPPNHILFLTNLPEET NELMLSMLEFNQFPGFKEVRLVPGRHDIAFVEFDNEVQAGAARD ALQGFKITQNNAMKISFAK	60
CIRT-15	MGIINTLQKYYCRVGRCAVLSCLPKEEIGKCSTRGRKCCR RKKGGSGSGSGSGSGSGSMVPEPTRPNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPYDKPMRIQYAKTDKRIPAKMKGTFVGSLLPL ERLTGGSGSGSGSGSGSGSLLRDNTGSSNLSLSPSP GAWPYASDNSFTNVHSTS AKFPDYKSTWSPDPIGHNPHTLSN KMWKNHISSRNTTPLPRPPGLTNPKPSSPWSSTAPRSVRGWT QDSRLASASTWSDGGSVRPSYWLVLHNLTPQIDGSTLRITCMQ HGPLLTFLHNLQTALIRYSTKQEAQAQTALHMCVLGNTTI LAEFATDDEVSRFLAQAPPTPAATPSAPAAGWQSLETGQNQS DPVGPALNLFGGSTGLGQWSSAGSSGADLAGASLWGPNNYS SSLWGVPTVEDPHRMGSPAPLLPGDLLGGGSDSIGSDYKDDDD K*	64
TNRC6B	QLLRDNTGSSNLSLSPSGAWPYASDNSFTNVHSTS AKFP DYKSTWSPDPIGHNPHTLSNKMWKNHISSRNTTPLPRPPGLTN PKPSSPWSSTAPRSVRGWTQDSRLASASTWSDGGSVRPSYWL VLHNLTPQIDGSTLRITCMQHGPLLTFLHNLQTALIRYSTKQE AAKAQTALHMCVLGNTTILAEFATDDEVSRFLAQAPPTPAAT PSAPAAGWQSLETGQNQSDPVGPALNLFGGSTGLGQWSSAG GSSGADLAGASLWGPNNYSSSLWGVPTVEDPHRMGSPAPLLPG DLLGGGSDSI	61
CIRT-16	MGIINTLQKYYCRVGRCAVLSCLPKEEIGKCSTRGRKCCR RKKGGSGSGSGSGSGSGSMVPEPTRPNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPYDKPMRIQYAKTDKRIPAKMKGTFVGSLLPL ERLTGGSGSGSGSGSGSGSMAANPETPNSTISREASTQS SSAAASQGWVLEPGKIVPNTVFGGIDARMDETEIGSCFGRYGS VKEVKIITNRTGVSKGYGFVSFVNDVDVQKIVGSQIHFGKKLK LGPAIRKQKLCARHVQPRPLVNNPPPPQFQNVWRNPNTETYL QPQITPNPVTQHVQSAANPETPNSTISREASTQSSAAASQGWV LPEGKIVPNTVFGGIDARMDETEIGSCFGRYGSVKEVKIITNRT GVSKGYGFVSFVNDVDVQKIVGSQIHFGKKKLGPAPKQKL CARHVQPRPLVNNPPPPQFQNVWRNPNTETYLQPQITPNPVTQ HVQAYSAYPHSPGQVITGCQLLVYNYQEYPTYDPAFQVTTGY QLPVYNYQFPFAYPRSPFQVTAGYQLPVYNYQAFPAYPNSPFQ VATGYQFPVYNYQFPFAYPSSPFQVTAGYQLPVYNYQAFPAYP NSPFQVATGYQFPVYNYQAFPAYPNSPVQVTTGYQLPVYNYQ AFPAYPSSPFQVTTGYQLPVYNYQAFPAYPNQVQVTTGYQFH VYNYQMPPQCPVGEQRRLWTEAYKWWYLVCLIQRRDGS DY KDDDDK*	65
DAZ4	MSAANPETPNSTISREASTQSSAAASQGWVLEPGKIVPNTVFG GGIDARMDETEIGSCFGRYGSVKEVKIITNRTGVSKGYGFVSFV NDVDVQKIVGSQIHFGKKKLGPAPKQKLCARHVQPRPLV NNPPPPQFQNVWRNPNTETYLQPQITPNPVTQHVQSAANPETPN STISREASTQSSAAASQGWVLEPGKIVPNTVFGGIDARMDET EIGSCFGRYGSVKEVKIITNRTGVSKGYGFVSFVNDVDVQKIVG SQIHFGKKKLGPAPKQKLCARHVQPRPLVNNPPPPQFQNV WRNPNTETYLQPQITPNPVTQHVQAYSAYPHSPGQVITGCQLL	62

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Description	Sequence	Seq ID
	VYNYQEYPTYPDSAFQVTTGYQLPVYNYQPPPAYPRSPFQVTA GYQLPVYNYQAFPAYPNSPFQVATGYQFPVYNYQPPPAYPSSP FQVTAGYQLPVYNYQAFPAYPNSPFQVATGYQFPVYNYQAFP AYPNSPVQVTTGYQLPVYNYQAFPAYPSSPFQVTTGYQLPVYN YQAFPAYPNSAVQVTTGYQPHVYNYQMPPQCPVGEQRRNLWT EAYKWWYLVCLIQRRD	
CIRT-17	MGIINTLQKYICRVRGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGGSGGSGGSGGSGGSMVETPRNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTFVGSGLQLPPL ERLTLGGSGGSGGSGGSGGSGMSSEAETQQPPAAPPAAPALS AADTKPGTTGSGAGSGGPGGLTSAAPAGGDKKVIATKVLGTV KWFNVRNGYGFINRNDTKEDVFVHQTAIKNNPRKYLRSVGD GETVEFDVVEGEKGAEAAVNTGPGGVPVQGSKYAADRNHYRR YPRRRGPPRNYQQNYQNSESGEKNESGESAPEGQAQORRPYRR RRFPPYMRPYGRRPQYSNPPVQGEVMEGADNQGAGEQGRP VRQNMRYRGYRPRFRRGPPRQRPREDGNEEDKENQGDETQGG QPPQRRYRRNFYRRRRPENPKPDGKETKAADPPAENSSAPE AEQGGAEAGSDYKDDDDK*	66
YB-1	MSSEAETQQPPAAPPAAPALSAAADTKPGTTGSGAGSGGPGGLT SAAPAGGDKKVIATKVLGTVKWFNVRNGYGFINRNDTKEDVF VHQTAIKNNPRKYLRSVGDGETVEFDVVEGEKGAEAAVNTG PGGVPVQGSKYAADRNHYRRYPRRRGPPRNYQQNYQNSESGE KNESGESAPEGQAQORRPYRRRRFPPYMRPYGRRPQYSNPP VQGEVMEGADNQGAGEQGRPVRQNMRYRGYRPRFRRGPPRQRP QPREDGNEEDKENQGDETQGGQPPQRRYRRNFYRRRRPENP KPQDGKETKAADPPAENSSAPEAEQGGAE	69
CIRT-18	MGIINTLQKYICRVRGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGGSGGSGGSGGSGGSGGSMVETPRNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTFVGSGLQLPPL ERLTLGGSGGSGGSGGSGGSGMSSEAETQQPPAAPPAAPALS AADTKPGTTGSGAGSGGPGGLTSAAPAGGDKKVIATKVLGTV KWFNVRNGYGFINRNDTKEDVFVHQTAIKNNPRKYLRSVGD GETVEFDVVEGEKGAEAAVNTGPGGVPVQGSKYAADRNHYRR YPRRRGPPRNYQQNYQNSESGEKNESGESAPEGQAQORRPYRR RRFPPYMRPYGRRPQYSNPPVQGEVMEGADNQGAGEQGRP VRQNMRYRGYRPRFRRGPPRQRPREDGNEEDKENQGDETQGG QPPQRRYRRNFYRRRRPENPKPDGKETKAADPPAENSSAPE AEQGGAEAGSGGSGGSGGSGGSGGSLHLDQTPSRQPIPSEGLQL HLPQVLADAVSRLVLGKFGDLTDNESSPHARRKVLGVVMTT GTDVKDAKVISVSTGTCINGEYMSDRGLALNDCHAEIISRRSL LRFLYTQLELYLNNKDDQKRSIFQKSERGGFRLKENVQFHLIYS TSPCGDARIFSPHEPILEPADRHPNRKARGQLRTKIESGQGTIPV RSNASIQTDGVLQGERLLTMSCDKIARWNVVGIQGSLLSIFV EPIYFSSIILGSLYHGDHLSRAMYQRISNIEDLPPLYTLNKPILLSGI SNAEARQPGKAPNFSVNWTVGDSAEVINATTGKDELGRASRL CKHALYCRWMRVHGKVP SHLLRSKITKPNVYHESKLAAKEYQ AAKARLFTAFIKAGLGAWVEKPTBQDQFSLTPGSDYKDDDDK*	67
YB-1- ADAR2 (E488Q)	MSSEAETQQPPAAPPAAPALSAAADTKPGTTGSGAGSGGPGGLT SAAPAGGDKKVIATKVLGTVKWFNVRNGYGFINRNDTKEDVF VHQTAIKNNPRKYLRSVGDGETVEFDVVEGEKGAEAAVNTG PGGVPVQGSKYAADRNHYRRYPRRRGPPRNYQQNYQNSESGE KNESGESAPEGQAQORRPYRRRRFPPYMRPYGRRPQYSNPP VQGEVMEGADNQGAGEQGRPVRQNMRYRGYRPRFRRGPPRQRP QPREDGNEEDKENQGDETQGGQPPQRRYRRNFYRRRRPENP KPQDGKETKAADPPAENSSAPEAEQGGAEAGSGGSGGSGGSGG SGGSLHLDQTPSRQPIPSEGLQLHLPQVLADAVSRLVLGKFGD LTDNFSSPHARRKVLGVVMTTGTVDKAKVISVSTGTCING EYMSDRGLALNDCHAEIISRRSLRFLYTQLELYLNNKDDQKRS	72

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Description	Sequence	Seq ID
	IFQKSERGGFRLKENVQFHLIYSTSPCGDARIFSPHEPILEEPADR HPNRKARGQLRTKIESGQGTIPVRSNASIQTDWGVLQGERLLT MSCSDKIARWNVVGIQGSLLSIFVEPIYFSSIILGSLYHGDHLSRA MYQRISNIEDLPPLYTLNKPILLSGISNAEARQPGKAPNFSVNWT VGDSAI EVINATTGKDELGRASRLCKHALYCRWMRVHGKVP HLRSKITKPNVYHESKLAKEYQAAKARLFTAFIKAGLGAWV EKPTEQDQFSLTP	
ADAR2 (E488Q)	LHLDQTPSRQPIPSEGLQLHLPQVLADAVSRLVLGKFGDLTDNF SSPHARRKVLAVVMTTGTVDKDAKVISVSTGTCINGEYMSD RGLALNDCHAEIISRRSLRFLYTQLELYLNNKDDQKRSIFQKSE RGGFRLKENVQFHLIYSTSPCGDARIFSPHEPILEEPADRHPNRK ARGQLRTKIESGQGTIPVRSNASIQTDWGVLQGERLLTMSCDK IARWNVVGIQGSLLSIFVEPIYFSSIILGSLYHGDHLSRAMYQRIS NIEDLPPLYTLNKPILLSGISNAEARQPGKAPNFSVNWTVGDSAI VINATTGKDELGRASRLCKHALYCRWMRVHGKVP SHLLRSKIT KPNVYHESKLAKEYQAAKARLFTAFIKAGLGAWVEKPTEQD QFSLTP	70
CIRT-19	MGIINTLQKYICRVRGRCVLSCLPKEEIGKCTRGRKCCR RKKGGSGSGSGSGSGSGSMVETPRNHTIYINNLSKIK KDELKSLYAIQSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPYDKPMRIQYAKTDKRIPAKMKGTFVGSLLQPLPL ERLTGGSGSGSGSGSGSGSGSMSSEAETQPPAAPPAAPALS AADTKPGTTGSGAGSGGPGGLTSAAPAGDKKVIATKVLGT KWFNVRNGYGFINRNDTKEDVFVHQTAKNNPRKYLRVSGD GETVEFDVVEGEKGAEANVTGPGGVFVQGSKYAADRNHYRR YPRRGPPRNYQNYQNSESGEKNESSESAPEGQAQRRPYRR RRFPYMYRRPYGRRPQYSNPPVQGEVMEGADNQGAGEQGRP VRQNMRYGYRPRFRGPPRQRPREDGNEEDKENQDETQGG QPPQRRYRRNFYRRRRPENPKPDGKETKAADPPAENSSAPE AEQGAEGSGSGSGSGSGSGSLHLDQTPSRQPIPSEGLQL HLPQVLADAVSRLVLGKFGDLTDNFSSPHARRKVLAVVMTT GTVDKDAKVISVSTGTCINGEYMSDRGLALNDCHAEIISRRSL RFLYTQLELYLNNKDDQKRSIFQKSERGGFRLKENVQFHLI YSTSPCGDARIFSPHEPILEEPADRHPNRKARGQLRTKIESG GTIPVRSNASIQTDWGVLQGERLLTMSCDKIARWNVVGIQGS LLSIFVEPIYFSSIILGSLYHGDHLSRAMYQRISNIEDLPPLY TLNKPILLSGISNAEARQPGKAPNFSVNWTVGDSAI EVINATT GKDELGRASRLCKHALYCRWMRVHGKVP SHLLRSKITKPNVY HESKLAKEYQAAKARLFTAFIKAGLGAWVEKPTEQDQFSLTP GSDYKDDDDK*	68
YB-1- ADAR2 (wt)	MSSEAETQPPAAPPAAPALS AADTKPGTTGSGAGSGGPGGLT SAAPAGDKKVIATKVLGTVKWFNVRNGYGFINRNDTKEDVF VHQTAKNNPRKYLRVSGDGETVEFDVVEGEKGAEANVTG PGGVFVQGSKYAADRNHYRRYPRRGPPRNYQNYQNSESGE KNESESAPEGQAQRRPYRRRRFPYMYRRPYGRRPQYSNPP VQGEVMEGADNQGAGEQGRPVRQNMRYGYRPRFRGPPRQR QPREDGNEEDKENQDETQGGQPPQRRYRRNFYRRRRPENP KPQDGKETKAADPPAENSSAPEAEQGAEGSGSGSGSGSG SGGSLHLDQTPSRQPIPSEGLQLHLPQVLADAVSRLVLGKFGD LTDNFSSPHARRKVLAVVMTTGTVDKDAKVISVSTGTCING EYMSDRGLALNDCHAEIISRRSLRFLYTQLELYLNNKDDQKRS IFQKSERGGFRLKENVQFHLIYSTSPCGDARIFSPHEPILEEPADR HPNRKARGQLRTKIESGQGTIPVRSNASIQTDWGVLQGERLLTM SCSDKIARWNVVGIQGSLLSIFVEPIYFSSIILGSLYHGDHLSRAM YQRISNIEDLPPLYTLNKPILLSGISNAEARQPGKAPNFSVNWT VGDSAI EVINATTGKDELGRASRLCKHALYCRWMRVHGKVP SHLLRSKITKPNVYHESKLAKEYQAAKARLFTAFIKAGLGAWVE KPTEQDQFSLTP	73
ADAR2 (wt)	LHLDQTPSRQPIPSEGLQLHLPQVLADAVSRLVLGKFGDLTDNF SSPHARRKVLAVVMTTGTVDKDAKVISVSTGTCINGEYMSD RGLALNDCHAEIISRRSLRFLYTQLELYLNNKDDQKRSIFQKSE RGGFRLKENVQFHLIYSTSPCGDARIFSPHEPILEEPADRHPNRK ARGQLRTKIESGQGTIPVRSNASIQTDWGVLQGERLLTMSCDK IARWNVVGIQGSLLSIFVEPIYFSSIILGSLYHGDHLSRAMYQRIS	71

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Description	Sequence	Seq ID
	NIEDLPPLYTLNKPILLSGISNAEARQPGKAPNFSVNWTVGDSAIE VINATTGKDELGRASRLCKHALYCRWMRVHGKVP SHLLRSKIT KPNVYHESKLAAKEYQAAKARLFTAFIKAGLGAWVEKPTQD QFSLTP	
CIRT-20	MGIINTLQKYYCRVGGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGGSGSGSGSGSGSGSGSMVPETRPNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTFVGSLLPPL ERLTLGGSGSGSGSGSGSGSGSMAELTVEVRGSGNGAFYKGF KDVEDSLTVFENNWQPERQVPFNEVRLPPPDIKKEISEGDE VEVSRANDQEPGWWLAKVRMMKGEFYVIEYAACDATYNE IVTFERLRPNQNKTVKNTFFKCTVDVPEDLREACANENAHK DFKKAVGACRIFYHPETTQLMILSASEATVKRVNILSDMHLRSI RTKLMLMSRNEEATKHLECTKQLAAAFHEEFVREDLMGLAI GTHGSNIQQARKVPGVTAIELDEDTGTFRITYGESADAVKKARGF LEFVEDFIQVPRNLVGKVIKNGKVIQEIVDKSGVVRVRIEGDN ENKLPRDGMVPFVFGTKESIGNVQVLEHYHAYLKEVEQLR MERLQIDEQLRQIGRSYSGRGRGRGPNTSGYGTNSSELSNPS ETESERKDELSDWSLAGEDDRSRHQDRRRRPGGRGRSVSGG RGRGGPRGKSSISSVLKDPDPSNPYSLLDNTESDQTADTDASES HHSTNRRRRSRRRTDEDAVLMDGMTESDTASVNENGLGKRC DGS DYKDDDDK	77
FXR1	MAELTVEVRGSGNGAFYKGFIKDVEDSLTVFENNWQPERQV PFNEVRLPPPDIKKEISEGDEVEVSRANDQEPGWWLAKVR MMKGEFYVIEYAACDATYNEIVTFERLRPNQNKTVKNTFFK CTVDVPEDLREACANENAHKDFKKAVGACRIFYHPETTQLMIL SASSEATVKRVNILSDMHLRSIRTKLMLMSRNEEATKHLECTKQ LAAAFHEEFVREDLMGLAIGTHGSNIQQARKVPGVTAIELDE DTGTFRITYGESADAVKKARGFLEFVEDFIQVPRNLVGKVIKNG KVIQEIVDKSGVVRVRIEGDNENKLPRDGMVPFVFGTKESIG NVQVLEHYHAYLKEVEQLRMERLQIDEQLRQIGRSYSGRGRG RRGPNYTSYGTNSSELSNPSSETESERKDELSDWSLAGEDDRS RHQRDSRRRPGGRGRSVSGGRGGPRGKSSISSVLKDPDPSN PYSLLDNTESDQTADTDASESHHSTNRRRRSRRRTDEDAVLMD GMTESDTASVNENGLGKRC	74
CIRT-21	MGIINTLQKYYCRVGGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGGSGSGSGSGSGSGSGSMVPETRPNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTFVGSLLPPL ERLTLGGSGSGSGSGSGSGSGSMADRAPRQPTSLGSSPQRCG SPPSAPPPPPDRGERIEDCLAPLCPVVGGRGGCALSRFLPVDLS LLRSSRTAAVELVAGGVCSPSQENGSDYKDDDDK	78
APPLE	MADRAPRQPTSLGSSPQRCGSPPSAPPPPPDRGERIEDCLAPLCP PVVGGRGGCALSRFLPVDLSLLRSSRTAAVELVAGGVCSPSQ EN	75
CIRT-22	MGIINTLQKYYCRVGGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGGSGSGSGSGSGSGSGSMVPETRPNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTFVGSLLPPL ERLTLGGSGSGSGSGSGSGSGSMNKFDAKDDSDGDHDQ EENSTQKDGKEKTERDKNQSSSKRKAVVPGPAEHLQYNYTF WYSRTPGRPTSSQSYEQNIKQIGTFASVEQFWRFYSHMVRPG DLTGHSDFHLFKEGIKPMWEDDANKNGGKWIIRLRKGLASRC WENLILAMLGEQFMVGEEICGAVSVRFQEDIISIWNKTASDQA TTARIRDTLRRVLNLPNTIMEYKTHTDISKMPGRLGPQRLLFQ NLWKPRNLNVPGSYKDDDDK	79
eIF4E2	MNKFDALKDDSDGDHDQNEENSTQKDGKEKTERDKNQSS KRKAVVPGPAEHLQYNYTFWYSRTPGRPTSSQSYEQNIKQIG TFASVEQFWRFYSHMVRPGDLTGHSDFHLFKEGIKPMWEDDA NKNKGKWIIRLRKGLASRCWENLILAMLGEQFMVGEEICGAVV SVRFQEDIISIWNKTASDQATTARIRDTLRRVLNLPNTIMEYKT HTDSIKMPGRLGPQRLLFQNLWKPRNLNVP	76

VIII. EXAMPLES

[0156] The following examples are given for the purpose of illustrating various embodiments of the invention and are not meant to limit the present invention in any fashion. One

skilled in the art will appreciate readily that the present invention is well adapted to carry out the objects and obtain the ends and advantages mentioned, as well as those objects, ends and advantages inherent herein. The present examples,

along with the methods described herein are presently representative of preferred embodiments, are exemplary, and are not intended as limitations on the scope of the invention. Changes therein and other uses which are encompassed within the spirit of the invention as defined by the scope of the claims will occur to those skilled in the art.

Example 1: KNOCKDOWN OF PMP22
TRANSCRIPT IN NIH/3T3 CELLS BY CIRTSPin
AND CIRTSPY2

[0157] NIH-3T3 mouse fibroblasts were plated on 96-well plates and transfected at 70% confluency 24 hours later. 125 ng of the indicated gRNA-CIRTS expression plasmids was transfected per well using Lipofectamine LTX and Opti-MEM I Reduced Serum Medium (ThermoFisher Scientific) according to the manufacture’s protocol. After 48 hours, total RNA was harvested and isolated using the RNeasy Mini Kit (QIAGEN). After RNA isolation, RNA was reverse transcribed to cDNA using the PrimeScript RT Reagent Kit (TaKaRa). All qPCR reactions were run at 20 µL volumes with at least three biological replicates using PowerUp SYBR Green Master Mix (Applied Biosystems) and amplified on a LightCycler 96 Instrument (Roche). All qPCR primers used were designed on NCBI Primer BLAST and validated by no measured amplification in the absence of cDNA, a sharp, single-product melting temperature, and an amplification efficiency within 95-105%.

[0158] Expression levels were calculated using the house-keeping control gene (GAPDH) cycle threshold (Ct) value and the gene of interest Ct value. The relative expression level of PMP22 was determined by 2dCt, where dCt=Ct (PMP22)–Ct(GAPDH). Relative expression level for targeted gene was obtained upon normalizing the targeted gene expression level of cells experiments treated with the on-target gRNA to those treated by the nontargeting (NT) gRNA. The qPCR primers used in FIG. 1 are shown in the table below:

Sequences of qPCR primers used in FIG. 1			SEQ ID NO:
mPMP22 Primer 1	cgtccaacactgctactcctca		83
mPMP22 Primer 2	gcctttggtgagagtgaagagc		84
mGAPDH Primer 1	CATCACTGCCACCCAGAAGACTG		85
mGAPDH Primer 2	ATGCCAGTGAGCTTCCCGTTTCTAG		86

Example 2: PMP22 ACTIVATION WITH eIF4E
AND Y1-BASED CIRTSPACTIVATORS

[0159] SK-MEL-28 cells were plated in a 12-well plate and transfected 24 hours later using Lipofectamine LTX

according to the manufacturer’s protocol. Each well was transfected with a CIRTS (Bdef-TBP) expression plasmid containing either Y1 or EIF4E as the effector domain, along with a gRNA expression plasmid which was either nontargeting (NT) or complementary to the PMP22 mRNA. Protein was isolated from the SK-MEL-28 cells 48 hours later by incubation with RIPA buffer and protease inhibitors.

[0160] For western blotting, 22.5 µg of total protein was loaded for each condition then PMP22 and GAPDH levels were detected by incubation with anti-PMP22-HRP or anti-GAPDH-HRP antibody followed by chemiluminescent detection. When compared to the GAPDH loading control, transfection with Bdef-TBP-EIF4E and either g15, g17, or g1 appears to increase PMP22 protein present compared to the NT gRNA. Transfection with Bdef-TBP-Y1 and either g15 or g17 may also demonstrate detectable increases in PMP22 protein levels. All other conditions show PMP22 levels comparable to the NT negative control.

[0161] As shown in FIG. 2, g15 (targets 5’UTR) and g17 (targets 3’UTR) activate PMP22 expression, especially with an eIF4E-based CIRTS system.

Example 3: P53 EDITING WITH CIRTSPADAR
EDITOR

[0162] HEK293 cells were transfected using Lipofectamine 2000 with CIRTS-ADAR expression plasmid (125 ng), a gRNA expression plasmid (100 ng), and a P53 R175H reporter (12 ng). The gRNA expression plasmids expressed either a non-targeting control guide or one of six editing guides tiling the p53 R175 position. Each of the editing guides were designed such that the mismatch position, which directs the intended editing of p53 R175H, occurred at 11, 13, 15, 17, 19, or 21 nt from the 5’ end of the gRNA sequence. Each gRNA included two TAR hairpins. Editing efficiency of the guides to generate p53 R175H edited mRNA was monitored by Sanger sequence of RT-qPCR products. FIG. 3 demonstrates that the CIRTS-ADAR editor delivered to cells along with gRNAs can edit p53 mismatch sites and the highest editing is observed when the mismatch occurs 15 nucleotides from the 5’ end of the gRNA sequence.

[0163] The gRNAs used in FIGS. 1-2 are shown in the tables below.

FIG. 1 mouse PMP22 targeting sequences		
gRNA Localization and Number	gRNA Sequence	SEQ ID NO:
Non- targeting guide (NT)	TGACAGCCCACATGGCATTCCACTTATCACTGGCATCCTT	87
5' Untranslated Region (UTR)		
g1	GTGGCTGGGAGTTGGGCTCGGGATCAGAGGAGCGGTGCCT	88
Coding Sequence (CDS)		
g1	CAGAAGAACAGGAACAGAGCCAGGACGCTGAAGATGACA G	89
g2	ATCACATAGATGATACCACTGAGGAGGGCTAGGGGAAAGG	90
g3	GAGACGAAGAGCAACACTAGCACCGCGATGTGCAGGAAC A	91
g4	CGATGTGCAGGAACAGGATCCCCAACAGAGTAGGAGCAT	92
g5	AATCCAGTGTATGTAACCGGCCCTTTGGTGAGAGTGA	93
g6	ATGCCACTCACTGTGCCCTCACTGTGTAGATGGCCGCTGCA	94
g7	TCAGTCGTGTGTCCATTGCCCCAGAGCCATTGGCTGACGA	95
g8	AGACTGCAGCCATTCGCTCACTGATGAGGAGTAGCAGTGT	96
g9	CTGAAGATGACAGACAGGATCATGGTGGCCTGGACAGACT	97
g10	GCGAAGCCATAGGAGTAGTCAGTGTGACATGCCACTCAC	98
3' UTR		
g1	CGCGCTCAGAGCCTAGACGGACGGTGCCTCGTCGGGCGCC	99
g2	CCACAGGGAGGGAGGAAGGAAACCAGAGAACAAACCAA C	100
g3	TTGGTGGCCAATACAAGTCATCGCCAGACAGTCCTTGGAG	101
g4	CAGTTGGTATAAAATCAGAGAAAGCAAGCAAAAACAAGA A	102
g5	GATTTATTTATTGTAGTAATAACAACAGCCTAGCTAGGTA	103
g6	GATATTATATACATCTTCAATCAACAGCAATCCCACTCA	104
g7	CAAAACAAAGATACTGAGCTTGGTTATAGTGTAGGATGT	105
g8	CTTGCTCTGACTTCTCATTAAACGTGTGGCCCCAGTAC	106
g9	TGCACTGGGTCACCCATAGTGTAAGGTCTTCAACTGTCT	107
FIG. 2 human PMP22 targeting sequences		
gRNA Localization and Number	gRNA Sequence	SEQ ID NO:
Non- targeting guide (NT)	TGACAGCCCACATGGCATTCCACTTATCACTGGCATCCTT	87
5' UTR		
g1	GCTCAGCGGAGTTTCTGCCCGGCCAAACAGCGTAACCCCT	108
g15	CTGGCGGCAAGTTCTGCTCAGCGGAGTTTCTGCCCGGCCA	109

-continued

Coding Sequence (CDS)		
g3	GGGTGAAGAGTTGGCAGAAGAACAGGAACAGAGACAGAA T	110
3' UTR		
g5	TTTTCCTTCCTCCCTTCCCTATGTACGCTCAGAGCCTCAG	111
g17	CCTATGTACGCTCAGAGCCTCAGACAGACCGTCTGGGCGC	112

Description	Sequences	SEQ ID NO:
CIRTS used in FIG 2.		
Bdef-TBP-Y1 (N-terminal truncation)	MGIINTLQKYICRVRGRCVLSCLPKEEQIGKCSTRGRKCCRRK KGGSGSGSGSGSGSGSGSMAPETRPNHTIYINNLSKIKKDEL KKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATNALRSM QGFFPFYDKPMRIQYAKTDKRIPAKMKGT FVGS LQLPPLERLTLGG SGSGSGSGSGSGSGSMSATSVD TQRTKGQDNKVQNGSLHQKD TVHDNDFEPYLTGQSNQSN SYPSMDFYLS SYPPSIGFPYSLNEA PWSTAGDPPIPYLT TYGQLSNGDHHFMHDAVFGQPGGLGNNIYQ HRFNFFPENPAFSAWGTSGSQGQQTQSSAYGSSYTYPPSSLGGTV VDGQPGFHS DTL SKAPGMNSLEQGMVGLKIGDVSSAVKT VGSV VSSVALTGVLSNGGTNVNMPVSKPTSWAAIASKPAKPQPKMKT KSGFVMGGGLPPPIKHNM DIGTWDNKGVPKAPVPQQAPSPQA APQPQQVAQPLPAQPPALAQPPQYQSPQQPPQTRWVAPRNNAAF GQSGGAGSDSNSPGNVQPN SAPSVESGSDYKDDDDK	80
Bdef-TBP-EIF4E	MGIINTLQKYICRVRGRCVLSCLPKEEQIGKCSTRGRKCCRRK KGGSGSGSGSGSGSGSGSMAPETRPNHTIYINNLSKIKKDEL KKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATNALRSM QGFFPFYDKPMRIQYAKTDKRIPAKMKGT FVGS LQLPPLERLTLGG SGSGSGSGSGSGSGSMATVEPETTPNPTTEEEKTESNQEVA NPEHYIKHPLQNRWALWFFKNDKSKTWQANLRLISKEDTVEDFW ALYNHIQLSSNLMFGCDYSLFKDGI EPMWEDEKNRGRWLITL NKQQRSDLD RFWLETLLCLIGESFDDYSDDVCGAVNVNRAKGD KIAIWTTECENREAVTHIGRVYKERLGLPKKIVIGYQSHADTATKS GSTTKNRFVVGSDYKDDDDK	81
CIRTS used in FIG 3.		
CIRTS-ADAR (Bdef-TBP-ADAR)	MGIINTLQKYICRVRGRCVLSCLPKEEQIGKCSTRGRKCCRRK KGGSGSGSGSGSGSGSGSMAPETRPNHTIYINNLSKIKKDEL KKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATNALRSM QGFFPFYDKPMRIQYAKTDKRIPAKMKGT FVGS LQLPPLERLTLGG SGGSLHLDQTPSRQPIPS EQLQLHLPQVLADAVSRLVLGKFGDLT DNFSSPHARKVLAGVVM TTGTDVDAKVISVSTGKTCINGEYM SDRGLALNDCHAETISRRSLLRFLYTQLELYLNNKDDQKRSIFQKS ERGGFRLEKENVQFHLIYSTSPCGDARIFSPHEPILEPADRHPRKA RGQLRTKIESGEGTIPVRSNASIQTDGVLQGERLLTMSCDKIAR WNVVGIQGSLLSIFVEPIYFSSIILGSLYHGDHLSRAMYQRISNIEDL PPLYTLNKPLLSGISNAEARQPGKAPNFSVNWTVGDSAIEVINATT GKDELGRASRLCKHALYCRWMRVHGKVP SHLLRSKITKPNVYHE SKLAAKEYQAARLFTAFIKAGLGAWVEKPT EQDQFSLTPKRP AATKKAGQAKKKKGS DYKDDDDK	82

Example 4: RNA Degraders

A. FIG. 4 Methods and Results

[0164] Methods: See qPCR and transfection methods described in Example 1.

[0165] Results: Several novel CIRTs degraders were tested in NIH-3T3 cells along with gRNAs validated in Example 1 that land in the coding sequence (g6 CDS), 3'

UTR (g2 and g3 3' UTR), and 5' UTR (g1 5' UTR). PMP22 transcript degradation was measured by RT-qPCR as described in Example 1. These experiments demonstrated PMP22 transcript degradation from each CIRTs tested with PMP22 expression levels depending on the effector and gRNA pair deployed. Additionally, we have targeted these degraders to the 5' UTR as well but found this gRNA landing site was generally suboptimal for CIRTs-mediated degradation.

TABLE		
Effectors		
Description	Sequence	SEQ ID NO:
GFP	VSKGEELFTGVVPIIIVELDGDVNGHKFSVSGEGEDATYGKLT LKFICTTGKLPVPWPTLVTTLTYGVCFSRYPDHMKQHDFFKS AMPEGYVQERTIFFKDDGNYKTRAEVKFEGDTLVNRIELKGIDF KEDGNILGHKLEYNHNHVMYIMADKQKNGIKVNFKIRHNIED GSVQLADHYQQNTPIGDGPVLLPDNHYLSTQSALS KDPNEKRD HMLLEFVTAAGITLGMDELYK	113
TOB2	MQL EIKVALNFIISVLYNKLPRRRADLFGEELERLLKKKYEGHW YPEKPLKSGSFRVHIGEMVDPVVELAAKRSGLAVEDVRANVP EELSVWIDPFVSYQIGEGAVKVLVLDLDDSGCGAPELDKEIKS SFNPDAQVFVPIGSQDSSLNSPSPSPSGQSPSPTFIPRSAQPIITFTT ASFAATKFGSTKMKKGGGAASGGGVASSGAGGQQPPQQPRM ARSPTNSLLKHKSLSLSMHS LNFI TANPAPQSQLSPNAKEFVYN GGGSPLS LFFDAADGQSGTGPFGGSGAGTCNSSSFDMAQVFG GGANSLFLEKTPFVEGLSYNLNTMQYPSQQFQPVVLAN	114
RNase4	MQDGM YQRFLRQH VHP EETGGSDRYCNLMMQRRKMTLYHC KRNTFIHEDIWNIRSI CSTTNIQCKNGKMNCHGVVKVTD CRD TGSSRAPNCRYRAIASTRRVVIACEGNPQVPVHFDG	115
RNaseL	WK PQSSHWGAALKDLHRIYRPMIGKLKFFIDEKYKIADTSEGGI YLG FYEKQEVAVKTFCEGSPRAQREVSC LQSSRENSHLVTFYG SESHRGHLFVCVTLCEQTLEACLDVHRGEDVENEDEFARNVL SSI FKAVQELHLSCGYTHQDLQPQNILIDSKKAHLADF KSIK WAGDPQEVKRDLEDLGRVLVYVVKKGSISFEDLKAQSNEEVV QLSPDEETKD LIHRLFHPGEHVRDCLSDLLGHPPFWTWESRYRT LRNVGNESDIKTRKSESEILRL LQPGPSEHSKSFDKWTKINECV MKMKNFY EKRGNFYQNTVGDLLKFIRNLGEHIDEKHKMKM LKI GDPSLYFQKTFPDLVIYVYTKLQNT EYRKHFQTHSPNKPQ CDGAGGASGLASPGC	116
RNaseA	ETSAEKFERQHMDADSSPASNSNYCNMMSRSMTTDRCKPV NTFIHEPLADVQAVCSQTNVACKNGQTNCYQSKAAMRI TDCRE TGSSKYPDCAYTTTEEEKHIIVACEGNPYVPVHFDATV	117
RNase1	MKESRAKKFQRQHMDSDSSPSSSSTYCNQMMRRRNMTQGR C KPVNTFVHEPLVDVQNVCFQEKVTC KNGQGN CYKSNSSMHIT DCRLINGSRYPNCAYRTSPKERHIIVACEGSPYVPVHFDASVED ST	118
RNaseU2	CDI PQSTNCGGNVYSNDDINTAIQGALDDVANGDRPDNYPHQY YDEASEDITLCCSGPWFSEFPLVYNGPYYSSRDNYVSPGDRVI YQNTNGEFCATVTHTGAASYDGFTQCS	119
HRSP12	MSSLIRRVISTAKAPGAIGPYSQAVLVDR TIYISGQIGMDPSSGQ LVSGGVAEEAKQALKNMGEILKAAGCDFTNVVKTTVLLADIN DPNTVNEIYKQYFKSNFPARAAYQVAALPKGSRIEIEAVAIQGP LTTASL	120
RNaseP (RPP25)	MENFRKVRSEEPAGCGAEGGPGSGPFADLAPGAVHMRVKE GSKIRNLMAFATASMAQPATRAIVFSGCGRATTKTVTCAEILKR RLAGLHQVTRLRYSVREVWQSLPPGPTQGQTGPGEPAASLSVL KNVPGLA ILLSKDALDPRQPGYQPPNPHPGPSSPPAAPASKRSL GEPAAEGSAKRSQPEPGVADEDQTA	121

TABLE

Full CIRT sequences used in FIG. 4: (Bdef-(GGG) ₆ -TBP6.7-NES-(GGG) ₆ -Effector-Tag		
Description	Sequence	SEQ ID NO:
GFP	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCSTRGRKC CRRKKGSGSGSGSGSGSGSGSGSMVAPETRPNHTIYINNLN SKIKKDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKE VSSATNALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTF VGSLLQLPPLERLTLGGSGSGSGSGSGSGSGSGSVSGEELFT GVVPILVELDGDVNGHKFSVSGEGEGDATYGLTLKFICTT GKLPVPWPTLVTTLTYGVQCFSRYPDHMKQHDFFKSAMPE GYVQERTIFFKDDGNKYKTRAEVKFEGDTLVNRIELKGIDFKE DGNILGHKLEYNYSNHVYIMADKQKNGIKVNFKIRHNIED GSVQLADHYQNTPIGDGPVLLPDNHYLSTQSALSKDPNEK RDHMLLEFVTAAGITLGMDELYKGSYKDDDDK	148
Y2 (1-191)	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCSTRGRKC CRRKKGSGSGSGSGSGSGSGSGSMVAPETRPNHTIYINNLN SKIKKDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKE VSSATNALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTF VGSLLQLPPLERLTLGGSGSGSGSGSGSGSGSGSASLLEQ RPGQGNKVQNGSVHQKDLNDDDFEYLSQARPNNAIT AMSDSYLPSYSPSIGFSYSLGEAAWSTGGDTAMPYLTSYG QLSNGEPHFLPDAMFGQPGALGSTPFLGQHGFNFFPSGIDFS AWGNNSQQGSTQSSGYSSNYAYAPSSLGGAMIDGQSAPFA NETLNKAPGMNTIDQGMAGSDYKDDDDK	149
CNOT7	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCSTRGRKC CRRKKGSGSGSGSGSGSGSGSGSMVAPETRPNHTIYINNLN SKIKKDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKE VSSATNALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTF VGSLLQLPPLERLTLGGSGSGSGSGSGSGSGSGMPAATVDHS QRICEVWACNLDEEMKKIRQVIRKYNVAMDETFPGVVAR PIGEFRSNADYQYQLLRCNVDLLKIQLGLTFMNEQGEYPPG TSTWQPNFKFNLTEDMYAQDSIELLTSGIQFKHEEEGIET QYFAELMTSGVVLCEGVKWLSPHSGYDFGYLIKILTNSNL PEEELDFFEILRLFFPVIYDVKYLMSCKNLKGLQEVABQL ELERIGPQHQAQSDSLTGMFAFFKMREMFEDHIDDAKYCG HLYGLSGSSYVQNGTGNAEYEEANKQSGSDYKDDDDK	150
SNRPA	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCSTRGRKC CRRKKGSGSGSGSGSGSGSGSGSMVAPETRPNHTIYINNLN SKIKKDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKE VSSATNALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTF VGSLLQLPPLERLTLGGSGSGSGSGSGSGSGSGMAVETRPN HTIYINNLEKIKKDELKKSLEYAIFSQFGQILDILVSRSLKMR GQAFVIFKEVSSATNALRSMQGFPPFYDKPMRIQYAKTDSII AKMKGTFVERDRKREKRPKSQETPATKKAVQGGGATPVV GAVQGPVPGMPMTQAPRIMHMPGQPPYMPPPGMIPPPG LAPGQIPPGAMPQQQLMPGQMPPAQPLSENPNHILFLTNLP EETNELMLMLFNQFPGFKEVRLVPGRHDI AFVEFDNEVQA GAARDALQGFKITQNNAMKISFAKKGSDYKDDDDK	63
ZFP36	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCSTRGRKC CRRKKGSGSGSGSGSGSGSGSGSMVAPETRPNHTIYINNLN SKIKKDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKE VSSATNALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTF VGSLLQLPPLERLTLGGSGSGSGSGSGSGSGSGMDLTAIYESL LSLSPDPVPVPSDHGTESSPGWSSSGPWSLSPSDSSPSGVS RLPGRSTSLVEGRSCGWVPPPPGFAPLAPRLGPELSPSPTSPT ATSTPSTRYKTELCTPSESSEGRCRYGAKQFAHGLGELRQA NRHPKYKTELCHKFYLGRCPYGSRCHFHNPSDELAAAPH PPVLRQISIFSGLPSGRRTSPPPPGLAGPSLSSSSSPSSPPPP GDLPLSPSAFSAAPGTPLARRDPTPVCCPSCRRATPISVWGP LGLVLRTPSVQSLGSDPDEYASSGSLGSDSPVFEAGVFAP PQPVAAPRRLPINFNISVSEGSYKDDDDK	151
TOB2	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCSTRGRKC CRRKKGSGSGSGSGSGSGSGSGSMVAPETRPNHTIYINNLN SKIKKDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKE VSSATNALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTF VGSLLQLPPLERLTLGGSGSGSGSGSGSGSGSGMQLEIKVAL NFIISYLYNKLPRRADLFGEELERLLKKYEGHWYPEKPLK SGGFRCVHIGEMVDPVVELAAKRSGLAVEDVRANVPEELS VWIDPFVVSQIGEGKAVKVLVLDSEGGCAPELDKEIKSSF	152

TABLE-continued

Full CITS sequences used in FIG. 4: (Bdef-(GGG) ₆ -TBP6.7-NES-(GGG) ₆ -Effector-Tag		
Description	Sequence	SEQ ID NO:
	NPDAQVFVPIGSQDSSLNSPSPSFGQSPSPTFIPRSAQPIITFTT ASFAATKFGSTKMKGGGAASGGGVASSGAGGQPPQOPR MARSPTNLLKHKSLSLSMHSLNFITANPAPQSQSPNAKEF VYNGGGSPSLFFDAADGQSGTGPFGGSGAGTCNSSSFD AQVFGGANSLEKTPFVEGLSYNLNTMQYPSQQFPVV ANGSDYKDDDDK	
RNase4	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCTRGRKC CRRKKGSGSGSGSGSGSGSGSMVETRPNHTIYINN SKIKDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKE VSSATNALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGT VGLQLPPLERLTLGGSGSGSGSGSGSGSMQDGMQYR FLRQHVHPEETGSDRYCNLMQRRKMTLYHCKRFNTFIH EDIWNIRSICSTTNIQCKNGKMNCHGVVKVTDCTDGTSSR APNCRYRAIASTRRVVIACEGNPQVPVHFDGSDYKDDDD K	153
RNaseL	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCTRGRKC CRRKKGSGSGSGSGSGSGSGSMVETRPNHTIYINN SKIKDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKE VSSATNALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGT VGLQLPPLERLTLGGSGSGSGSGSGSGSWKPSHW GAALKDLHRIYRPMIGKLKFFIDEKYKIADTSEGGIYLG KQEVAVKTFCEGSPRAQREVSLQSSRENSHLVTFYGS RGHLFVCVTLCEQTLACLVDVHRGEDVENEDEFARNV IFKAVQELHLSCGYTHQDLQPQNILIDSKAAHLADF WAGDPQEVKRDLEDLGRVLVYVVKGSISPEDLKAQ VVQLSPDEETKDLIHLRFHPGEHVRDCLDLLGH SRYRTLNRVGNESDIKTRKSESEILRLQLPGPSEH TTKINECVMKMKNKFYEKRGNFYQNTVGDLLKFI DEEKHKMKLKIIGDPSLYFQKTFPDLVIYVYTKLQ HFPQTHSPNKPQCDGAGGASGLASPGCGSDYKDDDD DK	154
RNaseA	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCTRGRKC CRRKKGSGSGSGSGSGSGSGSMVETRPNHTIYINN SKIKDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKE VSSATNALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGT VGLQLPPLERLTLGGSGSGSGSGSGSGSETSAEKFERQ HMDADSSPASNSNYCNKMMKSRSMTTDRCKPVNTFI ADVQAVCSQTNVACKNGQTNCYQSKAMRITDCRETS YPCAYTTTEEEKHIVACEGNPYVPVHFDATVGS DK	155
Pin nuclease	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCTRGRKC CRRKKGSGSGSGSGSGSGSGSMVETRPNHTIYINN SKIKDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKE VSSATNALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGT VGLQLPPLERLTLGGSGSGSGSGSGSGSMEIRPLFL VPDINGFIDHLASLARLLESRKYLIVVPLIVINELDL TDRAGGYARVVQEKARKSIEFLEQRFESRDSCLRALT NELESIAFRSEDTGQLGNDDLLSCCLHYCKDKAKDF SKEEPIRLLEVVLLTDDRNLRVKALTRNVVPRDIPAF QVGGSDYKDDDDK	156
RNase1	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCTRGRKC CRRKKGSGSGSGSGSGSGSGSMVETRPNHTIYINN SKIKDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKE VSSATNALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGT VGLQLPPLERLTLGGSGSGSGSGSGSGSMKESRAK QRQHMDSDSSPSSSTYCNQMMRRRNMTQGRCKPVNT EPLVDVQNVCFQEKVTCNGQNCYKSNSSMHITDCRL GSRYPNCAVRTSPKERHIVACEGSPYVPVHFDASVED DYKDDDDK	157

TABLE-continued

Full CIRTs sequences used in FIG. 4: (Bdef-(GGs) ₆ -TBP6.7-NES-(GGs) ₆ -Effector-Tag		
Description	Sequence	SEQ ID NO:
RNaseU2	MGIINTLQKYYCRVRGRC AVLSCLPKEEQIGKCSTRGRKC CRRKKGGSGGSGGSGGSGGSGGSMVPE TRPNHTIYINN LN SKIKKDELKKS LYAIFSQFGQILDILVPRQRTPRGQAFVIFKE VSSATNALRSMQGFPFYDKPMRIQYAKTDKRIPAKMKGTF VGS LQLPPLERLTLGGSGGSGGSGGSGGSGGSCDIPQSTNCG GNVYSNDDINTAIQ GALLDDVANGDRPDNYPHQYYDEASED ITLCCGSGPWSEFPLVYNGPYSSRDNYVSPGPDRVIYQNTNT GEFCATVTHTGAA SYDGF TQCSGSDYKDDDDK	158
HRSP12	MGIINTLQKYYCRVRGRC AVLSCLPKEEQIGKCSTRGRKC CRRKKGGSGGSGGSGGSGGSGGSMVPE TRPNHTIYINN LN SKIKKDELKKS LYAIFSQFGQILDILVPRQRTPRGQAFVIFKE VSSATNALRSMQGFPFYDKPMRIQYAKTDKRIPAKMKGTF VGS LQLPPLERLTLGGSGGSGGSGGSGGSGGSMSS LIRRVIS TAKAPGAIGPYSAVLVDRTIYISGQIGMDPSSGQLVSGGVA EEAKQALKNMGEILKAAGCDFTNVVKTTVLLADINDENTV NEIYKQYFKSNFPARAAYQVAALPKGSRIEIEAVAIQGPLTT ASLGS DYKDDDDK	159
RNaseP (RPP25)	MGIINTLQKYYCRVRGRC AVLSCLPKEEQIGKCSTRGRKC CRRKKGGSGGSGGSGGSGGSGGSMVPE TRPNHTIYINN LN SKIKKDELKKS LYAIFSQFGQILDILVPRQRTPRGQAFVIFKE VSSATNALRSMQGFPFYDKPMRIQYAKTDKRIPAKMKGTF VGS LQLPPLERLTLGGSGGSGGSGGSGGSGGSMENFRKVR S EEAPAGCGAEGGGPGSGPFADLAPGAVHMRVKEGSKIRNL MAFATASMAQPATRAIVESGCGRATTKTVTCAEILKRRLAG LHQVTRLRYSRVREVWQSLPPGP TQGGTPGEPAASLSVLKN VPG LAI LLSKDALDPRQPGYQPPNPHPGPSPPAAPASKRSL GEPAAGEGSAKRSQPEPGVADEDQTAGSDYKDDDDK	160

B. FIG. 5 Methods and Results

[0166] Methods: HEK293T cells were plated on 96-well plates and transfected at 60% confluency 24 hours later. 250 ng of the indicated gRNA-CIRTs expression plasmids was transfected per well using Lipofectamine 2000 and Opti-MEM I Reduced Serum Medium (ThermoFisher Scientific) according to the manufacture’s protocol. After 48 hours, total RNA was harvested and isolated using the RNeasy Mini Kit (QIAGEN). After RNA isolation, RNA was reverse transcribed to cDNA using the PrimeScript RT Reagent Kit (TaKaRa). All qPCR reactions were run at 20 AL volumes with at least three biological replicates using PowerUp SYBR Green Master Mix (Applied Biosystems) and amplified on a Quantstudio 6 (Applied Biosystems). All qPCR primers used were designed on NCBI Primer BLAST and validated by no measured amplification in the absence of

cDNA, a sharp, single-product melting temperature, and an amplification efficiency within 95-105%.

[0167] Expression levels were calculated using the house-keeping control gene (beta-actin) cycle threshold (Ct) value and the gene of interest Ct value. The relative expression level of SNCA was determined by 2dCt, where dCt=Ct (SNCA)–Ct(beta-actin). Relative expression level for targeted gene was obtained upon normalizing the targeted gene expression level of cells experiments treated with the on-target gRNA to those treated by the nontargeting (NT) gRNA.

[0168] Results: We deployed CIRTs-Y2 with a gRNA panel targeting SNCA in HEK293T cells and demonstrated that CIRTs-Y2 can degrade SNCA transcript levels in a gRNA-dependent manner.

[0169] gRNA sequences used in FIG. 5

Description	Sequence	SEQ ID NO:
g1	TAATGAATTCCTTTACACCACACTGTCGTCGAATGGCCAC	122
g2	TCTTCTCAGCCACTGTTGCCACACCATGCACCACTCCCTC	123
g3	AGAAGGCATTTCATAAGCCTCATTGTCAGGATCCACAGGC	124
g4	TTAACATCGTAGATTGAAGCCACAAAATCCACAGCACACA	125
g5	TTTCAATAATTAATCACAGCCACTTAAGGAACCAAGTGCAT	126
g6	CGAATATTATTATTGTGAGAAAGGTACAGCATTACACC	127

-continued

g7	GCAACATTTAAGGAGGCCATGAAATTTTATCAGTCTAG	128
g8	CAGAGTTTTTACCATCAAGTTTATTTTCAGCACCCATGG	129
g9	CTACTAGTGACTTTTAGAAATGACTATGCCCCAGAATAAT	130
g10	TAGATTTAAATATATCCTAACCGCCACTTTCTAACCTTC	131
Sequences of qPCR primers used in FIG. 5		SEQ ID NO:
SNCA Primer 1	CAACAGTGGCTGAGAAGACCA	132
SNCA Primer 2	GCTCCTTCTTCATTCTTGCCCA	133
Beta-actin Primer 1	CACCAACTGGGACGACAT	134
Beta-actin Primer 2	ACAGCCTGGATAGCAACG	135

Example 5: CIRTs Activators

[0170] FIG. 6 graphically represents the dual luciferase assays carried out in FIG. 7 and FIG. 8 to characterize the ability of novel CIRTs activators to increase Firefly luciferase levels in a gRNA-dependent manner.

A. FIG. 7 Methods and Results

[0171] Methods: Gibson Assembly was used to assemble all plasmids and plasmids were sequenced by the University of Chicago Comprehensive Cancer Center DNA Sequencing and Genotyping Facility. PCR fragments used for Gibson Assembly were amplified with Q5 DNA Polymerase (NEB). All transcript UTR regions were either cloned from gBlock (IDT) or amplified from reverse transcription reactions completed with total RNA isolated from Neuro 2a (ATCC) cells for SCN1a 3' UTR.

[0172] HEK293T cells were plated on 96-well plates and transfected at 70% confluency 24 hours later. 250 ng of the indicated gRNA-CIRTs expression plasmids and 12 ng reporter plasmid were transfected per well using Lipofectamine 2000 and Opti-MEM I Reduced Serum Medium (ThermoFisher Scientific) according to the manufacture's protocol. After 48 hours, luciferase activity for both Firefly

and nano luciferase was assayed according to protocols previously developed and described by Baker et. al (using the nano luciferase in place of *Renilla* luciferase). Firefly luciferase values were then normalized to the corresponding nano luciferase readout by dividing the firefly RLU by nano luciferase RLU for each individual well. Fold-changes were then calculated by averaging the luciferase ratio for the negative control and dividing all biological replicates by the negative control's luciferase ratio (see, for example, J. M. Baker and F. M. Boyce, J. Vis. Exp. JoVE, 2014, 50282). [0173] Results: Plasmids were constructed in order to develop a dual-luciferase reporter assay which expresses a luciferase transcript appended with the mouse PMP22 transcript 5' and 3' UTR regions. CIRTs targeting the PMP22 UTRs that affect translation create changes in the luminescence readout that can be readily normalized by a second, untargeted luciferase (nano luciferase) also expressed from the reporter construct. This reporter system was leveraged to test previously reported (Y1) and novel CIRTs activators for the ability to increase luciferase expression when directed to the reporter's PMP22 3' UTR (g3) compared to effector-matched constructs with non-targeting (NT) gRNA. These results demonstrate several novel CIRTs with the ability to increase firefly expression in a gRNA-dependent manner.

TABLE

Effectors		
Description	Sequence	SEQ ID NO.:
PABP	MNPSAPSYPMASLYVGDLPDVTEAMLYEFSPAGPILSIRVCR DMI TRRSLGYAVNFPQQPADAERALDTMNFVDVI KGKPVRIWMS QRDPSLRKSGVGNIFIKNLDKSIDNKALYDTFSAPGNILSKVVC DENGSKGYGFVHFETQEAARAIEKMNGMLLNDKRVFVGRPKS RKERAEELGARAKEFTNVYIKNFGEDMDDERLKDLPFGKFGPAL SVKVMTDESGKSGFGFVSFERHEDAQKAVIDEMNGKELNGKQI YVGRAQKQKVERQTELKRFQMKQDRITRYQGVNLYVKNLDD GIDDERLRKEFSPFGTITSKVMMEGGRSKGFGFVCFSSPEEATK AVTEMNGRIVATKPLYVALAQRKEERQAHLTNQYMQRMASVR AVPNPVINPYQPAPPSGYFMAAIPQTQNRAAAYPPSQIAQLRSP RWTAQGARPHFPQNMPGAIRPAAPRPPFSTMRPASSQVPRVMST QRVANTSQTMGPRPAAAAAATPAVRTVPQYKYAAGVRNPQ QHLNAQPQVTMQPPAVHVQGEPLTASMLASAPPEQKQMLG ERLFPLIQAMHPTLAGKITGMLEIDNSELHLMLESPELSLRKVD EAVAVLQAHQAKEAAQKAVNSATGVPTV	136

TABLE-continued

Effectors		SEQ ID NO.:
Description	Sequence	
eIF4G (711-1599)	EPRKIIATVLMTEDIKLNKA EKAWKPSSKRTAADKDRGEEDADG SKTQDLFRRVRSILNKLTPQMFPQQLMKQVTQLAIDTEERLKGVI DLIFEKAI SEPNFSVAYANMCRCLMALKVPTTEKPTVTNFRKL LLNRCQKEFEKDKDDDEVFEKKQKEMDEAATAEERGRLKEELE EARDIARRRSLGNIKFIGELFKLKM LTEAIMHDCVVKLLKNHDE ESLECLCRLLTTIGKDLDFEKAKPRMDQYFNQMEKI I KEKKTSSR IRFMLQDVLDLRGSNWVPRRGDQGPKTIDQIHKEAEME E HREHI KVQQLMAKGS DKRRGGPGPPISRGLPLVDDGGWNTVPI SKGSR PIDTSRLTKITKPGSIDSNQLFAPGGRLSWGKGSSGGSGAKPSD AASEAARPATSTLNRFSALQQAVPTESDNRVVRSSLSRERG EKAGDRGDRLESERGGDRGDR LDRARTPATKRSFSKEVEERSR ERPSQPEGLRKAASLTEDRDRGRDAVKREAAALPPVSPLKAALSE EELEKKSKAIIEEYLHLNDMKEAVQCVELASPSLLFIFVRHGV E STLSAISAREHMQLLHQLLCAGHLSTAQYYQGLYEILELAED MEIDIPHVWLYLAELVTPI LQEGGVPMGELFREITKPLRPLGKAA SLLEILGLLCKSMGPKKVGTLWREAGLSWKEPLPEGQDIGAFV AEQKVEYTLGESEAPQGRALPSEELNRQLEKLLKEGSSNQRVF DWIEANLSEQQIVSNTLVRLMTAVCYSAI IFETPLRVDVAVLK ARAKLLQKYLCEQKELQALYALQALVVLTLEQPPNLLRMFFDA LYDEDVVKEDAFYSWESSKDPAEQQGKGVALKSVTAFFKWLR EAEEESDHN	137
eIF4G (711-1105)	EPRKIIATVLMTEDIKLNKA EKAWKPSSKRTAADKDRGEEDADG SKTQDLFRRVRSILNKLTPQMFPQQLMKQVTQLAIDTEERLKGVI DLIFEKAI SEPNFSVAYANMCRCLMALKVPTTEKPTVTNFRKL LLNRCQKEFEKDKDDDEVFEKKQKEMDEAATAEERGRLKEELE EARDIARRRSLGNIKFIGELFKLKM LTEAIMHDCVVKLLKNHDE ESLECLCRLLTTIGKDLDFEKAKPRMDQYFNQMEKI I KEKKTSSR IRFMLQDVLDLRGSNWVPRRGDQGPKTIDQIHKEAEME E HREHI KVQQLMAKGS DKRRGGPGPPISRGLPLVDDGGWNTVPI SKGSR PIDTSRLTKITKPGSIDSNQLFAPGGRLSWGKGSSGGSGAK	138
Bo11	MQTDSLSPSPNPVSPVPLNNPTSAPRYGTVIPNRI FVGGIDFKTNE SDLRKFFSYQGSVKEVKIVNDRAGVSKGYGFVTFETQEDAQKIL QEAEKLNKYDKKLNIGPAIRKQQVGIPRSSIMPAAGTMYLTTST GYPYTYHNGVAYFHTPEVTSVPPWPSPRSVCSPPVMVAQPI YQQ PAYHYQATTQYLPQGWQWSVPQPSASSAPPLYLQPSSEVIYQPV E IAQDGGCVPPPLSLMETSVPPEPYSDHGVQATYHQVYAPSAITMP APVMQPEPIKTVWSIHY	139
ERa	MTMTLHTKASGMALLHQIQGNELEPLNRPQLKIPLERPLGEVYL DSSKPAVYNYPEGAAYEFNAAAAANAQVYGTLPGYGGSEA AAFGSNGLGGFPPLNSVSPSPLMLLHPPQLSPFLQPHGQQVPY Y LENEPSGYTVREAGPPAFYRPNSDNRQGGRERLASTNDKGS M AMESA KETRYCAVCNDYASGYHYGVWSCGCKAFFKRSIQGH NDYMC PATNQCTIDKNRRKSCQACRLRKCYEVGMMKG GIRKD RRGGRMLKHKRQRDDGEGRGVGSAGDMRAANLWPSPLMIKR SKKNSLALSLTADQMVSALDAEPPILYSEYDPTRPFS EASMMG LLTNLADRELVHMINWAKRVPGFVDLTLDHQVHLLCAWLEIL MIGLVWRSMEHPGKLLFAPNLLDRNQGKCEGMVEIFDMLLA TSSRFRMMNLQGEFVCLKSII LNSGVYTFLSSTLKSLEEKDHIH RVLDKI TDTLIHLMAKAGLTQQQHQLAQLLLI LSHIRHMSNK GMEHLYSMCKKNVVPLYD LLEMLDAHRLHAPTSRGGASVEE TDQSHLATAGTSSHSLQKYIITGEAEGFPATV	140
PCBP	MDTGVIEGGLNVTLTIRLLMHGKEVGSII GKKGESVKMREESG ARINISEGNCPERIITLAGPTNAIFKAFAMIIDKLEEDISSMTNST AASRPPVTLRLVVPASQCGSLIGKGGCKIKEIRESTGAQVQVAG DMLPNSTERAITIAIPQSIIECVKQICVVMLESPPKGV TIPYRKP SSSPVI FAGGQAYTIQGQYAI PQPDLTKLHQLAMQQSHFPMTHG NTGFSGIESSSPEVKGYWAGLDASAQTTSHELTIPNDLIGCIIGRQ GAKINEIRQMSGAIKIANPVEGSTDRQVTITGSAASISLAQYLIN VRLSSETGGMGSS	141

TABLE-continued

Effectors		
Description	Sequence	SEQ ID NO.:
HuR	MSNGYEDHMAEDCRGDIGRTNLIVNYLPQNMTQDELRSLESSIG EVESAKLIRDKVAGHSLGYGFVNYVTAKDAERAINTLNLRLQS KTIKVSYPSPSEVIKDANLYISGLPRTMTQKDVEDMFSRFGRII NSRVLDQTTGLSRGVAFIRFDRSEAEAAITSFNHGKPPGSSEPI TVKFAANPNQKNVALLSPLYHSPARRFGGPVHHQAQRFRFSP MGVDHMSGSLSGVNVPGNASSGWCIFIYNLGDQADEGILWQMF GPPGAVTNVKVIRDFNTNKCKGFGVMTMTNYEEAAMAIASLNG YRLGDKILQVSFKTNKSHK	142
RNSP1	MDLSGVKKKSLGKKNKKSSTRAPSPTKRKDRSDEKSKDRS KDKGATKESSEKDRGRDKTRKRRSASSGSSSTRSRSSSTSSSGSS TSTGSSSGSSSSASSRSGSSSTRSSSSSSSGSPSPRRRHDNRRR SRSKSKPPKRDEKERKRRSPSPKPTKVHIGRLTRNVTKDHIMEIFS TYGKIKMIDMPVERMHPHLSKGYAYVEFENPDEAEKALKHMD GGQIDGQEIATATAVLAPWPRPPRRFSPPRMLPPPPMWRSPPR MRRRSRSPRRSPVRRRSRSPGRRRHRSRSSSNSSR	143

TABLE

Full CITS sequences used in FIG. 7: (Bdef-(GGG)₆-TBP6.7-NES-(GGG)₆-Effector-Tag

Description	Sequence	SEQ ID NO.:
Y1 (1-364)	MGIINTLQKYICRVRGRCVLSCLPKKEEIGKCTRGRKCCR RKKGGSGSGSGSGSGSGSMVPEPTRPNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTfVGSLLQPLPL ERLTLGGSGSGSGSGSGSGSMSATSVDTQRTKGQDNKVQ NGSLHQKDTVHDNDFEPYLTGQSNQSNYSYMSDPYLSYYPP SIGFPYSLNEAPWSTAGDPPIPYLTTYGQLSNGDHHFMDAVFG QPGGLGNNIYQHRPNFFPENPAFSAWGTSGSQGQQTQSSAYGS SYTYPPSSLGTVVDGQPGFHSDDLTSKAPGMNLSLEQGMVGLKI GDVSSSAVKTGVSVVSSVALTGVLSGNGGTNVNMPVSKPTSW AAIASKPAKPQPKMKTKSGPVMGGGLPPPIKHNMDIGTWDNK GPVPKAPVPQQAQSPQAAPQQAQPLPAQPPALAPQPYQSP QPPQTRWVAPRNRNAAFQSGGAGSDSNSPGNVQPNAPS ESGSDYKDDDDK	80
eIF4a	MGIINTLQKYICRVRGRCVLSCLPKKEEIGKCTRGRKCCR RKKGGSGSGSGSGSGSGSMVPEPTRPNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTfVGSLLQPLPL ERLTLGGSGSGSGSGSGSGSMSASQDSRSRDNGPDGMEPE GVIESNWEIVDSFDDMNLSLRLRGIYAYGFEKPSAIQQRAILP CIKGYDVIAQAQSGTGKTATFAISILQQIELDLKATQALVLAPTR ELAQQIQKVVMAIGDYMAGSCHACIGGTNVRAEVQKLQMEA PHIIVGTPGRVFDMLNRRYLSPKYIKMFVLDDEADMLSRGFKD QIYDIFQKLSNNTQVLLSATMPSDVLEVTKKFMRDPRIPLVKK EELTLEGIRQFYINVEREEWKDLTLCDLYETLTIQAVIFINTRR KVDWLTEKMHARDFTVSAMHGDMDQKRDVIMREFRSGSSR VLIITDLLARGIDVQVSLVINYLPTNRENYIHRIGRGGRFGRK GVAINMVTEDKRTLRLDIETFYNTSIEEMPLNVADLIGSDYKDD DDK	161
eIF4e	MGIINTLQKYICRVRGRCVLSCLPKKEEIGKCTRGRKCCR RKKGGSGSGSGSGSGSGSMVPEPTRPNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTfVGSLLQPLPL ERLTLGGSGSGSGSGSGSGSMATVEPETTPPNPPTTEEEK TESNQEVANPEHYIKHPLQNRWALWFFKNDKSKTWQANLRLIS KFDTVDEFWALYNHQLSSNLMPCDYSLFKDGIEMWDEKKN KRGGRWLI TLNKQQRSDLRFWLETLLCLIGESFDDYSDDVC GAVNVRAKGDKIAIWTTECENREAVTHIGRVYKERLGLPPKI VIGYQSHADTATKSGSTTKNRFVVGSDYKDDDDK	81

TABLE-continued

Full CIRT sequences used in FIG. 7: (Bdef- (GGS) ₆ -TBP6.7-NES- (GGS) ₆ -Effector-Tag		
Description	Sequence	SEQ ID NO:
YB1	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGGSGSGSGSGSGSGSGSMVETPRNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTFVGSLLPLPL ERLTLGGSGSGSGSGSGSGSGSMSEAETQPPAAPPAAPALS AADTKPGTTGSGAGSGGPGGLTSAAPAGGDKKVIATKVLGTV KWFNVNRNGYGFINRNDTKEDVFVHQTAIKKNNPRKYLRVSGD GETVEFDVVEGEKGAEAAVNTGPGGVVQGSKYAADRNHYRR YPRRRGPPRNYQQNYQNSESGEKNESESAPEGQAQQRRPYRR RRFPPYYMRRPYGRRPQYSNPPVQGEVMEGADNQGAGEQGRP VRQNMRYGRYPRFRGPPRQRPREDGNEEDKENQGDDETQGG QPPQRRYRRNFYRRRRPENPKPDGKETKAADPPAENSSAPE AEQGAEGSDYKDDDDK	66
HSPB1	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGGSGSGSGSGSGSGSGSMVETPRNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTFVGSLLPLPL ERLTLGGSGSGSGSGSGSGSGSMTEERRVPSLLRGPSWDPR DWYPHSRLFDQAFGLPRLPEEWSQWLGGSSWPGYVRPLPPAAI ESPAVAAPAYSRALSRQLSSGVSEIRHTADRWRVSLDVNHFAP DELTVKTKDGVVEITGKHEERQDEHGYISRCFTRKYTLPPGVDP TQVSSSLSPGTLTVEAPMPKLATQSNBITIPVTFESRAQLGGPE AAKSDETAALGSDYKDDDDK	11
PABP	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGGSGSGSGSGSGSGSGSMVETPRNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTFVGSLLPLPL ERLTLGGSGSGSGSGSGSGSGSMNPSAPSYPMASLYVGDLP DVTEAMLYEKFSPAGPILSIRVCRDMI TRRSLGYAYVNFQQPAD AERALDTMNFVDIKGKPVRIWMSQRDPSLRKSGVGNIFIKNLD KSIDNKALYDTFSAFGNILSCKVVCDENGSKGYGFVHFETQEA AERAIEKMNGLMLNDRKVFVGRFKSRKEREAEELGARAKEFTN VYIKNFGEDMDDERLKDLPFGKFPALSVKMTDESGKSKGFGF VSFERHEDAQKAVDEMNGKELNGQIYVGRAQKKVERQTELK RKFEQMKQDRITRYQGVNLVYKLNDDGIDDERLRKEFSPFGTIT SAKVMMEGGRSKGFGFVCFSSPEEATKAVTEMNGRIVATKPLY VALAQRKEERQAHLTNQYMQRMASVRAVPNPVINPYQPAPPS GYFMAAIPQTQNRAYYPPSQIAQLRPSPRWTAQGAPHPFQN MPGAIRPAAPRPPFSTMRPASQVPRVMSTQRVANTSTQTMGP RPAAAAAAATPAVRTVPQKYAAGVRNPQHLNAQPQVTMQ QPAVHVQGEPLTASMLASAPPQEQKQMLGERLFLPLIQAMHPT LAGKITGMLLEIDNSELHMLESPESLRSKVDEAVAVLQAHQA KEAAQKAVNSATGVPTVGSYKDDDDK	162
eIF4G (711-1599)	MGIINTLQKYYCRVRGGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGGSGSGSGSGSGSGSGSMVETPRNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTFVGSLLPLPL ERLTLGGSGSGSGSGSGSGSGSEPRKIIATVLMTEDIKLNKAE KAWKPSKRTAADKDRGEEDADGSKTQDLFRVRVRSILNKLTPQ MFQQLMKQVTQLAIDTEERLKGVIDLIFEKAI SEP NFSVAYANM CRCLMALKVPTTEKPTVTNFRKLLLNRCQKEFEKD KDDDEVF EKKQKEMDEAATAEERGRLEKELEEARDIARRSLGNIKFIGEL FKLKMLTEAIMHDCVVKLLKNHDEESLECLCRLTTIGKDLDFE KAKPRMDQYFNQMEKIKEKKTSSRIRFMLQDVLDRGSNWVP RRGDQGPKTIDQIHKEAEMEEHREHIKVQQLMAKGS DKRRGGP PGPPISRGLPLVDDGGWNTVPISKGSRPIDTSRLTKITKPGSIDSN	163

TABLE-continued

Full CIRT sequences used in FIG. 7: (Bdef-(GGG) ₆ -TBP6.7-NES-(GGG) ₆ -Effector-Taq		
Description	Sequence	SEQ ID NO:
	NQLFAPGGRLSWGKSSGGSGGAKPSDAASEAARPATSTLNRFSLQQAQVPTSTDNRRVVQSSLSRERGEKAGDRGLERSERGGDRGDRLDARTPATKRSFSKEVEERSRERPSQPEGLRKAASLTEDRDRGRDAVKREAALPVSPKKAALSEEELEKSKAII EYHLNDMKEAVQCVELASPSLLFIFVRHGVSTLERSAIAREHMGQLLHQLLCAGHLSTAQYYQGLYEILELAEDMEIDIPHVWLYLAELVTPILQEGGVPMGELFREITKPLRPLGKAASLLEILGLLCKSMGPKKVGTLWREAGLSWKEFLPEGQDIGAFVAEQKVEYTLGEESEAPGQRALPSEELNRQLEKLLKEGSSNQRFVDWIEANLSEQQIVSNTLVRLMTAVCYSAIIFETPLRVDVAVLKARAKLLQKYLCEQKELQALYALQALVVTLEQPPNLLRMFFDALYDEDVVKEDAFYSWESSKDPAEQQGKGVALKSVTAFFKWLR EAEESDHNGSDYKDDDDK	
eIF4G (711-1105)	MGIINTLQKYYCRVRGGRCVAVLSCLPKKEEQIGKCSTRGRKCCR RKKGGSGSGSGSGSGSGSGSMVPEPTRPNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTfVGSLLQPLPL ERLTLGGSGSGSGSGSGSGSGSEPRKIIATVLMTEDIKLNKAE KAWKPPSSKRTAADKDRGEEDADGSKTQDLFRRVRSILNKLTPQ MFQQLMKQVTQLAIDTEERLKGVIDLIFEKAISEPNFSVAYANM CRCLMALKVPTTEKPTVTVNFRKLLLNRCQKEFEKD KDDDEVF EKKQKEMDEAATAEERGRLEKELEARDIARRRS LGNIKFIGEL FKLKMTEAIMHDCVVKLLKNHDEESLECLRLTTIGKDLDFE KAKPRMDQYFNQMEKIIKEKKTSSRI RFMLQDVLDRGSNWVP RRGDQGPKTIDQIHKEAEMEEREHIKVQQLMAKGS DKRRGGP PGPPI SRGLPLVDDGGWNTVPI SKGSRPIDTSRLTKITKPGSIDSN NQLFAPGGRLSWGKSSGGSGGAKGSDYKDDDDK	164
Bol1	MGIINTLQKYYCRVRGGRCVAVLSCLPKKEEQIGKCSTRGRKCCR RKKGGSGSGSGSGSGSGSGSMVPEPTRPNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTfVGSLLQPLPL ERLTLGGSGSGSGSGSGSGSGSMQTDLSLSPSPNPVSPVPLNNP TSAPRYGTVIPNRI FVGGIDFKTNESDLRKFFSQYGSVKEVKIVN DRAGVSKGYGFVTFETQEDAQKILQEAELKNYKDKKLNIGPAI RKQVQGI PRSSIMPAAGTMYLTTSTGYPTYHNGVAYFHTPEV TSVPPPWPSPRSVSSPVMVAQPIYQQPAYHYQATTQYLPQWQ WSVPQPSASSAPFLYLQPSFVIYQPV EIAQDGGCVPPPLSLMETS VPEPYSDHGVQATYHQVYAPSAITMPAPVMQPEPIKTVWSIHY GSDYKDDDDK	165
Era	MGIINTLQKYYCRVRGGRCVAVLSCLPKKEEQIGKCSTRGRKCCR RKKGGSGSGSGSGSGSGSGSMVPEPTRPNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTfVGSLLQPLPL ERLTLGGSGSGSGSGSGSGSGSMTMTLHTKASGMALLHQIQ GNELEPLNRPQLKIPLERPLGEVYLDSSKPAVYNYPEGAAYEFN AAAAANAQVYGQTGLPYGPGSEAAAFSGNGLGGFPPLNSVSPS PLMLLHPPPPQLSPFLQPHGQQVPYYLENESPGYTVREAGPPAFY RPNSDNRRQGRERLASTNDKGSMAMESAKETRYCAVCNDY ASGYHYGVWSCEGCKAFFKRSIQGHNDYMC PATNQCTIDKNR RKSCQACRLRKCYEVGMMKGIRKDRRGGRMLKHKRQDDG EGRGEVGSAGDMRAANLWPSPLMIKRSKKNLALSLTADQMV SALLDAEPPILYSEYDPTRFSEASMMGLLTNLADREL VHMINW AKRVPGFVDLTLDQVHLLCAWLEILMIGLVWRSMHEPGKL LFAPNLLDRNQGKCV EGMVEIFDMLLATSSRFMMNLQGEF VCLKSII LLNSGVYTFLLSSTLKSLEEKDHIHRVLDKIDTDLIHLMA KAGLTQQQHQLAQLLLILSHIRHMSNKGMEHLYSMKCKNV VPLYDLLEMLDAHRLHAPTSRGASVEETDQSHLATAGSTSS HSLQKYYITGEAEGFPATVGSYKDDDDK	166
PCBP	MGIINTLQKYYCRVRGGRCVAVLSCLPKKEEQIGKCSTRGRKCCR RKKGGSGSGSGSGSGSGSGSMVPEPTRPNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPPFYDKPMRIQYAKTDKRIPAKMKGTfVGSLLQPLPL ERLTLGGSGSGSGSGSGSGSGSMDTGVI EGGNLVTLTIRLLM HGKEVGSII GKKGESVKKMREESGARINISEGNCPERIITLAGPT NAIFKAFAMIIDKLEEDISSMTNSTAASRPVTLRLVVPASQCG SLIGKGCKIK EIRESTGAQVQVAGDMLPNSTERAITIAGIPQSI I	167

TABLE-continued

Full CIRT sequences used in FIG. 7: (Bdef- (GGS) ₆ -TBP6.7-NES- (GGS) ₆ -Effector-Tag		
Description	Sequence	SEQ ID NO:
	ECVKQICVVMLESPPKGVITIPYRKPSSSPVIFAGGQAYTIQQQY AIPQPDLTLLHQLAMQQSHFPMTHGNTGFSGIESSSPEVKGYW AGLDASAQTTSHELTI PNDLIGCIIGRQGAKEIRQMSGAIKIA NPVEGSTDRQVTITGSAASISLAQYLINVLRSSETGGMGSSGSDY KDDDDK	
HuR	MGIINTLQKYYCRVGGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGGSGSGSGSGSGSGSMVPETRPNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPFYDKPMRIQYAKTDKRIPAKMKGT FVGSLLQLPPL ERLTLGGSGSGSGSGSGSGSGMSNGYEDHMAEDCRGDIGR TNLIVNVLPMNTQDELRSLSFSSIGEVESAKLIRDKVAGHSLGY GFVNYVTAKDAERAINTLNLRLQSKTIKVSYPARPSSEVTKDAN LYISGLPRTMTQKDVEDMFSRFGRIINSRVLVDQTTGLSRGVAFI RFDKRS EAEEAITSFNGHKPPGSSEPI TVKFAANPNQKNVALL SQLYHSPARRFGGPVHHQAQRFRFSPMGVDHMSGLSGVNVPG NASSGWCI FIIYNLGQDADEGILWQMFPGFPAVTNVKVI RIDENT NKCKGFGFVTMTNYEEAAMIASLNGYRLGDKILQVSFKTNKS HKGSDYKDDDDK	168
RNSP1	MGIINTLQKYYCRVGGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGGSGSGSGSGSGSGSMVPETRPNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPFYDKPMRIQYAKTDKRIPAKMKGT FVGSLLQLPPL ERLTLGGSGSGSGSGSGSGSGMDLSGVKKKSLLGVKENNK KSSTRAPSPTRKDRSDEKSKDRSKDKGATKESSEKDRGRDKT RKRRSASSGSSSTRSRSSSTSSSGSSTSTGSSSGSSSSASSRSGSS STSRSSSSSSSGSPSPRRRHDNRSSRSKSKPKRDEKERKRR SPSPKPTKVHIGRLTRNVT KDHIMEIFSTYGIKIMIDMPVERMH PHLSKGYAYVEFENPDEAEKALKHMDGGQIDGQEITATAVLAP WPRPPRRRFSPPRRMLPPPPMWRRSPPMRRRRSRSPRRRSPVRR RSRSPGRRRRRSRSSNSSRSGDYKDDDDK	169
FXR1	MGIINTLQKYYCRVGGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGGSGSGSGSGSGSGSMVPETRPNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPFYDKPMRIQYAKTDKRIPAKMKGT FVGSLLQLPPL ERLTLGGSGSGSGSGSGSGSGMAELTVEVRGSNGAFYKGF KDVHEDSLTVVFENNWQPERQVPFNEVRLPPPD IKKEISEGDE VEVYSRANDQEP CGWWLAKVRMMKGEFYVIEYACDATYNE IVTFERLRPNVQNKTVKNTFFKCTVDVPEDLREACANENAHK DFKKAVGACRIFYHPETTQLMILSASEATVKRVNILSDMHLRSI RTKLMLMSRNEEATKHLECKQLAAAFHEEFVVREDLMGLAI GTHGSNIQQAARKVPVGTAI ELDEDTGTFRITYGESADAVKKARGF LEFVEDFIQVPRNLVGKVIKNGKVIQEIIVDKSGVVRVRIEGDN ENKLPRDGMVPFVFGTKESIGNVQVLLHYHIAYLKEVEQLR MERLQIDEQLRQIGSRSYSGRGRGRRPNYTSGYGTNS ELSNPS ETESERKDELSDWSLAGEDDRDSRHQRDSRRRPGGRGRSVSGG RGRGGPRGGKSSISSVLKDPDSNPYSLDNTESDQTADTDASES HHSTNRRRRSRRRRTDEDAVLMDGMTESDTASVNEGLGKRC DGS DYKDDDDK	77
DAZ4	MGIINTLQKYYCRVGGRCVLSCLPKEEQIGKCSTRGRKCCR RKKGGSGSGSGSGSGSGSMVPETRPNHTIYINNLSKIK KDELKKSLEYAIFSQFGQILDILVPRQRTPRGQAFVIFKEVSSATN ALRSMQGFPFYDKPMRIQYAKTDKRIPAKMKGT FVGSLLQLPPL ERLTLGGSGSGSGSGSGSGSGMAANPETPNSTISREASTQS SSAAASQGWLP EGTKIVPNTVFVGGIDARMDTEIGSCFGRYGS VKEVKIITNRTGVSKGYGFVSFVNDVDVQKIVGSQIHFHGGKLR LGPAIRKQKLCARHVQPRPLVNVNPPPPQFQNVWRNPNTETYL QPQITPNPVTQHVQSAANPETPNSTISREASTQSSAAASQGWV	65

TABLE-continued

Full CIRTs sequences used in FIG. 7: (Bdef-(GGG) ₆ -TBP6.7-NES-(GGG) ₆ -Effector-Tag		
Description	Sequence	SEQ ID NO:
	LPEGKIVPNTVFVGGIDARMDTEIGSCFGRYGSVKEVKIITNRT GVSKGYGFVSFVNDVDVQKIVGSQIHFGKKLKLGPARKQKL CARHVQPRPLVVNPPPPQFQNVWRNPNTETYLQPOITPNPVTQ HVQAYSAYPHSPGQVITGCQLLVYNYQEYPTYPDSPQVTTGY QLPVYNYQPPFPAYPRSPFQVTAGYQLPVYNYQAFPAYPNSPFQ VATGYQFPVYNYQPPFPAYPSSPFQVTAGYQLPVYNYQAFPAYP NSPFQVATGYQFPVYNYQAFPAYPNSPVQVTTGYQLPVYNYQ AFPAYPSSPFQVTTGYQLPVYNYQAFPAYPNSAVQVTTGYQFH VYNYQMPPQCPVGEQRRNLWTEAYKWWYLVCLIQRRDGS DY KDDDDK*	

B. FIG. 8 Methods and Results

[0174] Methods: See methods described for FIG. 7.

[0175] Results: Additional dual luciferase reporter plasmids were constructed with human Jagged1 (JAG1) 3' UTR and mouse SCN1a 3' UTR appended to the Firefly luciferase open-reading frame. Then CIRTs-Y1 was tested for the ability to increase Firefly luciferase expression in a gRNA-dependent manner when targeting the Jag1 and SCN1a 3' UTRs. At least one gRNA was identified by each reporter system to mediate CIRTs-Y1 and gRNA-dependent increase of Firefly expression. gRNA sequences used in FIG. 8.

[0178] Results: When compared to the alpha-tubulin loading control, transfection with Bdef-TBP-Y1 (SEQ ID NO:80) and SCN1a g1 appears to increase SCN1a protein present compared to the NT gRNA by approximately 1.4-fold.

[0179] All of the methods disclosed and claimed herein can be made and executed without undue experimentation in light of the present disclosure. While the compositions and methods of this invention have been described in terms of preferred embodiments, it will be apparent to those of skill in the art that variations may be applied to the methods and in the steps or in the sequence of steps of the method described herein without departing from the concept, spirit and scope of the invention. More specifically, it will be

Description	Sequence	SEQ ID NO:
Jag1 g1	GCCGGTAGAACTACGTAAGCTCAGAAGAGGCCACACTGTT	144
Jag1 g2	GAGTATGACACGACAGTTTAAAGAACTACAAGCCCTCAGA	145
Jag1 g3	TAAATTCTTCATTTTACCAGCAACTGCTGACATCAAAGTC	146
Jag1 g4	TTAATCATATTCAACCACTCACATTTTGGCTATGTTAAGT	147
SCN1a g1	CTCTCTGATTTTGAGGAAAAGAAATCTGAAGCCACATT	55

C. FIG. 9 Methods and Results

[0176] Methods: Neuro 2a cells were plated in a 12-well plate and transfected 24 hours later using Lipofectamine 3000 according to the manufacturer's protocol. Each well was transfected with a plasmid expressing CIRTs-Y1 and either a NT gRNA or SCN1a g1. Protein was isolated from the Neuro 2a cells 48 hours later by incubation with RIPA buffer and protease inhibitors.

[0177] For western blotting, 15 µg of total protein was loaded for each condition. SCN1a and alpha-tubulin (α-tubulin) levels were detected by incubation with anti-SCN1a, or anti-alpha-tubulin antibody followed by incubation with an appropriate secondary antibody conjugated to HRP then chemiluminescent detection.

apparent that certain agents which are both chemically and physiologically related may be substituted for the agents described herein while the same or similar results would be achieved. All such similar substitutes and modifications apparent to those skilled in the art are deemed to be within the spirit, scope and concept of the invention as defined by the appended claims. The references cited herein, to the extent that they provide exemplary procedural or other details supplementary to those set forth herein, are specifically incorporated herein by reference.

SEQUENCE LISTING

Sequence total quantity: 169

SEQ ID NO: 1 moltype = AA length = 641

FEATURE Location/Qualifiers

source 1..641
mol_type = protein
organism = synthetic construct

SEQUENCE: 1

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MGIINTLQKY YCRVRGGRCA VLSCLPKKEQ IGKCSTRGRK CRRKKKGGSG GSGSGSGSGG 60
SGGSMAVPET RPNHTIYINN LNSKIKKDEL KKSLEYAIFSQ FGQILDILVP RQRTPRGQAF 120
VIFKEVSSAT NALRSMQGFY FYDKPMRIQY AKTDKRIPAK MKGTFVGSQ LPPLERLTLG 180
GSGSGSGSGG GSGSGSGMDL TAIYESLLSL SPDVPVPSDH GGTESSPGWG SSGPWSLSPS 240
DSSPSGVTSR LPGRSTSLVE GRSCGWVPPP PGFAPLAPRL GPESLSPSPTS PTATSTTPSR 300
YKTELCTFES ESGRCRYGAK CQFAHGLGEL RQANRHPKYK TELCHKFYLO GRCPYGSRCH 360
FIHNPSEDLA APGHPPVLRQ SISFSGLPSSG RRTSPPPPG LAGPSLSSSS SPSSSPPPPG 420
DLPLSPSAFS AAPGTPLARR DTPVCCPSC RRATPISVWG PLGGLVVRTPS VQSLGSDPDE 480
YASSGSSSLGG SDSPVFEGAV FAPPQPVAA RRLPIFNRI VSEGGSGSGG GSGSGSGSGG 540
SDAMFGQPGA LGSTPFLGQH GFNFFPSGID FSAWGNSSQ GQSTQSSGYS SNYAYAPSSL 600
GGAMIDGQSA PANETLNKAP GMNTIDQGMA AGSDYKDDDD K 641

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SEQ ID NO: 2 moltype = AA length = 600

FEATURE Location/Qualifiers

source 1..600
mol_type = protein
organism = synthetic construct

SEQUENCE: 2

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MGIINTLQKY YCRVRGGRCA VLSCLPKKEQ IGKCSTRGRK CRRKKKGGSG GSGSGSGSGG 60
SGGSMAVPET RPNHTIYINN LNSKIKKDEL KKSLEYAIFSQ FGQILDILVP RQRTPRGQAF 120
VIFKEVSSAT NALRSMQGFY FYDKPMRIQY AKTDKRIPAK MKGTFVGSQ LPPLERLTLG 180
GSGSGSGSGG GSGSGSGMPA ATVDHSQRIC EVWACNLDEE MKKIRQVIRK YNYVAMDTF 240
PGVVARPIGE FRSNADYQYQ LLRCNVDDLK IIQLGLTFMN EQGEYPPGTS TWQFNFKFNL 300
TEDMYAQDSI ELLTTSIGIQ KHEEEEGIET QYFAELLMTS GVVLCCEGVKW LSPHSGYDFG 360
YLKILNTSN LPBEELDFFE ILRLFFPVIY DVKYLKMSCK NLKGGLQEVA EQLELERIGP 420
QHAGSDSLT TGMAFFKMRE MFFEDHIDDA KYCGHLYGLG SGSSYVQNGT GNAYEEEAANK 480
QSGSGSGSGG GSGSGSGSGS DAMFGQPGAL GSTPFLGQHG FNFFPSGIDF SAWGNSSSQ 540
QSTQSSGYSS NYAYAPSSLG GAMIDGQSAF ANETLNKAPG MNTIDQGMAA GSDYKDDDDK 600

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SEQ ID NO: 3 moltype = AA length = 836

FEATURE Location/Qualifiers

source 1..836
mol_type = protein
organism = synthetic construct

SEQUENCE: 3

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MGIINTLQKY YCRVRGGRCA VLSCLPKKEQ IGKCSTRGRK CRRKKKGGSG GSGSGSGSGG 60
SGGSMAVPET RPNHTIYINN LNSKIKKDEL KKSLEYAIFSQ FGQILDILVP RQRTPRGQAF 120
VIFKEVSSAT NALRSMQGFY FYDKPMRIQY AKTDKRIPAK MKGTFVGSQ LPPLERLTLG 180
GSGSGSGSGG GSGSGSGMPA ATVDHSQRIC EVWACNLDEE MKKIRQVIRK YNYVAMDTF 240
PGVVARPIGE FRSNADYQYQ LLRCNVDDLK IIQLGLTFMN EQGEYPPGTS TWQFNFKFNL 300
TEDMYAQDSI ELLTTSIGIQ KHEEEEGIET QYFAELLMTS GVVLCCEGVKW LSPHSGYDFG 360
YLKILNTSN LPBEELDFFE ILRLFFPVIY DVKYLKMSCK NLKGGLQEVA EQLELERIGP 420
QHAGSDSLT TGMAFFKMRE MFFEDHIDDA KYCGHLYGLG SGSSYVQNGT GNAYEEEAANK 480
QSGSGSGSGG GSGSGSGSGS MDLTAIYESL LSLSPDVVP SDHGGTESSP GWGSSGPWSL 540
SPSDSSPSGV TSRLPGRSTS LVEGRSCGWV PPPPGFAPLA PRLGPESLPS PTSPTATSTT 600
PSRYKTELCR TFSESGRCRY GAKCQFAHGL GELRQANRHP KYKTELCHKF YLQGRCPYGS 660
RCHFIHNPSE DLAAPGHPPV LRQSISSGL PSGRRTSPPP PGLAGPSLSS SSFSPSSSPP 720
PPGDLPLSPS AFSAAPGTPL ARRDPTPVCC PSCRRATPIS VWGPLGLVLR TPSVQSLGSD 780
PDEYASSGSS LGGSDSPVFE AGVFAPPQV AAPRRLPIFN RISVSEGS DYKDDDDK 836

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SEQ ID NO: 4 moltype = AA length = 627

FEATURE Location/Qualifiers

source 1..627
mol_type = protein
organism = synthetic construct

SEQUENCE: 4

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MGIINTLQKY YCRVRGGRCA VLSCLPKKEQ IGKCSTRGRK CRRKKKGGSG GSGSGSGSGG 60
SGGSMAVPET RPNHTIYINN LNSKIKKDEL KKSLEYAIFSQ FGQILDILVP RQRTPRGQAF 120
VIFKEVSSAT NALRSMQGFY FYDKPMRIQY AKTDKRIPAK MKGTFVGSQ LPPLERLTLG 180
GSGSGSGSGG GSGSGGSTRS RYASDFEEIA VLQGQAFQV VKARNALDSR YYAIKKIRHT 240
EEKLSTILSE VMLASLNHQ YVRYAAAWL EEDSMDENVF ESTDESDLS ESSSDFEEND 300
LLDQSSIFKN RTNHDLDNSN WDFISGSGYP DIVFENS SRD DENEDLDHDT SSTSSSESQD 360
DTDKESKSIQ NVPRRRNFVK PMTAVKKKST LFIQMEYCEN GTLYDLIHSE NLNQQRDEYW 420
RLFRQILEAL SYIHSQGIH RDLKPMNLI DESRNVKIGD FGLAKNVHRS LDILKLSQDN 480
LPGSSDNLTS AIGTAMVYAT EVLDGTGHYN EKIDMYSLGI IFFEMIYPFS TGMERNVILK 540
KLRVSIEFP PDFDDNKMKV EKKIIRLLID HDPNKRPGAR TLLNSGWLPV KHQDEVIKEA 600
LKSLSNPSSP WQQQVREGSD YKDDDDK 627

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-continued

SEQ ID NO: 5 moltype = AA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 5

MGIINTLQKY	YCRVRGGRCA	VLSCLPKEEQ	IGKCSTRGRK	CCRKKKGGSG	SGSGSGSGSG	60
SGGSMVAVPET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120
VIFKEVSSAT	NALRSMQGF	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSLO	LPPLERLTLG	180
SGSGSGSGSG	SGSGSGMSG	GSSCSQTPSR	AIPTRVALG	DGVQLPPGDY	STAPGGTLFS	240
TAPGGTRIY	DRKFLMECRN	APVAKAPPKD	LPTIPGVTSP	TSDEPPMQAS	QSHLHSSPED	300
KRAGGEESQF	EMDIGSDYKD	DDDK				324

SEQ ID NO: 6 moltype = AA length = 399
 FEATURE Location/Qualifiers
 source 1..399
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 6

MGIINTLQKY	YCRVRGGRCA	VLSCLPKEEQ	IGKCSTRGRK	CCRKKKGGSG	SGSGSGSGSG	60
SGGSMVAVPET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120
VIFKEVSSAT	NALRSMQGF	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSLO	LPPLERLTLG	180
SGSGSGSGSG	SGSGSGSGMT	FDLWTDYGL	AHLVRALSGK	EGPETRLSPQ	PEPEPMLPEV	240
SALEPMPAPE	SVPVPGPKDQ	KRSLESSPAP	ERLCSFCKHN	GESRAIYQSH	VLKDEAGRVL	300
CPILRDYVCP	QCGATREAH	TRRFCLTGO	GYTSVYSHTT	RNSAGKKLVR	PDKAKTQDTG	360
HRRGGGGGAG	FRGAGKSEPS	PSCSPSMSTG	SDYKDDDDK			399

SEQ ID NO: 7 moltype = AA length = 848
 FEATURE Location/Qualifiers
 source 1..848
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 7

MGIINTLQKY	YCRVRGGRCA	VLSCLPKEEQ	IGKCSTRGRK	CCRKKKGGSG	SGSGSGSGSG	60
SGGSMVAVPET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120
VIFKEVSSAT	NALRSMQGF	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSLO	LPPLERLTLG	180
SGSGSGSGSG	SGSGSGSGMT	VEPETPTPN	PPTTEEEKTE	SNQEVANPEH	YIKHPLQNRW	240
ALWFFKNDKS	KTWQANLRLI	SKFDTVEDFW	ALYNHIQLSS	NLMPGCDYSL	FKDGIEMWE	300
DEKNKRGGWR	LITLNKQRR	SDLDRFWLET	LLCLIGESFD	DYSDDVCAGV	VNVRAKGDKI	360
AIWTTTECNR	EAVTHIGRVY	KERLGLPPKI	VIGYQSHADT	ATKSGSTTKN	RFVVGSGSGS	420
GGSGSGSGSG	GSMSASQDSR	SRDNGPDGME	PEGVIESNWN	EIVDSFDDMN	LSSELLRGIY	480
AYGFEKPSAI	QQRAILPCIK	GYDVIAQAQS	GTGKTATPAI	SILQQIELDL	KATQALVLP	540
TRELAQQIQK	VVMALGDYMG	ASCHACIGGT	NVRAEVQKLQ	MEAPHIIVGT	PGRVFDMLNR	600
RYLSPKYIKM	FVLDEADEML	SRGFKDQIYD	IFQKLNSNTQ	VVLLSATMPS	DVLEVTKKFM	660
RDPIRILVKK	EELTLEGIR	FYINVEREEW	KLDTLCDLYE	TLTITQAVIF	INTRRKVDWL	720
TEKMHARDFT	VSAMHGMDQ	KERDVIMREF	RSGSSRVLIT	TDLLARGIDV	QQVSLVINYD	780
LPTNRENIYH	RIGRGGFRGR	KGVAINMVTE	EDKRTLRLDIE	TFYNTSIEEM	PLNVADLIGS	840
DYKDDDDK						848

SEQ ID NO: 8 moltype = AA length = 1110
 FEATURE Location/Qualifiers
 source 1..1110
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 8

MGIINTLQKY	YCRVRGGRCA	VLSCLPKEEQ	IGKCSTRGRK	CCRKKKGGSG	SGSGSGSGSG	60
SGGSMVAVPET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120
VIFKEVSSAT	NALRSMQGF	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSLO	LPPLERLTLG	180
SGSGSGSGSG	SGSGSGMSA	SQDSRSRDNG	PDGMEPEGVI	ESNWNEIVDS	FDDMNLSESL	240
LRGIYAYGFE	KPSAIQQRAI	LPCIKGYDVI	AQAQSGTGKT	ATPAISILQQ	IELDLKATQA	300
LVLAPTRELA	QQIQKVVMAL	GDYMGASCHA	CIGGTNVRAE	VQKLQMEAPH	IIVGTGPRVF	360
DMLNRRYLSP	KYIKMFVLDE	ADEMLSRGFK	DQIYDIFQKL	NSNTQVVLLS	ATMPSDVLEV	420
TKKFMRDPIR	ILVKKEELTL	EGIRQFYINV	EREWKLDLTL	CDLYETLTIT	QAVIFINTRR	480
KVDWLTEKMH	ARDFTVSAMH	GDMDQKERDV	IMREFRSGSS	RVLITTDLLA	RGIDVQQVSL	540
VINYDLPTNR	ENYIHRIGRG	GRFGRKGVAI	NMVTEEDKRT	LRDIETFYNT	SIEEMPLNVA	600
DLIGSGSGSG	SGSGSGSGSG	SMSDGFDRAP	GAGRGRSRGL	GRGGGGPEGG	GFPNGAGPAE	660
RARHQPPQPK	APGFLQPPPL	RQPRTTTPPG	AQCEVPASPQ	RPSRPGALPE	QTRPLRAPP	720
SQDKIPQQNS	ESAMAKPQV	VAPVLSKLS	VNAPEFYPSG	YSSSYTESYE	DGCEDYPTLS	780
EYVQDFLNHL	TEQPGSFETE	IEQFAETLNG	CVTTDDALQE	LVELIYQQAT	SIPNFSYMGA	840
RLCNVLSHHL	TISQSGNFR	QLLLQRCRTE	YEYKQDAAKG	DEVTRKRFAH	FVLFLGELYL	900
NLEIKGTNGQ	VTRADILQVG	LRELLNALFS	NPMDDNLICA	VKLLKLTGSV	LEDAWKEKKG	960
MDMEEIIQRI	ENVVLDANCS	RDVKQMLLKL	VELRSSNWGR	VHATSTYREA	TPENDPNYFM	1020
NEPTFYTSDG	VPFTAADPDY	QEKYQELLER	EDFFPDYEEN	GTDLGSGAGDP	YLDIDDEMD	1080
PEIEEAYEKF	CLESERKRKQ	GSDYKDDDDK				1110

SEQ ID NO: 9 moltype = AA length = 1004

-continued

FEATURE
source

Location/Qualifiers
1..1004
mol_type = protein
organism = synthetic construct

SEQUENCE: 9

MGIIINTLQKY	YCRVRGGRC	VLSCLPKEEQ	IGKCSTRGRK	CCRRKKGGSG	GSGGSGGSGG	60
SGGSMVAVPET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120
VIFKEVSSAT	NALRSMQGFP	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSQ	LPPLERLTLG	180
GSGGSGGSGG	GSGGSGGMSA	TSVDTQRTKG	QDNKVQNGSL	HQKDTVHDND	FEPYLTGQSN	240
QNSYSPMSD	PYLSSYPPS	IGFPYSLNEA	PWSTAGDPPI	PYLTTYGQLS	NGDHHFMHDA	300
VFGQPGGLGN	NIYQHRFNFF	PENPAFSAWG	TSGSQGQQQT	SSAYGSSYTY	PPSSLGGTVV	360
DGQPGFHSDT	LSKAPGMNSL	EQGMVGLKIG	DVSSSAVKTV	GSVVSSVALT	GVLSGNGGTN	420
VNMPVSKPTS	WAAIASPKAK	PQPKMKTSG	PVMGGGLPPP	PIKHNDIGT	WDNKGVPVKA	480
PVPQQAPSPQ	AAPQPPQVAQ	PLPAQPPALA	QPQYQSPQQP	PQTRWVAPRN	RNAAFQGGSG	540
AGSDSNSPQN	VQPNAPSVE	SGGSGGSGGS	GSGGSGGSGM	SATSVDRPK	GQGNKVSQVN	600
GSIHQKDAVN	DDDPEPYLSS	QTNQNSYPP	MSDPYMPSY	APSIGFPYSL	GEAAWSTAGD	660
QMPPLYTTYG	QMSNGEHYI	PDGVFSQPGA	LGNTPPFLGQ	HGFNFFPGNA	DFSTWGTSGS	720
QQQSTQSSAY	SSSYGYPPSS	LGRAITDGQA	GFNDTLTKV	PGISSIEQGM	TGLKIGGDLT	780
AAVTKTVGTA	LSSSGMTSIA	TNSVPPVSSA	APKPTSWAAI	ARKPAKPQPK	LKPKGNVIG	840
GSAVPPPIIK	HNMNIGTWDE	KGSVVKAPPT	QPVLPQTII	QQPQPLIQPP	PLVQSQLPQQ	900
QPPQPPQPPQ	QGPQQAQPH	QVQPPQQQLQ	NRWVAPNRNG	AGFNQNNAG	SENFGLGVVP	960
VSASPSSVEV	HPVLEKLKAI	NNYNPKDFDW	NLKNQSDYKD	DDDK		1004

SEQ ID NO: 10

FEATURE
source

Location/Qualifiers
1..540
mol_type = protein
organism = synthetic construct

SEQUENCE: 10

MGIIINTLQKY	YCRVRGGRC	VLSCLPKEEQ	IGKCSTRGRK	CCRRKKGGSG	GSGGSGGSGG	60
SGGSMVAVPET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120
VIFKEVSSAT	NALRSMQGFP	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSQ	LPPLERLTLG	180
GSGGSGGSGG	GSGGSGGMSG	DEMIFDPTMS	KKKKKKKKKF	MLDEEGDTQT	EETQPSSETKE	240
VEPEPTEDKD	LEADEEDTRK	KDASDDLDDL	NFFNQKKKKK	KTKKIFDIDE	AEEGVKDLKI	300
ESDVQEPTPE	EDDLDIMLGN	KKKKKKNVKF	PDEDEILEKD	EALDEDEDNK	DDGISFSNQ	360
GPAWAGSERD	YTYEELNLRV	FNIMREKNPD	MVAGEKRKFV	MKPPQVVRVG	TKKTSFVNFT	420
DICKLLHRQP	KHLLAFLLAE	LGTSGSIDGN	NQLVIKGRFQ	QKQIENVLRR	YIKEYVTCHT	480
CRSPDTILQK	DTRLFYFLQCE	TCHSRCSVAS	IKTGFQAVTG	KRAQLRAKAN	GSDYKDDDDK	540

SEQ ID NO: 11

FEATURE
source

Location/Qualifiers
1..413
mol_type = protein
organism = synthetic construct

SEQUENCE: 11

MGIIINTLQKY	YCRVRGGRC	VLSCLPKEEQ	IGKCSTRGRK	CCRRKKGGSG	GSGGSGGSGG	60
SGGSMVAVPET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120
VIFKEVSSAT	NALRSMQGFP	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSQ	LPPLERLTLG	180
GSGGSGGSGG	GSGGSGGSMTE	RRVPFSLLRG	PSWDPPRDWY	PHSRLFDQAF	GLPRLPEEWS	240
QWLGGSSWPG	YVRPLPPAAI	ESPAAAPAY	SRALSRQLSS	GVSEIRHTAD	RWRVSLDVNH	300
FAPDELTVKT	KDGVVEITGK	HEERQDEHGY	ISRCFTRKYT	LPPGVDPQTQV	SSSLSPGEGTL	360
TVEAMPKLA	TQSNBITIPV	TFESRAQLGG	PEAAKSDETA	AKLGSQDYKDD	DDK	413

SEQ ID NO: 12

FEATURE
source

Location/Qualifiers
1..462
mol_type = protein
organism = synthetic construct

SEQUENCE: 12

MGIIINTLQKY	YCRVRGGRC	VLSCLPKEEQ	IGKCSTRGRK	CCRRKKGGSG	GSGGSGGSGG	60	
SGGSMVAVPET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120	
VIFKEVSSAT	NALRSMQGFP	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSQ	LPPLERLTLG	180	
GSGGSGGSGG	GSGGSGGSMDD	REDLVYQAKL	AEQAERYDEM	VESMKKVAGM	DVELTVEERN	240	
LLSVAYKNVI	GARRASWRII	SSIEQKEENK	GGEDKLMIR	EYRQMVETEL	KLICCDILDV	300	
LDKHLIPAA	TGESKVFYIK	MKGDYHRYLA	EFATGNDRKE	AAENSLVAYK	AASDIAMTEL	360	
PPTHPIRLGL	ALNFSVFYIE	ILNSPDRA	LAKAAFD	DAI	AELDTLSEES	YKDSLIMQL	420
LRDNLTLWTS	DMQGDGEEQN	KEALQDVEDE	NQGSQDYKDD	DK		462	

SEQ ID NO: 13

FEATURE
source

Location/Qualifiers
1..606
mol_type = protein
organism = synthetic construct

SEQUENCE: 13

MGIIINTLQKY	YCRVRGGRC	VLSCLPKEEQ	IGKCSTRGRK	CCRRKKGGSG	GSGGSGGSGG	60
SGGSMVAVPET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120

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VIFKEVSSAT	NALRSMQGF	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSQ	LPPLERLTG	180
GGSGSGSGG	SGSGSGSMAE	EMVEAAGEDE	RELAAEEMAA	FLNENLPESI	FGAPKAGNGQ	240
WASVIRVMNP	IQGNLTLVQ	LEQNEAAFSV	AVCRFSNIGE	DWYVLVGVA	DLILNPRSV	300
GGFVYTYKLV	NNGEKLEFLH	KTPVEEVPAA	IAPFQGRVLI	GVGKLLRVYD	LGKKKLLRKC	360
ENKHIANYIS	GIQTIGHRVI	VSDVQESFIW	VRYKRNNQNL	IIFADDTYPR	WVTASLLDY	420
DTVAGADKFG	NICVVRLLPN	TNDEVEDPT	GNKALWDRGL	LNGASQKAEV	IMNYHVGETV	480
LSLQKTTLIP	GGSESLVYTT	LSGGIGILVP	FTSHEDHDF	QHVEMHLRSE	HPPLCGRDHL	540
SFRSYFVPVK	NVIDGDLCEQ	FNSMEPNKQK	NVSEELDRT	PEVSKKLEDI	RTRYAFGSDY	600
KDDDDK						606
SEQ ID NO: 14	moltype = AA length = 46					
FEATURE	Location/Qualifiers					
source	1..46					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 14						
MGIIINTLQKY	YCRVRGGRCA	VLSCLPKEEQ	IGKCTRGRK	CCRRKK		46
SEQ ID NO: 15	moltype = AA length = 18					
FEATURE	Location/Qualifiers					
source	1..18					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 15						
GGSGSGSGSG	GGSGSGGS					18
SEQ ID NO: 16	moltype = AA length = 102					
FEATURE	Location/Qualifiers					
source	1..102					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 16						
MAVPETRPNH	TIYINNLSK	IKKDELKKS	LDAIFSQFGQI	LDILVPRQRT	PRGQAFVIFK	60
EVSSATNALR	SMQGFPPYDK	PMRIQYAKTD	KRIPAKMKGT	FV		102
SEQ ID NO: 17	moltype = AA length = 11					
FEATURE	Location/Qualifiers					
source	1..11					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 17						
LQLPPLERLT	L					11
SEQ ID NO: 18	moltype = AA length = 8					
FEATURE	Location/Qualifiers					
source	1..8					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 18						
DYKDDDDK						8
SEQ ID NO: 19	moltype = AA length = 434					
FEATURE	Location/Qualifiers					
source	1..434					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 19						
MDLTAIYESL	LSLSPDVPVP	SDHGGTESSP	GWGSSGPWSL	SPSDSSPSGV	TSRLPGRSTS	60
LVEGRSCGWV	PPPPGFAPLA	PRLGPPLSPS	PTSPTATSTT	PSRYKTELCR	TFSESGRCRY	120
GAKCQFAHGL	GELRQANRHP	KYKTELCHKF	YLQGRCPYGS	RCHFPHNPSE	DLAAPGHPPV	180
LRQISIFSGL	PSGRRTSPPP	PGLAGPSLSS	SSFSPSSSPP	PPGDLPLSPS	AFSAAPGTPL	240
ARRDPTPVCC	PSCRRATPIS	VWGPLGGLVR	TPSVQSLGSD	PDEYASSGSS	LGGSDSPVFE	300
AGVFAPPQPV	AAPRRLPIFN	RISVSEGGSG	GGSGSGSGSG	SGGSDAMFGQ	PGALGSTPFL	360
GQHGFFNFFPS	GIDFSAWGNN	SSQGQSTQSS	GYSSNYAYAP	SSLGGAMIDG	QSAFANETLN	420
KAPGMNTIDQ	GMAA					434
SEQ ID NO: 20	moltype = AA length = 393					
FEATURE	Location/Qualifiers					
source	1..393					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 20						
MPAATVDHSQ	RICEVWACNL	DEEMKKIRQV	IRKYNVYAMD	TEFPGVVARP	IGEFRSNADY	60
QYQLLRNCVD	LLKIIQLGLT	FMNEQGEYPP	GTSTWQFNFK	FNLTEDMYAQ	DSIELLTTS	120
IQFKKHEEG	IETQYFAELL	MTSGVVLCEG	VKWLSEFHSY	DFGYLIKILT	NSNLPEEELD	180
FPEILRLFPF	VIYDVKYLK	SCKNLKGGLO	EVAEQLELER	IGPQHAGSD	SLLTGMAFFK	240

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MREMPFEDHI	DDAKYCGHLY	GLGSGSSSYVQ	NGTGNAYEEE	ANKQSGGSGG	SGGSGGSGGS	300
GGSDAMFGQP	GALGSTPFLG	QHGFNFFPSG	IDFSAWGNNS	SQQQSTQSSG	YSSNYAYAPS	360
SLGGAMIDGQ	SAFANETLNK	APGMNTIDQG	MAA			393

SEQ ID NO: 21 moltype = AA length = 629
 FEATURE Location/Qualifiers
 source 1..629
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 21

MPAATVDHSQ	RICEVWACNL	DEEMKKIRQV	IRKYNVYVAM	TEFPGVVARP	IGEFRSNADY	60
QYQLLRNCVD	LLKIIQLGLT	FMNEQGEYPP	GTSTWQFNFK	FNLTEDMYAQ	DSIELLTTS	120
IQFKKHEEBG	IETQYFAELL	MTSGVVLCEG	VKWLSPHSGY	DFGYLIKILT	NSNLPEEELD	180
FFEILRLFFP	VIYDVKYLK	SCKNLKGGLO	EVAEQLELER	IGPQHAGSD	SLLTGMAFFK	240
MREMPFEDHI	DDAKYCGHLY	GLGSGSSSYVQ	NGTGNAYEEE	ANKQSGGSGG	SGGSGGSGGS	300
GGMDLTAIY	ESLLSLSPDV	PVPSDHGGTE	SSPGWGSSGP	WSLSPSDSSP	SGVTSRLPGR	360
STSLVEGRSC	GWVPPPPGFA	PLAPRLGPGL	SPSPTSPTAT	STTPSRYKTE	LCRTFSESGR	420
CRYGAKCQFA	HGLGELRQAN	RHPKYKTELC	HKFYLGQRCP	YGSRCHFIHN	PSEDLAAPGH	480
PPVLRQSI SF	SGLPSGRRTS	PPPPGLAGPS	LSSSSFSFSS	SPPPPGDLPL	SPSAFSAAPG	540
TPLARRDPTP	VCCPSCRAT	PISVWGPLGG	LVRTPSVQSL	GSDPDEYASS	GSSLGGSDSP	600
VFEAGVFAPP	QPVAAPRRLP	IFNRISVSE				629

SEQ ID NO: 22 moltype = AA length = 420
 FEATURE Location/Qualifiers
 source 1..420
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 22

TRSRYSDFE	EIAVLGQGA	GQVVKARNAL	DSRYIAIKKI	RHTEEKLSTI	LSEVMLLASL	60
NHQYVVRYYA	AWLEEDSMDE	NVFESTDEES	DLSESSSDFE	ENDLLDQSSI	FKNRTNHDLD	120
NSNWDIFSIS	GYPDIVFENS	SRDDENEDLD	HTSSTSSSE	SQDDTDKESK	SIQNVPRRRN	180
FVKPMTAVKK	KSTLFIQMEY	CENGTLYDLI	HSENLNQQRD	EYWRLEFRQIL	EALSYIHSQG	240
IIHRDLKPMN	ILIDESRNVK	IGDFGLAKNV	HRSLDILKLD	SQNLPGSSDN	LTSAGTAMY	300
VATEVLDGTG	HYNEKIDMYS	LGIIFFEMIV	PFSTGMERVN	ILKKLRSVSI	EFPPDPPDNK	360
MKVEKKIIRL	LIDHDPNKR	GARTLLNSGW	LPVKHQDEVI	KEALKSLSNP	SSPWQQQVRE	420

SEQ ID NO: 23 moltype = AA length = 117
 FEATURE Location/Qualifiers
 source 1..117
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 23

MSGGSSCSQT	PSRAIPTRRV	ALGDGVQLPP	GDYSTAPGGT	LFSTAPGGTR	IIYDRKFLME	60
CRNAPVAKAP	PKDLPTIPGV	TSPTSDEPPM	QASQSHLHSS	PEDKRAGGEE	SQFEMDI	117

SEQ ID NO: 24 moltype = AA length = 192
 FEATURE Location/Qualifiers
 source 1..192
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 24

MGTFDLWTDY	LGLAHLVRAL	SGKEGPETRL	SPQPEPEPML	EPVSALEPMP	APESVPVPGP	60
KDQKRSLESS	PAPERLCSFC	KHNGESRAIY	QSHVLKDEAG	RVLCPILRDY	VCPQCAGATRE	120
RAHTRRFCLP	TGQGYTSVYS	HTTRNSAGKK	LVRPKAKATQ	DTGHRRGGGG	GAGFRGAGKS	180
EPSPSCSPSM	ST					192

SEQ ID NO: 25 moltype = AA length = 641
 FEATURE Location/Qualifiers
 source 1..641
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 25

MATVEPETTP	TPNPPTTEEE	KTESNQEVAN	PEHYIKHPLQ	NRWALWFFKN	DKSKTWQANL	60
RLISKFDTVF	DFWALYNHIO	LSSNLMPCGD	YSLFKDGI EP	MWEDEKNKRG	GRWLITLNKQ	120
QRRSLLDRFW	LETLLCLIGE	SFDDYSDDVC	GAVVNVRAKG	DKIAIWTTEC	ENREAVTHIG	180
RVYKERLGLP	PKIVIGYQSH	ADTATKSGST	TKNRFVVGGS	GGSGGSGSGG	GGSGGMSASQ	240
DSRSRDNGPD	GMEPEGVIES	NWNEIVDSFD	DMNLSESLLR	GIYAYGFEKP	SAIQQRAILP	300
CIKGYDVIAQ	AQSGTGKTAT	FAISILQQIE	LDLKATQALV	LAPTRELAQQ	IQKVVMALGD	360
YMGASCHACI	GGTNVRAEVQ	KLQMEAPHII	VGTPGRVFDM	LNRRYLSPKY	IKMFVLEAD	420
EMLSRGFKDQ	IYDIFQKLNS	NTQVVLSSAT	MPSDVLEVTK	KFMRDPIRIL	VKKEELTLEG	480
IRQFYINVER	BEWKLDLDCD	LYETLTITQA	VIFINTRRKV	DWLTEKM HAR	DFTVSAMHGD	540
MDQKERDVIM	REFRSGSSRV	LITTDLLARG	IDVQQVSLVI	NYDLPTNREN	YIHRIGRGGGR	600
FGRKGVAINM	VTEEDKRTL	DIETFYNTSI	EEMPLNVADL	I		641

SEQ ID NO: 26 moltype = AA length = 903
 FEATURE Location/Qualifiers

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source 1..903
mol_type = protein
organism = synthetic construct

SEQUENCE: 26

MSASQDSRSR	DNGPDGMEPE	GVIESNWNNEI	VDSFDDMNLS	ESLLRGIYAY	GFEKPSAIQQ	60
RAILPCIKGY	DVIAQAQSGT	GKTATFAISI	LQQIELDLKA	TQALVLAPTR	ELAQQIQKV	120
MALGDYMGAS	CHACIGGTNV	RAEVQKLQME	APHIIVGTPG	RVFDMNLRRY	LSPKYIKMFV	180
LDEADEMLSR	GFKDQIYDIF	QKLNSNTQVV	LLSATMPSDV	LEVTKKFMRD	PIRILVKKEE	240
LTLEGIRQFY	INVEREEWKL	DTLCDLYETL	TITQAVIFIN	TRRKVDWLTE	KMHARDFTVS	300
AMHGDMDQKE	RDVIMREFRS	GSSRVLITTD	LLARGIDVQQ	VSLVINYDLP	TNRENYIHRI	360
GRGGRFRKRG	VAINMVTEED	KRTLRLDIET	YNTSIEEMPL	NVADLIGSGG	GSGGSGSGGG	420
SGGSMDSGFD	RAPGAGRGRS	RGLGRGGGGP	EGGGFPNGAG	PAERARHOPP	QPKAPGFLQP	480
PPLRQPRRTT	PPGAQCEVPA	SPQRPSRPGA	LPEQTRPLRA	PPSSQDKIPQ	QNSSESAMAKP	540
QVVVAPVLMS	KLSVNAPEFY	PSGYSSSYTE	SYEDGCEDYP	TLSEYVQDFL	NHLTEQPGSF	600
ETEIEQFAET	LNGCVTTDDA	LQELVELIYQ	QATSIPNFSY	MGARLCNYLS	HHLTISPQSG	660
NFRQLLLQRC	RTEYEVKQQA	AKGDEVTRKR	FHAFVLFLGE	LYLNLEIKGT	NGQVTRADIL	720
QVGLRELLNA	LFSNPMDDNL	ICAVKLLKLT	GSVLEDAWKE	KGKMDMEEII	ORIENVVLLA	780
NCSRVDVKQML	LKLVELRSSN	WGRVHATSTY	REATPENDPN	YFMNEPTFYT	SDGVPPFTAAD	840
PDYQEKYQEL	LEREDFFPDY	EENGTDLSGA	GDPYLDIDD	EMDPEIEEAY	EKFCLESERK	900
RKQ						903

SEQ ID NO: 27 moltype = AA length = 797
FEATURE Location/Qualifiers
source 1..797
mol_type = protein
organism = synthetic construct

SEQUENCE: 27

MSATSVDTQR	TKGQDNKVQN	GSLHQKDTVH	DNDFEPYLTG	QSNQSNSYPS	MSDPYLSSYY	60
PPSIGFPYSL	NEAPWSTAGD	PPIPYLTTYG	QLSNGDHHFM	HDAVFGQPGG	LGNNIYQHRF	120
NFFPENPAFS	AWGTSGSQGQ	QTQSSAYGSS	YTYPPSSSLG	TVVDGQPGFH	SDTLKAPGM	180
NSLEQGMVGL	KIGDVSSSAV	KTVGSSVSSV	ALTGVLSGNG	GTNVNMPVSK	PTSWAAIASK	240
PAKPQPKMKT	KSGPVMGGGL	PPPIKHNMD	IGTWDNKGVP	PKAPVPQQAP	SPQAAPQPQQ	300
VAQPLPAQPP	ALAQPYQSP	QPPQTRWVA	PRNRNAAFQ	SGGAGSDSNS	PGNVQPNAP	360
SVESGSGSGS	GSGSGSGSGS	GSMATSVDQ	RPKGQGNKVS	VQNGSIHQKD	AVNDDDFEPY	420
LSSQTNQSNS	YPPMSDPYMP	SYIAPSIGFP	YSLGEAAWST	AGDQPMPLYT	TYGQMSNGEH	480
HYIPDGVSQ	PGALGNTPPF	LGQHGFNFFP	GNADFSTWGT	SGSQGSTQS	SAYSSSYGYP	540
PSSLGRAITD	GQAGFGNDTL	SKVPGISSIE	QGMTGLKIGG	DLTAAVTKTV	GTALSSSGMT	600
SIATNSVPPV	SSAAPKPTSW	AAIARKPAKP	QPKLKPKNV	GIGGSVPPPP	PIKHNMNIGT	660
WDEKGSVKA	PPTQPVLPQ	TIIQQPQLI	QPPPLVQSOL	QQQPPQPPQ	QQQQGPPQQA	720
QPHQVQPQQQ	QLQNRWVAPR	NRGAGFNQNN	GAGSENFGLG	VVPVSASPSS	VEVHPVLEKL	780
KAINNYNPKD	FDWNLKN					797

SEQ ID NO: 28 moltype = AA length = 333
FEATURE Location/Qualifiers
source 1..333
mol_type = protein
organism = synthetic construct

SEQUENCE: 28

MSGDEMIFDP	TMSKKKKKKK	KPFMLDEEGD	TQTEETQPS	EKEVEPEPTE	DKDLEADEED	60
TRKKDASDDL	DDLNFNQQK	KKKTKKIFD	IDEAEEGVKD	LKIESDVQEP	TEPEDDLDIM	120
LGKKKKKKKN	VKFPDEDEIL	EKDEALEDED	NKKDDGISFS	NQTGPAAWGS	ERDYTEEELL	180
NRVFNIMRRE	KPDVMAVEGR	KFVMMKPPQV	RVGTTKTSFV	NFTDICKLLH	RQPKHLLAFL	240
LDELGTSGSI	DGNQNLVIK	RFQKQKIENV	LRRIYKEYVT	CHTCRSPDTI	LQKDLRLYFL	300
QCETCHSRCS	VASIKTGQQA	VTGKRAQLRA	KAN			333

SEQ ID NO: 29 moltype = AA length = 206
FEATURE Location/Qualifiers
source 1..206
mol_type = protein
organism = synthetic construct

SEQUENCE: 29

MTERRVPFSL	LRGPSWDPPR	DWYPHSRLFD	QAFGLPRLPE	EWSQWLGGSS	WPGYVRPLPP	60
AAIESPAVAA	PAYSRAISRQ	LSSGVSEIRH	TADRNRVSLD	VNHFAPELDT	VKTGDGVVEI	120
TKGHEERQDE	KYIISRCFTR	KYTLPPGVDP	TQVSSSLSP	GTLTVEAPMP	KLATQSNIEIT	180
IPVTFESRAQ	LGGPEAAKSD	ETAAKL				206

SEQ ID NO: 30 moltype = AA length = 255
FEATURE Location/Qualifiers
source 1..255
mol_type = protein
organism = synthetic construct

SEQUENCE: 30

MDDRDLVYQ	AKLAEQAERY	DEMVESMKKV	AGMDVELTVE	ERNLLSVAYK	NVIGARRASW	60
RIISSIEQKE	ENKGGEDKLG	MIREYRQMV	TELKLIICDI	LDVLDKHLIP	AANTGESKVF	120
YYKMGDYHR	YLAEFATGND	RKEAAENSLV	AYKAASDIAM	TELPPTHPIR	LGLALNFSVF	180
YYEILNSPDR	ACRLAKAAFD	DAIAELDTLS	EESYKDISTLI	MQLLRDNLTL	WTSMDMQGDGE	240

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EQNKEALQDV EDENQ 255

SEQ ID NO: 31 moltype = AA length = 399
 FEATURE Location/Qualifiers
 source 1..399
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 31

MAEEMVEAAG	EDERELAEM	AAAFLENENLP	ESIFGAPKAG	NGQWASVIRV	MNPIQGNTLD	60
LVQLEQNEAA	FSVAVCRFSN	IGEDWYVLVG	VAKDLILNPR	SVAGGFVYTY	KLVNNGEKL	120
FLHKTPVEEV	PAAIAPQGR	VLIGVGKLLR	VYDLGKKLL	RKCNKHIAN	YISGIQTIGH	180
RVIVSDVQES	FIWVRYKRNE	NQLIIFADDT	YPRWVTASL	LDYDTVAGAD	KFGNICVVRL	240
PPNTNDEVDE	DPTGNKALWD	RGLNGASQK	AEVIMNYHVG	ETVLSLQKTT	LIPGGSSESLV	300
YTTLSGGIGI	LVPFTSHEDH	DFPQHVMHL	RSEHPPLCGR	DHLSFRSYYP	PVKNVIDGDL	360
CEQFNSMEPN	KQKNVSEELD	RTPEVSKKL	EDIRTRYAF			399

SEQ ID NO: 32 moltype = AA length = 326
 FEATURE Location/Qualifiers
 source 1..326
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 32

MDLTAIYESL	LSLSPDVPVP	SDHGGTESSP	GWGSSGPWSL	SPSDSSPSGV	TSRLPGRSTS	60
LVEGRSCGWV	PPPGFAPLA	PRLGPELSPS	PTSPTATSTT	PSRYKTELCR	TFSESGRCRY	120
GAKCQFAHGL	GELRQANRHP	KYKTELCHKF	YLQGRCPYGS	RCHFIHNPSE	DLAAPGHPPV	180
LRQSISSFGL	PSGRRTSPPP	PGLAGPSLSS	SSFSPSSSPP	PPGDLPLSPS	AFSAAPGTPL	240
ARRDPTPVCC	PSCRATPIS	VWGPLGGLVR	TPSVQSLGSD	PDEYASSGSS	LGGSDSPVFE	300
AGVFAPPQPV	AAPRRLPIFN	RISVSE				326

SEQ ID NO: 33 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 33

DAMFGQPGAL	GSTPFLGQHG	FNFFPSGIDF	SAWGNSSQG	QSTQSSGYSS	NYAYAPSSLG	60
GAMIDGQSAF	ANETLNKAPG	MNTIDQMAA				90

SEQ ID NO: 34 moltype = AA length = 285
 FEATURE Location/Qualifiers
 source 1..285
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 34

MPAATVDHSQ	RICEVWACNL	DEEMKKIRQV	IRKYNIVAMD	TEFPGVVARP	IGEFRSNADY	60
QYQLLRCNVD	LLKIIQLGLT	FMNEQGEYPP	GTSTWQFNFK	FNLTEDMYAQ	DSIELLTSG	120
IQFKKHEEEG	IETQYFAELL	MTSGVVLCEG	VKWSLFHSGY	DFGYLIKILT	NSNLPEEELD	180
PFILRLFFP	VIYDVXYLMK	SKNLKGGLQ	EVAEQLELER	IGPQHQAGSD	SLLTGMAFFK	240
MREMPFEDHI	DDAKYCGHLY	GLGSGSSYVQ	NGTGNAYEEE	ANKQS		285

SEQ ID NO: 35 moltype = AA length = 217
 FEATURE Location/Qualifiers
 source 1..217
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 35

MATVEPETTP	TPNPPTTEEE	KTESNQEVAN	PEHYIKHPLQ	NRWALWFFKN	DKSKTWQANL	60
RLISKEDTVE	DFWALYNHIO	LSSNLMPGCD	YSLFKDGIEP	MWEDEKNKRG	GRWLITLNKQ	120
QRRSDLDRFW	LETLLCLIGE	SFDDYSDDVC	GAVVNVRAKG	DKIAIWTTEC	ENREAVTHIG	180
RVYKERLGLP	PKIVIGYQSH	ADTATKSGST	TKNRFVV			217

SEQ ID NO: 36 moltype = AA length = 406
 FEATURE Location/Qualifiers
 source 1..406
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 36

MSASQDSRSR	DNGPDGMEPE	GVIESNWNNEI	VDSFDDMNLS	ESLLRGIYAY	GFEKPSAIQQ	60
RAILPCIKGY	DVIAQAQSGT	GKTATFAISI	LQQIELDLKA	TQALVLAPTR	ELAQQIQKV	120
MALGDYMGAS	CHACIGGTNV	RAEVQKLQME	APHIIVGTPG	RVFDMLNRRY	LSPKYIKMFV	180
LDEADEMLSR	GFKDQIYDIF	QKLSNNTQVV	LLSATMPSDV	LEVTKKFMRD	PIRILVKKEE	240
LTLEGIRQFY	INVEREEWKL	DTLCDLYETL	TITQAVIFIN	TRRKVDWLTE	KMHARDFTVS	300
AMHGDMQKE	RDVIMREFRS	GSSRVLITTD	LLARGIDVQQ	VSLVINVDLP	TNRENYIHRI	360
GRGGRFGRKG	VAINMVTEED	KRTLRLDIET	YNTSIEEMPL	NVADLI		406

SEQ ID NO: 37 moltype = AA length = 479

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FEATURE Location/Qualifiers
 source 1..479
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 37
 MSDGFDRAFG AGRGRSRGLG RGGGGPEGGG FPNGAGPAER ARHQPQPKA PGFLQPPPLR 60
 QPRTTPPPGA QCEVPASPQR PSRPGALPEQ TRPLRAPPSS QDKIPQQNSE SAMAKPOVVV 120
 APVLM SKLSV NAPEFYPSGY SSSYTESYED GCEDYPTLSE YVQDFLNHLT EQPGSFETEI 180
 EQFAETLNGC VTDDALQEL VELIYQQATS IPNFSYMGAR LCNYLSHHLT ISPGSGNFRQ 240
 LLLQRCRTEY EVKDAQAKGD EVTRKRHFHAF VLFLGELYLN LEIKGTNGQV TRADILQVGL 300
 RELNALFSN PMDDNLICAV KLLKLTGSLV EDWKEKGM DMEEIIQRIE NVVLDANCSR 360
 DVKQMLLKLIV ELRSSNWGRV HATSTYREAT PENDPNYFMN EPTFYTSDGV PFTAADPDYQ 420
 EKYQELLERE DFFPDYEENG TDLSGAGDPY LDDIDDEMDP EIEEAYEKFC LESERKRKQ 479

SEQ ID NO: 38 moltype = AA length = 364
 FEATURE Location/Qualifiers
 source 1..364
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 38
 MSATSVDTQR TKGQDNKVQN GSLHQKDTVH DNDFEPLYTG QSNQNSYPS MSDPYLSSYY 60
 PPSIGFPYSL NEAPWSTAGD PPIPYLTTYG QLSNGDHFM HDAVFGQPGG LGNNIYQHRF 120
 NFFPENPAPS AWGTSGSQGQ QTQSSAYGSS YTYPPSSLGG TVVDGQPGFH SDTLKAPGM 180
 NSLEQGMVGL KIGDVSSSAV KTVGSVVSVV ALTGVLSGNG GTNVNMPVSK PWSAAIASK 240
 PAKPQPKMT KSGPVMGGGL PPPIKHNMD IGTWDNKGVP PKAPVPQQAQ SPQAAPQPQQ 300
 VAQPLPAQPP ALAQPOYQSP QQPPQTRWVA PRNRNAAPGQ SGGAGSDSNS PGNVQPNSAP 360
 SVES 364

SEQ ID NO: 39 moltype = AA length = 415
 FEATURE Location/Qualifiers
 source 1..415
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 39
 MSATSVDTQR KGGQGNKVSQV NGSIHQKDAV NDDDFEPLYLS SQTNQNSYPS PMSDPYMPSY 60
 YAPSIGFPYSL LGEAAWSTAG DQPMPLYLTYG QMSNGEHY IPDGVFSQPG ALGNTPPFLG 120
 QHGFNFPPGN ADFSTWGTSG SQGSTQSSA YSSSYGYPPS SLGRAITDQ AGFGNDTLK 180
 VPGISSIEQG MTGLKIGGDL TAAVTKTVGT ALSSSGMTSI ATNSVPPVSS AAPKPTSWAA 240
 IARKPAKQP KLKPKGNVGI GGSVAVPPPPI KHNMNIGTWD EKSVMVAPP TQPVLPPTI 300
 IQPQPLIQP PPLVQSLPQ QQPQPQQAQ HQVQPQQQL QNRWVAPRNR 360
 GAGFNQNGA GSENFGLGVV PVSASPSSVE VHPVLEKLKA INNYNPKDFD WNLKN 415

SEQ ID NO: 40 moltype = RNA length = 31
 FEATURE Location/Qualifiers
 source 1..31
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 40
 ggccagatct gagcctggga gctctctggc c 31

SEQ ID NO: 41 moltype = RNA length = 26
 FEATURE Location/Qualifiers
 source 1..26
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 41
 ccaaaggctc ttctcagagc caccaca 26

SEQ ID NO: 42 moltype = RNA length = 21
 FEATURE Location/Qualifiers
 source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 42
 aaacatgagg attacccatg t 21

SEQ ID NO: 43 moltype = RNA length = 21
 FEATURE Location/Qualifiers
 source 1..21
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 43
 aaacatgagg atcaccatg t 21

SEQ ID NO: 44 moltype = RNA length = 15
 FEATURE Location/Qualifiers

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source	1..15 mol_type = other RNA organism = synthetic construct	
SEQUENCE: 44		
gccctgaaga agggc		15
SEQ ID NO: 45	moltype = RNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15 mol_type = other RNA organism = synthetic construct	
SEQUENCE: 45		
gccctgaaaa agggc		15
SEQ ID NO: 46	moltype = RNA length = 25	
FEATURE	Location/Qualifiers	
source	1..25 mol_type = other RNA organism = synthetic construct	
SEQUENCE: 46		
taaggagttt atatggaac cctta		25
SEQ ID NO: 47	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other RNA organism = synthetic construct	
SEQUENCE: 47		
aaacatgagg attacccatg t		21
SEQ ID NO: 48	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other RNA organism = synthetic construct	
SEQUENCE: 48		
aaacatgagg atcaccatg t		21
SEQ ID NO: 49	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other RNA organism = synthetic construct	
SEQUENCE: 49		
atgcatgtct aagacagcat		20
SEQ ID NO: 50	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other RNA organism = synthetic construct	
SEQUENCE: 50		
aaacatgagg attacccatg t		21
SEQ ID NO: 51	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other RNA organism = synthetic construct	
SEQUENCE: 51		
aaacatgagg atcaccatg t		21
SEQ ID NO: 52	moltype = RNA length = 23	
FEATURE	Location/Qualifiers	
source	1..23 mol_type = other RNA organism = synthetic construct	
SEQUENCE: 52		
aaaacataag gaaaacctat gtt		23
SEQ ID NO: 53	moltype = RNA length = 28	
FEATURE	Location/Qualifiers	
source	1..28 mol_type = other RNA organism = synthetic construct	
SEQUENCE: 53		

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ggctcgtgta gctcattagc tccgagcc	28
SEQ ID NO: 54	moltype = RNA length = 43
FEATURE	Location/Qualifiers
source	1..43
	mol_type = other RNA
	organism = synthetic construct
SEQUENCE: 54	
ggcgteccctc ccgaagctgc gcgctcggtc gaacaggacg acc	43
SEQ ID NO: 55	moltype = DNA length = 40
FEATURE	Location/Qualifiers
source	1..40
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 55	
ctctctgatt ttgaggaaaa gaaattctga agccacatt	40
SEQ ID NO: 56	moltype = RNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = other RNA
	organism = synthetic construct
SEQUENCE: 56	
ggccgaaatc ccgaatgagg cc	22
SEQ ID NO: 57	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other RNA
	organism = synthetic construct
SEQUENCE: 57	
ggatgcctcc cgagtgcac c	21
SEQ ID NO: 58	moltype = AA length = 90
FEATURE	Location/Qualifiers
source	1..90
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 58	
MDDPSFLTGR STYAKRRRAR RMNVCKCGAI LHNKDCRSS TISGHKLDRL RFVKEGRVAL	60
EGETPVYRTW VKWVETEHYH NILETSDDEE	90
SEQ ID NO: 59	moltype = AA length = 76
FEATURE	Location/Qualifiers
source	1..76
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 59	
MRVTLSKPKQ ALATPNKEEH GKRRKKKGKL GKRRDPCLRK YKDFCIHGEC KYVKELRAP	60
CICHPGYHGE RCHGLS	76
SEQ ID NO: 60	moltype = AA length = 282
FEATURE	Location/Qualifiers
source	1..282
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 60	
MAVPETRPNH TIYINNLEK IKKDELKKS L YAIQSQFGQI LDILVSRSLK MRGQAFVIFK	60
EVSSATNALR SMQGFPPFYDK PMRIQYAKTD SDIIAKMKGT FVERDRKREK RPKKSQETPA	120
TKKAVQGGGA TPVVGAVQGP VPGMPPMTQA PRIMHHMPGQ PPYMPPPGMI PPPGLAPGQI	180
PPGAMPPQQL MPGQMPPAQP LSENPPNHIL FLTNLPEETN ELMLSMLEFNQ FPGFKEVRLV	240
PGRHDIAFVE FDNEVQAGAA RDALQGFKIT QNNAMKISFA KK	282
SEQ ID NO: 61	moltype = AA length = 314
FEATURE	Location/Qualifiers
source	1..314
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 61	
QLLRDNTTGS NSSLNTSLPS PGAWPYSASD NSFTNVHSTS AKFPDYKSTW SPDPIGHNPT	60
HLSNMWKNH ISSRNTTPLP RPPPGLTNPK PSSPWSSTAP RSVRGWGTQD SRLASASTWS	120
DGGSVRPSYW LVLHNLTPQI DGSTLRTICM QHGPLLTFHL NLTQGTALIR YSTKQEAACA	180
QTAHMCVLG NTTILAEFAT DDEVSRFLAQ AQPTPTAATP SAPAAGWQSL ETGQNSQSDPV	240
GPALNLFGGG TGLGQWSSSA GGSSGADLAG ASLWGPPNYS SSLWGVPTVE DPHRMGSPAP	300
LLPGDLLGGG SDIS	314

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SEQ ID NO: 66      moltype = AA  length = 531
FEATURE            Location/Qualifiers
source             1..531
                  mol_type = protein
                  organism = synthetic construct
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SEQUENCE: 66

MGIINTLQKY	YCRVRGGRC	VLSCLPKEEQ	IGKCSTRGRK	CCRRKKGGSG	SGSGSGSGSG	60
SGGSMAVPET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120
VIFKEVSSAT	NALRSMQGF	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSQ	LPPLERLTLG	180
SGSGSGSGSG	SGSGSGSMSS	EAETQPPAA	PPAAPALSAA	DTKPGTTGSG	AGSGGPGGLT	240
SAAPAGGDKK	VIATKVLGT	KWFNVRNGY	FINRNDTKED	VFVHQTAIKK	NNPRKYLRV	300
GDGETVEFDV	VEGEKGAEAA	NVTGPGGVPV	QGSKYAADRN	HYRRYPRRRG	PPRNYQQNYQ	360
NSESGEKNEG	SESAPEGQAQ	QRRPYRRRRF	PPYMRPYG	RRPQYSNPPV	QGEVMEGADN	420
QGAGEQGRPV	RQNMVRYR	RFRRGPPRQR	QPREDGNEED	KENQGDETQG	QPPQRRYRR	480
NFNYYRRRPE	NPKPDGKET	KAADPPAENS	SAPEAEQGGA	EGSDYKDDDD	K	531

SEQ ID NO: 67

moltype = AA length = 952

FEATURE Location/Qualifiers

source

1..952

mol_type = protein

organism = synthetic construct

SEQUENCE: 67

MGIINTLQKY	YCRVRGGRC	VLSCLPKEEQ	IGKCSTRGRK	CCRRKKGGSG	SGSGSGSGSG	60
SGGSMAVPET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120
VIFKEVSSAT	NALRSMQGF	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSQ	LPPLERLTLG	180
SGSGSGSGSG	SGSGSGSMSS	EAETQPPAA	PPAAPALSAA	DTKPGTTGSG	AGSGGPGGLT	240
SAAPAGGDKK	VIATKVLGT	KWFNVRNGY	FINRNDTKED	VFVHQTAIKK	NNPRKYLRV	300
GDGETVEFDV	VEGEKGAEAA	NVTGPGGVPV	QGSKYAADRN	HYRRYPRRRG	PPRNYQQNYQ	360
NSESGEKNEG	SESAPEGQAQ	QRRPYRRRRF	PPYMRPYG	RRPQYSNPPV	QGEVMEGADN	420
QGAGEQGRPV	RQNMVRYR	RFRRGPPRQR	QPREDGNEED	KENQGDETQG	QPPQRRYRR	480
NFNYYRRRPE	NPKPDGKET	KAADPPAENS	SAPEAEQGGA	EGSGSGSGSG	SGSGSGSGSL	540
HLDQTPSRQP	IPSEGLQLHL	PQVLADAVSR	LVLGKFGDLT	DNFSSPHARR	KVLGVVMTT	600
GTDVVDKAKVI	SVSTGTCIN	GEYMSDRGLA	LNDCHAEIIS	RRSLRLFLYT	QLELYLNNKD	660
DQKRSIFQKS	ERGGFRLKEN	VQPHLYISTS	PCGDARIFSP	HEPILEEPAD	RHPNRKARGQ	720
LRTKIESGGG	TIPVRSNASI	QTDGVLQGE	RLTMTSCSDK	IARWNVVGIQ	GSLLSIFVEP	780
IYFSSIILGS	LYHGDHLSRA	MYQRISNIED	LPPLYTLNKP	LLSGISNAEA	RQPGKAPNFS	840
VNWTVGDSAI	EVINATTGKD	ELGRASRLCK	HALYCRWMRV	HGKVPShLLR	SKITKPNVYH	900
ESKLAAYEQ	AAKARLFTAF	IKAGLGAWVE	KPTEQDQFSL	TPGSDYKDDDD	DK	952

SEQ ID NO: 68

moltype = AA length = 952

FEATURE Location/Qualifiers

source

1..952

mol_type = protein

organism = synthetic construct

SEQUENCE: 68

MGIINTLQKY	YCRVRGGRC	VLSCLPKEEQ	IGKCSTRGRK	CCRRKKGGSG	SGSGSGSGSG	60
SGGSMAVPET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120
VIFKEVSSAT	NALRSMQGF	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSQ	LPPLERLTLG	180
SGSGSGSGSG	SGSGSGSMSS	EAETQPPAA	PPAAPALSAA	DTKPGTTGSG	AGSGGPGGLT	240
SAAPAGGDKK	VIATKVLGT	KWFNVRNGY	FINRNDTKED	VFVHQTAIKK	NNPRKYLRV	300
GDGETVEFDV	VEGEKGAEAA	NVTGPGGVPV	QGSKYAADRN	HYRRYPRRRG	PPRNYQQNYQ	360
NSESGEKNEG	SESAPEGQAQ	QRRPYRRRRF	PPYMRPYG	RRPQYSNPPV	QGEVMEGADN	420
QGAGEQGRPV	RQNMVRYR	RFRRGPPRQR	QPREDGNEED	KENQGDETQG	QPPQRRYRR	480
NFNYYRRRPE	NPKPDGKET	KAADPPAENS	SAPEAEQGGA	EGSGSGSGSG	SGSGSGSGSL	540
HLDQTPSRQP	IPSEGLQLHL	PQVLADAVSR	LVLGKFGDLT	DNFSSPHARR	KVLGVVMTT	600
GTDVVDKAKVI	SVSTGTCIN	GEYMSDRGLA	LNDCHAEIIS	RRSLRLFLYT	QLELYLNNKD	660
DQKRSIFQKS	ERGGFRLKEN	VQPHLYISTS	PCGDARIFSP	HEPILEEPAD	RHPNRKARGQ	720
LRTKIESGGG	TIPVRSNASI	QTDGVLQGE	RLTMTSCSDK	IARWNVVGIQ	GSLLSIFVEP	780
IYFSSIILGS	LYHGDHLSRA	MYQRISNIED	LPPLYTLNKP	LLSGISNAEA	RQPGKAPNFS	840
VNWTVGDSAI	EVINATTGKD	ELGRASRLCK	HALYCRWMRV	HGKVPShLLR	SKITKPNVYH	900
ESKLAAYEQ	AAKARLFTAF	IKAGLGAWVE	KPTEQDQFSL	TPGSDYKDDDD	DK	952

SEQ ID NO: 69

moltype = AA length = 324

FEATURE Location/Qualifiers

source

1..324

mol_type = protein

organism = synthetic construct

SEQUENCE: 69

MSSEAEQPP	PAAPPAAPAL	SAADTKPGTT	GSAGSGGPG	GLTSAAPAGG	DKKVIATKVL	60
GTVKWFNVRN	GYGFINRNDT	KEDVFVHQTA	IKKNNPRKYL	RSVGDGETVE	FDVVEGEKGA	120
EAANVTGPGG	VPVQGSKYAA	DRNHYYRYP	RRGPPRNYQQ	NYQNSSESGEK	NEGSESAPEG	180
QAQQRPPYRR	RRPPPYMR	PYGRPPQYSN	PPVQGEVMEG	ADNQGAGEQG	RPVRQNMVYR	240
YRPRFRRGPP	RQRQPREDG	EEDKENQGDE	TQGQPPQRR	YRRNFNYRRR	RPENPKPDG	300
KETKAADPPA	ENSSAPEAEQ	GGAE				324

SEQ ID NO: 70

moltype = AA length = 403

FEATURE Location/Qualifiers

source

1..403

mol_type = protein

organism = synthetic construct

SEQUENCE: 70

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LHLDQTPSRQ	PIPSEGLQLH	LPQVLADAVS	RLVLGKFGDL	TDNFSSPHAR	RKVLAGVVM	60
TGTDVKDAKV	ISVSTGTKCI	NGEYMSDRGL	ALNDCHAEII	SRRLRLRFLY	TQLELYLNK	120
DDQKRSIFQK	SERGGFRLKE	NVQFHLYIST	SPCGDARIFS	PHEPILEEPA	DRHPNRKARG	180
QLRTKIESQG	GTIPVRSNAS	IQTWDGVLQG	ERLLTMSCS	KIARWNVVGI	QGSLLSIFVE	240
PIYFSSIILG	SLYHGDHLSR	AMYQRISNIE	DLPPPLYTLNK	PLLSGISNAE	ARQPGKAPNF	300
SVNWTVGDSA	IEVINATTGK	DELGRASRLC	KHALYCRWMR	VHGKVP SHLL	RSKITKPNVY	360
HESKLAKEY	QAAKARLFTA	FIKAGLGAWV	EKPTQDQFS	LTP		403

SEQ ID NO: 71 moltype = AA length = 403
 FEATURE Location/Qualifiers
 source 1..403
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 71

LHLDQTPSRQ	PIPSEGLQLH	LPQVLADAVS	RLVLGKFGDL	TDNFSSPHAR	RKVLAGVVM	60
TGTDVKDAKV	ISVSTGTKCI	NGEYMSDRGL	ALNDCHAEII	SRRLRLRFLY	TQLELYLNK	120
DDQKRSIFQK	SERGGFRLKE	NVQFHLYIST	SPCGDARIFS	PHEPILEEPA	DRHPNRKARG	180
QLRTKIESQE	GTIPVRSNAS	IQTWDGVLQG	ERLLTMSCS	KIARWNVVGI	QGSLLSIFVE	240
PIYFSSIILG	SLYHGDHLSR	AMYQRISNIE	DLPPPLYTLNK	PLLSGISNAE	ARQPGKAPNF	300
SVNWTVGDSA	IEVINATTGK	DELGRASRLC	KHALYCRWMR	VHGKVP SHLL	RSKITKPNVY	360
HESKLAKEY	QAAKARLFTA	FIKAGLGAWV	EKPTQDQFS	LTP		403

SEQ ID NO: 72 moltype = AA length = 745
 FEATURE Location/Qualifiers
 source 1..745
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 72

MSSEAEQQP	PAAPPAAPAL	SAADTKPGTT	GSAGSGGPG	GLTSAAPAGG	DKKVIATKVL	60
GTVKWFNVN	GYGFINRNDT	KEDVFVHQTA	IKKNNPRKYL	RSVGDGETVE	FDVVEGEKGA	120
EAANVTGPGG	VPVQGSKYAA	DRNHRYRYP	RRGPPRNYQQ	NYQNSSEGEK	NEGSESAPEG	180
QAQQRPPYR	RRFPPIYMRR	PYGRRPQYSN	PPVQGEVMEG	ADNQGAGEQG	RPVRQNMVRG	240
YRPRFRGPP	RQRQPREDN	EEDKENQGD	TQQQPPQRR	YRRNFNRYRR	RPENPKPDG	300
KETKAADPPA	ENSSAPEAEQ	GGAEGSGSGS	GSAGSGGSGS	GSLHLDQTPS	RQPIPSEGLQ	360
LHLPQVLADA	VSRLVLGKFG	DLTDNFSSPH	ARRKVLAVV	MTTGTVDKDA	KVISVSTGTK	420
CINGEYMSDR	GLALNDCHAE	IISRRSLLRF	LYTQLELYLN	NKDDQKRSIF	QKSERGGFRL	480
KENVQFHLYI	STSPCGDARI	FSPHEPILEE	PADRHPNRKA	RGQLRTKIES	GQGTIPVRSN	540
ASIQTDWGV	QGERLLTMS	SDKIARWNV	GIQGSLLSIF	VEPIYFSSII	LGSLYHGDHL	600
SRAMYQRISN	IEDLPPLYTL	NKPLLSGISN	AEARQPGKAP	NFSVNWTVDG	SAIEVINATT	660
GKDELGRASR	LCKHALYCRW	MRVHGKVP SH	LLRSKITKPN	VYHESKLA	EYQAAKARLF	720
TAFIKAGLGA	VWEKPTQDQ	FSLTP				745

SEQ ID NO: 73 moltype = AA length = 745
 FEATURE Location/Qualifiers
 source 1..745
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 73

MSSEAEQQP	PAAPPAAPAL	SAADTKPGTT	GSAGSGGPG	GLTSAAPAGG	DKKVIATKVL	60
GTVKWFNVN	GYGFINRNDT	KEDVFVHQTA	IKKNNPRKYL	RSVGDGETVE	FDVVEGEKGA	120
EAANVTGPGG	VPVQGSKYAA	DRNHRYRYP	RRGPPRNYQQ	NYQNSSEGEK	NEGSESAPEG	180
QAQQRPPYR	RRFPPIYMRR	PYGRRPQYSN	PPVQGEVMEG	ADNQGAGEQG	RPVRQNMVRG	240
YRPRFRGPP	RQRQPREDN	EEDKENQGD	TQQQPPQRR	YRRNFNRYRR	RPENPKPDG	300
KETKAADPPA	ENSSAPEAEQ	GGAEGSGSGS	GSAGSGGSGS	GSLHLDQTPS	RQPIPSEGLQ	360
LHLPQVLADA	VSRLVLGKFG	DLTDNFSSPH	ARRKVLAVV	MTTGTVDKDA	KVISVSTGTK	420
CINGEYMSDR	GLALNDCHAE	IISRRSLLRF	LYTQLELYLN	NKDDQKRSIF	QKSERGGFRL	480
KENVQFHLYI	STSPCGDARI	FSPHEPILEE	PADRHPNRKA	RGQLRTKIES	GQGTIPVRSN	540
ASIQTDWGV	QGERLLTMS	SDKIARWNV	GIQGSLLSIF	VEPIYFSSII	LGSLYHGDHL	600
SRAMYQRISN	IEDLPPLYTL	NKPLLSGISN	AEARQPGKAP	NFSVNWTVDG	SAIEVINATT	660
GKDELGRASR	LCKHALYCRW	MRVHGKVP SH	LLRSKITKPN	VYHESKLA	EYQAAKARLF	720
TAFIKAGLGA	VWEKPTQDQ	FSLTP				745

SEQ ID NO: 74 moltype = AA length = 539
 FEATURE Location/Qualifiers
 source 1..539
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 74

MAELTVEVRG	SNGAFYKGF	KDVHEDSLTV	VFENNWQPER	QVPFNEVRLP	PPPDIKKEIS	60
EGDEVEVYSR	ANDQEPGQW	LAKVRMMKGE	FYVIEYAACD	ATYNEIVTFE	RLRPVNQKNT	120
VKNTTFKCT	VDVPEDLREA	CANENAHKDF	KKAVGACRIF	YHPETTQLMI	LSASEATVKR	180
VNILDMLHLR	SIRTKLMLMS	RNEEATKHLE	CTKQLAAAPH	EEFVREDLM	GLAIGTHGSN	240
IQQARKVPGV	TAIELDEDTG	TFRIYGESAD	AVKKARGFLE	FVEDFIQVPR	NLVGKVIIGN	300
GKVIQEIYDK	SGVVRVRIE	DNENKLPRD	GMVPPVFVGT	KESIGNVQVL	LEYHIAYLKE	360
VEQLRMERLQ	IDEQLRQIGS	RSYSGRGRGR	RGPNTSGY	TNSELNPSSE	TESERKDELS	420
DWSLAGEDDR	DSRHQDSRR	RPGRGRSVS	GGRGRGPRG	GKSSISSVLK	DPDSNPYSL	480

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DNTESDQTAD TDASESHHST NRRRRSRRRR TDEDVLMMDG MTESDTASVN ENGLGKRC D 539

SEQ ID NO: 75 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 75
 MADRAPRQPT SLGSSPQRCG SPSPAPPPPP DRGERIEDCL APLCPPVVG RGGCALSRL 60
 PVDLSLLRSS RTAAVELVAG GVCRRSPSQEN 90

SEQ ID NO: 76 moltype = AA length = 245
 FEATURE Location/Qualifiers
 source 1..245
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 76
 MNNKFDALKD DDSGDHDQNE ENSTQKDGEK EKTERDKNQS SSKRKAVVPG PAEHPLQYNY 60
 TFWYSRRTPG RPTSSQSYEQ NIKQIGTFAS VEQFWRFYSH MVRPGDLTGH SDFHLPKEGI 120
 KPMWEDDANK NGGKWIIRLR KGLASRCWEN LILAMLGEQF MVGEEICGAV VSVRFQEDII 180
 SIWNKTASDQ ATTARIRDTL RRVNLPPNT IMEYKTHTDS IKMPGRLGPQ RLLFQNLWKP 240
 RLNV P 245

SEQ ID NO: 77 moltype = AA length = 746
 FEATURE Location/Qualifiers
 source 1..746
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 77
 MGIINTLQKY YCRVGGRC A VLSCLPKEEQ IGKCSTRGRK CCRKKKGGSG GSGSGSGSGG 60
 SGGSMVAPET RPNHTIYINN LNSKIKKDEL KKSLYAIFSQ FGQILDILVP RQRTPRGQAF 120
 VIFKEVSSAT NALRSMQGF FDKPMRIQY AKTDKRIPAK MKGTFVGS LQ LPPLERLTLG 180
 GSGSGSGSGG SGGSGGMAE LTVFVRSNG AFYKGFIDV HEDSLTVFE NNWQPERQVP 240
 FNEVRLPPPP DIKKEISEG DEVEVSRAND QEPGWWLAK VRMMKGFEYV IEYAACDATY 300
 NEIVTFERLR PVNQNTVKK NTFFKCTVDV PEDLREACAN ENAHKDFKKA VGACRIFYHP 360
 ETQMLILSA SEATVVRVNI LSDMHLRSIR TKLMLMSRNE EATKHLECK QLAAAFHEEF 420
 VVEDLMLGLA IGTHGSNIQ ARKVPGVTAI ELDEDTGTFR IYGESADAVK KARGFLEFVE 480
 DFIQVPRNLV GKVIGKNGKV IQEIVDKSGV VVRVIEGDNE NKLPRDGMV PFVFGTKES 540
 IGVNQVLLLEY HIAYLKEVEQ LRMERLQIDE QLRQIGRSY SGRGRGRRG NYTSGYGTNS 600
 ELSNPSETES ERKDELSDS LAGEDDRSR HQDRSRRRPG GRGRSVSGGR GRGGPRGGKS 660
 SISSVLKDPD SNPYSLLDNT ESDQTADTDA SESHSTNRR RSRRRRTDE DAVLMDGMTE 720
 SDTASVNENG LGKRCDSYD KDDDDK 746

SEQ ID NO: 78 moltype = AA length = 297
 FEATURE Location/Qualifiers
 source 1..297
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 78
 MGIINTLQKY YCRVGGRC A VLSCLPKEEQ IGKCSTRGRK CCRKKKGGSG GSGSGSGSGG 60
 SGGSMVAPET RPNHTIYINN LNSKIKKDEL KKSLYAIFSQ FGQILDILVP RQRTPRGQAF 120
 VIFKEVSSAT NALRSMQGF FDKPMRIQY AKTDKRIPAK MKGTFVGS LQ LPPLERLTLG 180
 GSGSGSGSGG SGGSGGSMAD RAPRQPTSLG SSPQRCGSP SAPPPPPDRG ERIEDCLAPL 240
 CPPVVGGRG CALSRFLPVD LSLRSSRTA AVELVAGGVC RSPSQENGSD YKDDDDK 297

SEQ ID NO: 79 moltype = AA length = 452
 FEATURE Location/Qualifiers
 source 1..452
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 79
 MGIINTLQKY YCRVGGRC A VLSCLPKEEQ IGKCSTRGRK CCRKKKGGSG GSGSGSGSGG 60
 SGGSMVAPET RPNHTIYINN LNSKIKKDEL KKSLYAIFSQ FGQILDILVP RQRTPRGQAF 120
 VIFKEVSSAT NALRSMQGF FDKPMRIQY AKTDKRIPAK MKGTFVGS LQ LPPLERLTLG 180
 GSGSGSGSGG SGGSGGSMN KFDALKDDDS GDHDQNEENS TQKDGEKEKT ERDKNQSSSK 240
 RKAVVGPAAE HPLQYNTFW YSRRTGRPT SSQSYEQNIK QIGTFASVEQ FWRFYSHMVR 300
 PGDLTGHSDF HLPKEGKPM WEDDANKNGG KWIIRLRKGL ASRCWENLIL AMLGEQFMVG 360
 EEICGAVSV RFQEDIIISIW NKTASDQATT ARIRDTLRRV LNLPPNTIME YKTHTDSIKM 420
 PGRLPQRL L QNLWKPRLN VPGSDYKDDD DK 452

SEQ ID NO: 80 moltype = AA length = 571
 FEATURE Location/Qualifiers
 source 1..571
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 80

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MGIINTLQKY YCRVRGGRC A VLSCLPKEEQ IGKCSTRGRK CCRKKKGSGS GSGSGSGSGG 60
SGGSMAVPET RPNHTIYINN LNSKIKKDEL KKSLEYAIFSQ FGQILDILVP RQRTPRGQAF 120
VIFKEVSSAT NALRSMQGFY FYDKPMRIQY AKTDKRIPAK MKGTFVGSLSQ LPPLERLTLG 180
GSGSGSGSGS GSGSGSGMSA TSVDTRQTKG QDNKVQNGSL HQKDTVHDND FEPYLTGQSN 240
QSNSYPSMSD PYLSSYYPPS IGFPYSLNEA PWSAGDPPI PYLTTYGQLS NGDHHFMHDA 300
VFGQPGGLGN NIYQHRFNFF PENPAFSAWG TSGSQGQQTQ SSAYGSSYTY PPSSLGGTVV 360
DQQPGFHSDT LSKAPGMNSL EQGMVGLKIG DVSSSAVKTV GSVVSSVALT GVLSGNGGTN 420
VMMPVSKPTS WAAIASKPAK PQPKMKTSG PVMGGGLPPP PIKHNDIGT WDNKGVPVKA 480
PVPQQAQSPQ AAPQPPQVAQ PLPAQPPALA QPQYQSPQPP QPTRWVAPRN RNAAFQSGSG 540
AGSDSNSPGN VQPNAPSVE SGSDYKDDDD K 571

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SEQ ID NO: 81      moltype = AA length = 424
FEATURE           Location/Qualifiers
source            1..424
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 81
MGIINTLQKY YCRVRGGRC A VLSCLPKEEQ IGKCSTRGRK CCRKKKGSGS GSGSGSGSGG 60
SGGSMAVPET RPNHTIYINN LNSKIKKDEL KKSLEYAIFSQ FGQILDILVP RQRTPRGQAF 120
VIFKEVSSAT NALRSMQGFY FYDKPMRIQY AKTDKRIPAK MKGTFVGSLSQ LPPLERLTLG 180
GSGSGSGSGS GSGSGSGMAT VEPETTPTPN PPTTEEEKTE SNQEVANPEH YIKHPLQNRW 240
ALWFFKNDKS KTWQANLRLI SKFDTVEDFW ALYNHIQLSS NLMPCDYSL FKDGIEPMWE 300
DEKNKRGGRW LITLNKQRR SLDLRFWLET LLCLIGESPD DYSDDVCGAV VNVRAKGDKI 360
AIWTECENR EAVTHIRGVY KERLGLPPKI VIGYQSHADT ATKSGSTTKN RFVVGSDYKD 420
DDDK 424

```

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SEQ ID NO: 82      moltype = AA length = 613
FEATURE           Location/Qualifiers
source            1..613
                  mol_type = protein
                  organism = synthetic construct

```

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SEQUENCE: 82
MGIINTLQKY YCRVRGGRC A VLSCLPKEEQ IGKCSTRGRK CCRKKKGSGS GSGSGSGSGG 60
SGGSMAVPET RPNHTIYINN LNSKIKKDEL KKSLEYAIFSQ FGQILDILVP RQRTPRGQAF 120
VIFKEVSSAT NALRSMQGFY FYDKPMRIQY AKTDKRIPAK MKGTFVGGSG GSGSGSGSGG 180
SGGSLHLDQT PSRQPIPSG LQLHLPOVLA DAVSRLVLGK FGDLTDNFSS PHARRKVLG 240
VVMTTGTDVK DAKVISVSTG TKCINGEYMS DRGLALNDCH AEIISRRSL RFLYTQLELY 300
LNNKDDQKRS IFQKSERGGF RLKENVQFHL YISTSPCGDA RIFSPHEPIL EEPADRHPNR 360
KARGQLRTKI ESGEGTIPVR SNASIQTWDG VLQGERLLTM SCSDKIARWN VVGIGSLLS 420
IFVEPIYFSS IILGSLYHGD HLSRAMYQRI SNIEDLPLY TLNKPLLSGI SNAEARQPGK 480
APNFSVNWTV GDSAIEVINA TTGKDELGRA SRLCKHALYC RWMRVHGVKVP SHLLRSKITK 540
PNVYHESKLA AKEYQAAKAR LFTAFIKAGL GAWVEKPTEQ DQFSLTPKRP AATKKAGQAK 600
KKKGSYKDD DDK 613

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SEQ ID NO: 83      moltype = DNA length = 22
FEATURE           Location/Qualifiers
source            1..22
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 83
cgccaacac tgctactcct ca 22

```

```

SEQ ID NO: 84      moltype = DNA length = 22
FEATURE           Location/Qualifiers
source            1..22
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 84
gccttgggtg agagtgaaga gc 22

```

```

SEQ ID NO: 85      moltype = DNA length = 23
FEATURE           Location/Qualifiers
source            1..23
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 85
catcactgcc acccagaaga ctg 23

```

```

SEQ ID NO: 86      moltype = DNA length = 23
FEATURE           Location/Qualifiers
source            1..23
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 86
atgccagtga gcttccggtt cag 23

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SEQ ID NO: 87	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 87		
tgacagccca catggcattc cacttatcac tggcatcctt		40
SEQ ID NO: 88	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 88		
gtggctggga gttgggctcg ggatcagagg agcggtgctt		40
SEQ ID NO: 89	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 89		
cagaagaaca ggaacagagc caggacgctg aagatgacag		40
SEQ ID NO: 90	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 90		
atcacataga tgataccact gaggagggct aggggaaagg		40
SEQ ID NO: 91	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 91		
gagacgaaga gcaacactag caccgcatg tgcaggaaca		40
SEQ ID NO: 92	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 92		
cgatgtgcag gaacaggatc cccaacaaga gtaggagcat		40
SEQ ID NO: 93	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 93		
aatccagtga tgtaaaaccg gccgccttg gtgagagtga		40
SEQ ID NO: 94	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 94		
atgccactca ctgtgctca ctgtgtagat ggccgctgca		40
SEQ ID NO: 95	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 95		
tcagtcgtgt gtccattgcc cacgagccat tggctgacga		40
SEQ ID NO: 96	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	

-continued

SEQUENCE: 96	organism = synthetic construct	
agactgcagc cattcgctca ctgatgagga gtagcagtgt		40
SEQ ID NO: 97	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 97		
ctgaagatga cagacaggat catggtggcc tggacagact		40
SEQ ID NO: 98	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 98		
gcgaagccat aggagtagtc agtgttgaca tgccactcac		40
SEQ ID NO: 99	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 99		
cgcgctcaga gcctagacgg acggtgcgtc gtcgggcgcc		40
SEQ ID NO: 100	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 100		
ccacagggag ggaggaagga aaccagagaa caaaaccaac		40
SEQ ID NO: 101	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 101		
ttggtggcca atacaagtca tcgccagaca gtccttgag		40
SEQ ID NO: 102	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 102		
cagttggtat aaaatcagag aaagcaagca aaaacaagaa		40
SEQ ID NO: 103	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 103		
gatttattta ttgtagtaat aacaacagcc tagctaggta		40
SEQ ID NO: 104	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 104		
gatattatat acatcttcaa tcaacagcaa tccccactca		40
SEQ ID NO: 105	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 105		
caaaacaaag atactgagct tggttatagt gttaggatgt		40

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SEQ ID NO: 106	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 106		
ctgtgtctctg acttctcatt taaacgtgtg gccccagtac		40
SEQ ID NO: 107	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 107		
tgcaactgggt cacccatagt gtaaaggctct tcaactgtct		40
SEQ ID NO: 108	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 108		
gctcagcggga gtttctgccc ggccaaacag cgtaaccctt		40
SEQ ID NO: 109	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 109		
ctggcggcaa gttctgctca gcggagtctt tgcccggcca		40
SEQ ID NO: 110	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 110		
gggtgaagag ttggcagaag aacaggaaca gagacagaat		40
SEQ ID NO: 111	moltype = DNA length = 41	
FEATURE	Location/Qualifiers	
source	1..41	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 111		
ttttcccttc ctcccttccc tatgtacgct cagagcctca g		41
SEQ ID NO: 112	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 112		
cctatgtacg ctcagagcct cagacagacc gtctggggcg		40
SEQ ID NO: 113	moltype = AA length = 238	
FEATURE	Location/Qualifiers	
source	1..238	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 113		
VSKGEELFTG VVPILVELDG DVNGHKFSVS GEGEGDATYG KLTLKFICTT GKLPVPWPTL	60	
VITLTLYGVQC FSRYPDHMKQ HDPFKSAMPE GYVQERTIFF KDDGNYKTRA EVKFEGDTLV	120	
NRIELKGIDF KEDGNILGHK LEYNYNSHNV YIMADKQKNG IKVNFKIRHN IEDGSVQLAD	180	
HYQQNTPIGD GPVLLPDNHY LSTQSALSKD PNEKRDHMVL LEFVTAAGIT LGMDELYK	238	
SEQ ID NO: 114	moltype = AA length = 344	
FEATURE	Location/Qualifiers	
source	1..344	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 114		
MQLEIKVALN FIISYLYNKL PRRRADLFGE ELERLLKKKY EGHWYPEKPL KSGGFRVCHI	60	
GEMVDPVVEL AAKRSGLAVE DVRANVPEEL SVWIDPFEVS YQIGEKGAVK VLYLDDSEGC	120	
GAPELDKKIK SSFNPDAQVF VPIGSQDSSL SNSPSPSFGQ SPSPTFIPRS AQPITFTTAS	180	

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FAATKFGSTK	MKKGGAASG	GGVASSGAGG	QPPQPPRMA	RSPTNSLLKH	KSLSLSMHSL	240
NFITANPAPQ	SQLSPNAKEF	VYNGGGSPSL	FFDAADGQGS	GTPGPFGGSG	AGTCNSSSFD	300
MAQVFGGGAN	SLFLEKTPFV	EGLSYNLNTM	QYPSQQFQPV	VLAN		344

SEQ ID NO: 115 moltype = AA length = 120
 FEATURE Location/Qualifiers
 source 1..120
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 115
 MQDGMYQRFL RQHVHPEETG GSDRYCNLMM QRRKMTLYHC KRFNTFIHED IWNIRSICST 60
 TNIQCKNGKM NCHEGVVKVT DCRDTGSSRA PNCRYRAIAS TRRVVIACEG NPQVPVHFDG 120

SEQ ID NO: 116 moltype = AA length = 406
 FEATURE Location/Qualifiers
 source 1..406
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 116
 WKPQSSHWGA ALKDLHRIYR PMIGKLKFFI DEKYKIADTS EGGIYLGFPYE KQEVAVKTFC 60
 EGSPRAQREV SCLQSSRENS HLVTFYGSSES HRGHLFVCVT LCEQTLEACL DVHRGEDVEN 120
 EEDEFARNVL SSIFKAVQEL HLCSCGYTHQD LQPQNILIDS KKAHLADFD KSIKWAGDPQ 180
 EVKRDLEDLG RLVLYVVKKG SISFEDLKAQ SNEEVVQLSP DEETKDLIHR LFHPGEHVRD 240
 CLSDLLGHFP FWTWESRYRT LRNVGNESDI KTRKSESEIL RLLQPGPSEH SKSFDKWTTK 300
 INECVMKKMN KFYEKRGNFY QNTVGDLLKF IRNLGEHIDE EKHKKMKLKI GDPSLYFQKT 360
 FPDLYIVVYT KLQNTFYRKH FPQTHSPNKP QCDGAGGASG LASPGC 406

SEQ ID NO: 117 moltype = AA length = 123
 FEATURE Location/Qualifiers
 source 1..123
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 117
 ETSAEKFERQ HMDADSSPAS NSNYCNKMMK SRSMTDRCK PVNTFIHEPL ADVQAVCSQT 60
 NVACKNGQTN CYQSKAAMRI TDCRETGSSK YPDCAYTTE EEKHIIIVACE GNPYPVPVHFD 120
 ATV 123

SEQ ID NO: 118 moltype = AA length = 129
 FEATURE Location/Qualifiers
 source 1..129
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 118
 MKESRAKKFQ RQHMDSDSSP SSSSTYCNQM MRRRNMTQGR CKPVNTFVHE PLVDVQNVCF 60
 QEKVTCKNGQ GNCYKSNSM HIIDCRLTNG SRYPNCAVRT SPKERHIIIVA CEGSPYVPVH 120
 FASVEDST 129

SEQ ID NO: 119 moltype = AA length = 114
 FEATURE Location/Qualifiers
 source 1..114
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 119
 CDIPQSTNCG GNVYSNDDIN TAIQGALDDV ANGDRPDNYP HQYYDEASED ITLCCGSGPW 60
 SEFPLVYNGP YYSSRDNYVS PGPDRVIYQT NTGEFCATVT HTGAASYDGF TQCS 114

SEQ ID NO: 120 moltype = AA length = 137
 FEATURE Location/Qualifiers
 source 1..137
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 120
 MSSLIIRRVIS TAKAPGAIGP YSQAVLVDRIT IYISGQIGMD PSSGQLVSGG VAEAAKQALK 60
 NMGEILKAAG CDFTNVVKTT VLLADINDFN TVNEIYKQYF KSNFPARAAY QVAALPKGSR 120
 IEIEAVAIQG PLTTASL 137

SEQ ID NO: 121 moltype = AA length = 199
 FEATURE Location/Qualifiers
 source 1..199
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 121
 MENFRKVRSE EAPAGCGAEG GPGSGPFPAD LAPGAVHMRV KEGSKIRNLM AFATASMAQP 60
 ATRAIVFSGC GRATTKTVTC AEILKRRLAG LHQVTRLRYR SVREVWQSLP PGPTQGQTPG 120
 EPAASLSVLK NVPLGALLLS KDALDPRQPG YQPPNPHPGP SSPPAAPASK RSLGEPAGE 180
 GSAKRSQPEP GVADEDQTA 199

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SEQ ID NO: 122	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 122		
taatgaattc ctttacacca cactgtcgtc gaatggccac		40
SEQ ID NO: 123	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 123		
tcttctcagc cactgttgcc acaccatgca ccactccctc		40
SEQ ID NO: 124	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 124		
agaaggcatt tcataagcct cattgtcagg atccacaggc		40
SEQ ID NO: 125	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 125		
ttaacatcgt agattgaagc cacaaaatcc acagcacaca		40
SEQ ID NO: 126	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 126		
tttcaataat taatcacagc cacttaagga accagtgcac		40
SEQ ID NO: 127	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 127		
cgaatattat ttattgtcag aaaggtacag cattcacacc		40
SEQ ID NO: 128	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 128		
gcaacattta aaggaggcca tgaaattttt atcagtctag		40
SEQ ID NO: 129	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 129		
cagagttttt caccatcaag ttatttttca gcacccatgg		40
SEQ ID NO: 130	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 130		
ctactagtga cttttagaaa tgactatgcc ccagaataat		40
SEQ ID NO: 131	moltype = DNA length = 40	
FEATURE	Location/Qualifiers	
source	1..40	

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mol_type = other DNA
organism = synthetic construct

SEQUENCE: 131
tagatttaaa atatataccta accgccactt tctaaccttc 40

SEQ ID NO: 132      moltype = DNA length = 21
FEATURE            Location/Qualifiers
source              1..21
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 132
caacagtggc tgagaagacc a 21

SEQ ID NO: 133      moltype = DNA length = 22
FEATURE            Location/Qualifiers
source              1..22
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 133
gtctcttctt cattcttggc ca 22

SEQ ID NO: 134      moltype = DNA length = 18
FEATURE            Location/Qualifiers
source              1..18
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 134
caccaactgg gacgacat 18

SEQ ID NO: 135      moltype = DNA length = 18
FEATURE            Location/Qualifiers
source              1..18
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 135
acagcctgga tagcaacg 18

SEQ ID NO: 136      moltype = AA length = 636
FEATURE            Location/Qualifiers
source              1..636
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 136
MNPSAPSYPM ASLYVGLDHP DVTEAMLYEK FSPAGPILSI RVC RDMITRR SLGYAYVNFQ 60
QPADAERALD TMNFDVIK GK PVRIMWSQRD PSLRKSGVGN IFIKNLDKSI DNKALYDTFS 120
AFGNILSCKV VCDENGSKGY GFVHFETQEA AERAIEKMNG MLLNDRKV FV GRFKSRKERE 180
AELGARAKEF TNVYIKNFGE DMDDERL KDL FGKFGPALS V KVM TDES GKS KGFGFVS FER 240
HEDAQKAVDE MNGKELNGKQ IYVGRAQKKV ERQTELK RKF EQMKQDRITR YQGVNLYVK N 300
LDDGIDDERL RKEFSPFGTI TSAKVMMEGG RSKGFGFVCF SSPEEATKAV TEMNGRIVAT 360
KPLYVALAQR KEERQAHLTN QYMQRMASVR AVPNPVINPY QPAPPSGYFM AAIPQTQ NRA 420
ATYPPSQIAQ LRPSPRWTAQ GARPHFPQNM PGAIRPAAPR PPFSTMRPAS SQVPRVMSTQ 480
RVANTSTQTM GPRPAAAAAA ATPAVRTVPQ YKYAAGVRNP QQHLNAQPQV TMQQPAVHVQ 540
GQEPLTASML ASAPPQEQKQ MLGERLFPLI QAMHPTLAGK ITGMLLEIDN SELLHMLESP 600
ESLRSKVDEA VAVLQAHQAK EAAQKAVNSA TGVPTV 636

SEQ ID NO: 137      moltype = AA length = 889
FEATURE            Location/Qualifiers
source              1..889
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 137
EPRKIIATVL MTEDIKL NKA EKAWKPSSKR TAADKDRGEE DADGSKTQDL FRRVRSILNK 60
LTPQMFQQLM KQVTQLAIDT EERLKGVIDL IFEKAISEPN FSVAYANMCR CLMALKVPTT 120
EKPTVTVNFR KLLLNRCQKE FEKDKDDDEV FEKKQKEMDE AATAEERGR L KEELEEARDI 180
ARRRSLGNIK FIGELFKLKM LTEAIMHDCV VKLLKNHDEE SLECLCRLLT TIGKDLDFEK 240
AKPRMDQYFN QMEKIIKEKK TSSRIREFMLQ DVLDLRGSNW VPRRGDQGP K TIDQIHKEAE 300
MEEHREHIKV QQLMAKGS DK RRGGPPGPPI SRGLPLVDDG GWNTVPISK G SRPIDTSRLT 360
KITKPGSIDS NNQLFAPGGR LSWGKGSSGG SGAKPSDAAS EAARPATSTL NRFSALQQAV 420
PTESTDNRRV VQRSSLSRER GEKAGDRGDR LERSERGGDR GDR LDRARTP ATKRSFSKEV 480
EERSRERPSQ PEGLRKAASL TEDRDGRDA VKREALPPV SPLKAALSEE ELEKKS KAI I 540
EYVLHNDMK EAVQCQQLA SPSLLFIFVR HGVESTLERS AIAREHMGQL LHQLLCAGHL 600
STAQYYQGLY EILELAEDME IDIPHVWLYL AELVTPILQE GGVPMGELFR EITKPLRPLG 660
KAASLLLEIL GLLCKSMGPK KVGTLWREAG LSWKEFLPEG QDIGAFVAEQ KVEYTLGEES 720
EAPGQRALPS EELNRQLEKL LKEGSSNQRV FDWIEANLSE QQIVSNTLVR ALMTAVCYSA 780
IIFETPLRVD VAVLKARAKL LQKYLCEQK ELQALYALQA LVVTLEQPPN LLRMFFDALY 840
DEDVVKEDAF YSWESSK DPA EQQKGVALK SVTAFKWL R EAEESDHN 889

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SEQ ID NO: 138 moltype = AA length = 394
 FEATURE Location/Qualifiers
 source 1..394
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 138

EPRKIIATVL	MTEDIKLNKA	EKAWKPSSKR	TAADKDRGEE	DADGSKTQDL	FRVRVSILNK	60
LTPQMFQQLM	KQVTQLAIDT	EERLKGVIDL	IFEKAISEPN	FSVAYANMCR	CLMALKVPTT	120
EKPTVTVNFN	KLLLNRCQKE	FEKDKDDDEV	FEKKQKEMDE	AATAEERGL	KEELEEARDI	180
ARRRSLGNIK	FIGELFKLKM	LTEAIMHDCV	VKLLKNHDEE	SLECLCRLLT	TIGKDLDFEK	240
AKPRMDQYFN	QMEKIIKEKK	TSSRIRFMLQ	DVLDLRGSNW	VPRRGDQGP	TIDQIHKEAE	300
MEEHREHIKV	QQLMAKGS DK	RRGGPPGPPI	SRGLPLVDDG	GWNTVPISK	SRPIDTSRLT	360
KITKPGSIDS	NNQLFAPGGR	LSWKGSSSGG	SGAK			394

SEQ ID NO: 139 moltype = AA length = 283
 FEATURE Location/Qualifiers
 source 1..283
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 139

MQTDLSLSPSP	NPVSPVPLNN	PTSAPRYGTV	IPNRIFVGGI	DFKTNESDLR	KFFSQYGSVK	60
EVKIVNDRAG	VSKGYGVFTF	ETQEDAQKIL	QEAEKLNKYD	KKLNIGPAIR	KQQVGIPRSS	120
IMPAAGTMYL	TTSTGYPTY	HNGVAYFHTP	EVTSVPPPWP	SRSVCSFVM	VAQPIYQPPA	180
YHYQATTQVL	PGQWQWSVPQ	PSASSAPFLY	LQPSSEVIYQ	VEIAQDGGCV	PPPLSLMETS	240
VPEPYSDHGV	QATYHQVYAP	SAITMPAPVM	QPEPIKTVWS	IHY		283

SEQ ID NO: 140 moltype = AA length = 595
 FEATURE Location/Qualifiers
 source 1..595
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 140

MTMTLHTKAS	GMALLHQIQG	NELEPLNRPQ	LKIPLERPLG	EVYLDSSKPA	VYNYPEGAAAY	60
EFNAAAAANA	QVYQGTGLPY	GPGESEAAFG	SNGLGGFPPL	NSVSPSPML	LHPPPQLSPF	120
LQPHGQQVY	YLENEPSGYT	VREAGPPAFY	RPNSDNRRQG	GRERLASTND	KGSMAMESAK	180
ETRYCAVCND	YASGYHYGVW	SCEGCKAFFK	RSIQGHNDYM	CPATNQCTID	KNRRKSCQAC	240
RLRKCYEVGM	MKGIRKDRR	GGRMLKHKRQ	RDDGEGRGEV	GSAGDMRAAN	LWPSPLMIKR	300
SKKNSLALSL	TADQMVSALL	DAEPPILYSE	YDPTRPFFSEA	SMMGLLTNLA	DRELVHMINW	360
AKRVPGFVDL	TLHQVHLLLE	CAWLEILMIG	LVWRSMEHPG	KLLFAPNLLL	DRNQGKCEVG	420
MVEIFDMLLA	TSSRFRMMNL	QGEFVCLKS	IILLNSGVYT	FLSSTLKSLE	EKDHIHRVLD	480
KITDTLIHLM	AKAGLTQQQ	HQRLAQLLLI	LSHIRHMSNK	GMEHLYSMKC	KNVVPPLYDL	540
LEMLDAHRLH	APTSRGASV	EETDQSHLAT	AGSTSSHSLO	KYYITGEAEG	FPATV	595

SEQ ID NO: 141 moltype = AA length = 331
 FEATURE Location/Qualifiers
 source 1..331
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 141

MDTGVI EGG	NVTLTIRLLM	HGKEVGSII	G	KGESVKKMR	EESGARINIS	EGNCPERIIT	60
LAGPTNAIFK	AFAMIIDKLE	EDISSMTNS	TAASRPPVTL	RLVVPASQCG	SLIGKGCKI		120
KEIRESTGAQ	VQVAGDMLPN	STERAITIAG	IPQSIIECVK	QICVVMLES	PKGVITIPYRP		180
KPSSSPVIFA	GGQAYTIQGG	YAIQPDLT	LHQLAMQSSH	FPMTHTGNTGF	SGIESSSPEV		240
KGYWAGLDAS	AQTTSHELTI	PNDLIGCIIG	RQGAKEINEIR	QMSGAQIKIA	NPVEGSTDRQ		300
VTITGSAASI	SLAQYLINVR	LSSETGGMGS	S				331

SEQ ID NO: 142 moltype = AA length = 326
 FEATURE Location/Qualifiers
 source 1..326
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 142

MSNGYEDHMA	EDCRGDIGRT	NLIVNYLPQN	MTQDELRSLF	SSIGEVESAK	LIRDKVAGHS	60
LGYGFNYYVT	AKDAERAIN	LNLRLQSKT	IKVSYPARPSS	EVIKDANLYI	SGLPRMTMQK	120
DVEDMFSRFG	RIINSRLVLD	QTTGLSRGVA	FIRFDKRSEA	EEAITSFNHG	KPPGSSEPI	180
VKFAANPNQN	KNVALLSQLY	HSPARRFGGP	VHHQAQRFRF	SPMGVDHMSG	LSGVNVPGNA	240
SSGWCIFIYN	LGQDADEGIL	WQMFPGPGAV	TNVKVIRDFN	TNKCKGPGFV	TMTNYEEAAM	300
AIASLNGYRL	GDKILQVSFK	TNKSHK				326

SEQ ID NO: 143 moltype = AA length = 305
 FEATURE Location/Qualifiers
 source 1..305
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 143

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MDLSGVKKKS	LLGVKENKK	SSTRAPSPTK	RKDRSDEKSK	DRSKDKGATK	ESSEKDRGRD	60
KTRKRRSASS	GSSSTRSRSS	STSSSGSSTS	TGSSSGSSSS	SASSRSGSSS	TSRSSSSSSS	120
SGSPSPSRRR	HDNRRRSRSK	SKPPKRDEKE	RKRRSPSPKP	TKVHIGRLTR	NVTKDHIMEI	180
FSTYGKIKMI	DMPVERMHPH	LSKGYAYVEF	ENPDEAEKAL	KHMDGGQIDG	QEITATAVLA	240
PWPRPPPRRF	SPPRRMLPPP	PMWRRSPPRM	RRRSRSPRRR	SPVRRRSRSP	GRRRHRSRSS	300
SNSSR						305

SEQ ID NO: 144 moltype = DNA length = 40
 FEATURE Location/Qualifiers
 source 1..40
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 144
 gccggtagaa ctacgtaagc tcagaagagg ccacactgtt 40

SEQ ID NO: 145 moltype = DNA length = 40
 FEATURE Location/Qualifiers
 source 1..40
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 145
 gagtatgaca cgacagttaa aagaactaca agccctcaga 40

SEQ ID NO: 146 moltype = DNA length = 40
 FEATURE Location/Qualifiers
 source 1..40
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 146
 taaattcttc attttaccag caactgctga catcaaagtc 40

SEQ ID NO: 147 moltype = DNA length = 40
 FEATURE Location/Qualifiers
 source 1..40
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 147
 ttaatcatat tcaaaccttc acattttggc tatgttaagt 40

SEQ ID NO: 148 moltype = AA length = 445
 FEATURE Location/Qualifiers
 source 1..445
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 148
 MGIINTLQKY YCRVRGGRC A VLSCLPKEEQ IGKCTRGRK CCRKKKGGSG GSGSGSGSGG 60
 SGGMAVPET RPNHTIYINN LNSKIKKDEL KKSLEYAIFSQ FGQILDILVP RQRTPRGQAF 120
 VIFKEVSSAT NALRSMQGF FDKPMRIQY AKTDKRIPAK MKGTFVGSQ LPPLERLTLG 180
 GSGSGSGSGG SGGSGGSVSK GEELFTGVVP ILVELDGDVN GHKFSVSGEG EGDATYGKLT 240
 LKFICTTGKL PVPWPTLVTT LTYGVQCFSR YPDHMKQHDF FKSAMPEGYV QERTIFFKDD 300
 GNYKTRAEVK FEGDTLVNRI ELKGIDFKED GNILGHKLEY NYNSHNVYIM ADKQKNGIKV 360
 NFKIRHNIED GSVQLADHYQ QNTPIGDPV LLPDNHYLST QSALSKDPNE KRDMVLLLEF 420
 VTAAGITLGM DELYKGS DYK DDDDK 445

SEQ ID NO: 149 moltype = AA length = 398
 FEATURE Location/Qualifiers
 source 1..398
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 149
 MGIINTLQKY YCRVRGGRC A VLSCLPKEEQ IGKCTRGRK CCRKKKGGSG GSGSGSGSGG 60
 SGGMAVPET RPNHTIYINN LNSKIKKDEL KKSLEYAIFSQ FGQILDILVP RQRTPRGQAF 120
 VIFKEVSSAT NALRSMQGF FDKPMRIQY AKTDKRIPAK MKGTFVGSQ LPPLERLTLG 180
 GSGSGSGSGG SGGSGGMSA SSLLEQRPKG QGNKVQNGSV HQKDLNDDD FEPYLSQAR 240
 PINATYAMD SYLPSYSPS IGFSYSLGEA AWSTGGDTAM PYLTSYGQLS NGEPHFLPDA 300
 MFGQPGALGS TPFLQGHGFN FFPSSGIDFSA WGNSSSQQS TQSSGYSSNY AYAPSSLGGA 360
 MIDGQSAPAN ETLNKAPGMN TIDQMAAGS DYKDDDDK 398

SEQ ID NO: 150 moltype = AA length = 492
 FEATURE Location/Qualifiers
 source 1..492
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 150
 MGIINTLQKY YCRVRGGRC A VLSCLPKEEQ IGKCTRGRK CCRKKKGGSG GSGSGSGSGG 60
 SGGMAVPET RPNHTIYINN LNSKIKKDEL KKSLEYAIFSQ FGQILDILVP RQRTPRGQAF 120

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VIFKEVSSAT	NALRSMQGF	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSQ	LPPLERLTLG	180
SGSGSGSGSG	SGSGSGSMPA	ATVDHSQRIC	EVWACNLDEE	MKKIRQVIRK	YNYVAMDTF	240
PGVVARPIGE	FRSNADYQYQ	LLRCNVDDLK	IIQLGLTFMN	EQGEYPPGTS	TWQFNFKFNL	300
TEDMYAQDSI	ELLTTSGIQF	KKHEEEGIET	QYFAELLMTS	GVVLCGEVKK	LSFHSGYDFG	360
YLIKILTNNS	LPEEELDFFE	ILRLFFPVIY	DVKYLMKSCK	NLKGGLQEVA	EQLELERIGP	420
QHQAGSDSL	TGMAFFKMRE	MPFEDHIDDA	KYCGHLYGLG	SGSSYVQNGT	GNAYEEEANK	480
QSGSDYKDD	DK					492

SEQ ID NO: 151 moltype = AA length = 533
 FEATURE Location/Qualifiers
 source 1..533
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 151

MGIINTLQKY	YCRVRGGRCA	VLSCLPKEEQ	IGKCSTRGRK	CCRKKKGSGG	SGSGSGSGSG	60
SGGSMAPVET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120
VIFKEVSSAT	NALRSMQGF	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSQ	LPPLERLTLG	180
SGSGSGSGSG	SGSGSGSMDL	TAIYESLLSL	SPDVPVPSDH	GGTESSPGWG	SSGPWSLSPS	240
DSGPSGVTSR	LPGRSTSLVE	GRSCGWVPPP	PGFAPLAPRL	GPELSPSPTS	PTATSTTPSR	300
YKTELCTFES	ESGRCRYGAK	CQFAHGLGEL	RQANRHPKYK	TELCHKFYLO	GRCPYGSRCH	360
FIHNPSEDLA	APGHPPVLRQ	SISFSGLPSS	RRTSPPPPG	AGPSLSSSS	SPSSSPPPPG	420
DLPLSPSAFS	AAPGTPLARR	DPTPVCCPSC	RRATPISVWG	PLGGLVRTPS	VQSLGSDPDE	480
YASSGSSLGG	SDSPVFEAGV	FAPPQPVAAP	RRLPIFNRI	VSEGSYKDD	DDK	533

SEQ ID NO: 152 moltype = AA length = 551
 FEATURE Location/Qualifiers
 source 1..551
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 152

MGIINTLQKY	YCRVRGGRCA	VLSCLPKEEQ	IGKCSTRGRK	CCRKKKGSGG	SGSGSGSGSG	60
SGGSMAPVET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120
VIFKEVSSAT	NALRSMQGF	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSQ	LPPLERLTLG	180
SGSGSGSGSG	SGSGSGSMQL	EIKVALNFII	SYLYNKLPRR	RADLFGEEL	RLLKKKVEGH	240
WYPEKPLKGS	GFRCVHIGEM	VDPVVELAAK	RSLGAVEDVR	ANVPEELSVW	IDPFEVSQI	300
GEKGAVKVLY	LDDSEGCAP	ELDKKSSSF	NPDAQVFP	GSQDSSLSNS	PSPSFGQSPS	360
PTFIPRSAPQ	ITFTTASFAA	TKFGSTKMKK	GGGAASGGGV	ASSGAGGQOP	PQPRMARSP	420
TNSLLKHKSL	SLSMHSNFI	TANPAQSQL	SPNAKEFVYN	GGGSPSLFFD	AADGQSGTPT	480
GPFGSGGAGT	CNSSFDMQAQ	VFGGGANSLF	LEKTPFVEGL	SYNLNTMQYP	SQQFQPVVLA	540
NGSDYKDDDD	K					551

SEQ ID NO: 153 moltype = AA length = 327
 FEATURE Location/Qualifiers
 source 1..327
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 153

MGIINTLQKY	YCRVRGGRCA	VLSCLPKEEQ	IGKCSTRGRK	CCRKKKGSGG	SGSGSGSGSG	60
SGGSMAPVET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120
VIFKEVSSAT	NALRSMQGF	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSQ	LPPLERLTLG	180
SGSGSGSGSG	SGSGSGSMQD	GMYQRFLRQH	VHPEETGSD	RYCNLMQRR	KMTLYHCKRF	240
NTFIHEDLWN	IRSICSTTNI	QCKNGKMNCH	EGVVKVTDRC	DTGSSRAPNC	RYRAIASTRR	300
VVIACEGNPQ	VPVHFDGGSD	YKDDDDK				327

SEQ ID NO: 154 moltype = AA length = 613
 FEATURE Location/Qualifiers
 source 1..613
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 154

MGIINTLQKY	YCRVRGGRCA	VLSCLPKEEQ	IGKCSTRGRK	CCRKKKGSGG	SGSGSGSGSG	60
SGGSMAPVET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120
VIFKEVSSAT	NALRSMQGF	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSQ	LPPLERLTLG	180
SGSGSGSGSG	SGSGSGSWKP	QSSHWGAALK	DLHRIYRPMI	GKLFKFIDEK	YKIADTSEGG	240
IYLGFIYKQE	VAVKTFCEGS	PRAQREVSL	TFYGSSEHRL	HLFVVCVTLC		300
QTLAECCLDVH	RGEVENEED	EFARNVLSSI	FKAVQELHLS	CGYTHQDLQP	QNILIDSKKA	360
AHLADFDKSI	KWAGDPQEVK	RDLEDLGRLV	LYVVKKSGIS	FEDLKAQSN	EVVQLSPDEE	420
TKDLIHLRLH	PGEHVRDCLS	DLGHPPFFWT	WESRYRTLRL	VGNESDIKTR	KSESEILRL	480
QPGPSEHSKS	PKWTTKINE	CVMKKMNKFI	EKRGNFYQNT	VGDLLKFIRN	LGEHIDEKHE	540
KKMKLKIGDP	SLYQKTFPD	LVIVVYTKLQ	NTEYRKHFPO	THSPNKQPCD	GAGGASGLAS	600
PGCGSDYKDD	DDK					613

SEQ ID NO: 155 moltype = AA length = 330
 FEATURE Location/Qualifiers
 source 1..330
 mol_type = protein

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                                organism = synthetic construct
SEQUENCE: 155
MGIINTLQKY YCRVRGGRC A VLSCLPKKEEQ IGKCSRTRGRK CCRKKKGSGG GSGGSGSGG 60
SGGSMVAVPET RPNHTIYINN LNSKIKKDEL KKSLEYAIFSQ FGQILDILVP RQRTPRGQAF 120
VIFKEVSSAT NALRSMQGFY FYDKPMRIQY AKTDKRIPAK MKGTFVGSQ LPLERLTLG 180
GSGGSGSGG SGGSGGSET AEFERQHM ADSSPASNSN YCNKMMKSR MTTDRCKPVN 240
TFIHEPLADV QAVCSQTNVA CKNGQTNVQ SKAAMRITDC RETGSSKYPD CAYTTTEEEK 300
HIIVACEGNP YVPVHFADTV GSDYKDDDDK 330

SEQ ID NO: 156      moltype = AA length = 388
FEATURE            Location/Qualifiers
source              1..388
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 156
MGIINTLQKY YCRVRGGRC A VLSCLPKKEEQ IGKCSRTRGRK CCRKKKGSGG GSGGSGSGG 60
SGGSMVAVPET RPNHTIYINN LNSKIKKDEL KKSLEYAIFSQ FGQILDILVP RQRTPRGQAF 120
VIFKEVSSAT NALRSMQGFY FYDKPMRIQY AKTDKRIPAK MKGTFVGSQ LPLERLTLG 180
GSGGSGSGG SGGSGGSMEL EIRPLFLVPD TNGFIDHLAS LARLLESRY ILVVLIVIN 240
ELDGLAKGQE TDHRAGGYAR VVQEKARKSI EFLEQRFESR DSCRLALTSR GNELESIAFR 300
SEDIITGLGN NDDLILSCCL HYCKDKAKDF MPASKEEPIR LLREVVLTD DRNLRVKALT 360
RNVFVRDIPA FLTWAQVGGG DYKDDDDK 388

SEQ ID NO: 157      moltype = AA length = 336
FEATURE            Location/Qualifiers
source              1..336
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 157
MGIINTLQKY YCRVRGGRC A VLSCLPKKEEQ IGKCSRTRGRK CCRKKKGSGG GSGGSGSGG 60
SGGSMVAVPET RPNHTIYINN LNSKIKKDEL KKSLEYAIFSQ FGQILDILVP RQRTPRGQAF 120
VIFKEVSSAT NALRSMQGFY FYDKPMRIQY AKTDKRIPAK MKGTFVGSQ LPLERLTLG 180
GSGGSGSGG SGGSGGSMKE SRAKKFQRQH MDSDSPPSS STYCNQMRR RNMTQGRCKP 240
VNTFVHEPLV DVQNVCFQEK VTCKNGQNC YKSNSSMHIT DCRLTNGSRY PNCAYRTSPK 300
ERHIIVACEG SPYVPVHFDA SVEDSTGSDY KDDDDK 336

SEQ ID NO: 158      moltype = AA length = 321
FEATURE            Location/Qualifiers
source              1..321
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 158
MGIINTLQKY YCRVRGGRC A VLSCLPKKEEQ IGKCSRTRGRK CCRKKKGSGG GSGGSGSGG 60
SGGSMVAVPET RPNHTIYINN LNSKIKKDEL KKSLEYAIFSQ FGQILDILVP RQRTPRGQAF 120
VIFKEVSSAT NALRSMQGFY FYDKPMRIQY AKTDKRIPAK MKGTFVGSQ LPLERLTLG 180
GSGGSGSGG SGGSGGSCDI PQSTNCGGNV YSNDDINTAI QGALDDVANG DRPDNYPHQY 240
YDEASEDITL CCGSGPWSEF PLVYNGPYYS SRDNYVSPGP DRVIYQNTNG EFCATVHTG 300
AASYDGFTQC SGSDYKDDDD K 321

SEQ ID NO: 159      moltype = AA length = 344
FEATURE            Location/Qualifiers
source              1..344
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 159
MGIINTLQKY YCRVRGGRC A VLSCLPKKEEQ IGKCSRTRGRK CCRKKKGSGG GSGGSGSGG 60
SGGSMVAVPET RPNHTIYINN LNSKIKKDEL KKSLEYAIFSQ FGQILDILVP RQRTPRGQAF 120
VIFKEVSSAT NALRSMQGFY FYDKPMRIQY AKTDKRIPAK MKGTFVGSQ LPLERLTLG 180
GSGGSGSGG SGGSGGSMSS LIRRIVISTAK APGAIGPYSQ AVLVDRTIYI SGQIGMDPSS 240
GQLVSGGVAE EAKQALKNMG EILKAAGCDF TNVVKTTVLL ADINDFNTVN EIYKQYFKSN 300
FPARAAQVA ALPKGSRIEI EAVAIQGLT TASLGSDYKD DDDK 344

SEQ ID NO: 160      moltype = AA length = 406
FEATURE            Location/Qualifiers
source              1..406
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 160
MGIINTLQKY YCRVRGGRC A VLSCLPKKEEQ IGKCSRTRGRK CCRKKKGSGG GSGGSGSGG 60
SGGSMVAVPET RPNHTIYINN LNSKIKKDEL KKSLEYAIFSQ FGQILDILVP RQRTPRGQAF 120
VIFKEVSSAT NALRSMQGFY FYDKPMRIQY AKTDKRIPAK MKGTFVGSQ LPLERLTLG 180
GSGGSGSGG SGGSGGSMEN FRKVRSEEAP AGCGAEGGGP GSGPFADLAP GAVHMRVKEG 240
SKIRNLMAFA TASMAQPATR AIVFSGCGRA TTKTVTCAEI LKRRLAGLHQ VTRLRYRSVR 300
EVWQSLPPGP TQQTPEGPA ASLSVLKNVP GLAILLSKDA LDPRQPGYQP PNPHPGPSSP 360
PAAPASKRSL GEPAAGEGSA KRSQPEPGVA DEDQTAGSDY KDDDDK 406

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SEQ ID NO: 161 moltype = AA length = 613
 FEATURE Location/Qualifiers
 source 1..613
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 161

MGIINTLQKY	YCRVRGGRC	VLSCLPKEEQ	IGKCSTRGRK	CCRKKKGGSG	SGSGSGSGSG	60
SGGSMVAVPET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120
VIFKEVSSAT	NALRSMQGFP	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSLSQ	LPPLERLTLG	180
SGSGSGSGSG	SGSGSGSMSA	SQDSRSRDNG	PDGMEPEGVI	ESNWNIEVDS	FDDMNLSESL	240
LRGIYAYGFE	KPSAIQQRAI	LPCIKGYDVI	AQAQSGTGKT	ATFAISILQQ	IELDLKATQA	300
LVLAPTRELA	QQIQKVVMAL	GDYMGASCHA	CIGGTNVRAE	VQKLQMEAPH	IIVGTPGRVF	360
DMLNRRLSP	KYIKMFVLDE	ADEMLSRGFK	DQIYDIFQKL	NSNTQVVLLS	ATMPSDVLEV	420
TKKFMRDPIR	ILVKKEELTL	EGIRQFYINV	EREWKLDLTL	CDLYETLTIT	QAVIFINTRR	480
KVDWLTEKMH	ARDFTVSAMH	GDMDQKERV	IMREFRSGSS	RVLITTDLLA	RGIDVQQVSL	540
VINYDLPTNR	ENYIHRIGRG	GRFGRKGVAI	NMVTEDDKRT	LRDIETFYNT	SIEEMPLNVA	600
DLIGSDYKDD	DDK					613

SEQ ID NO: 162 moltype = AA length = 843
 FEATURE Location/Qualifiers
 source 1..843
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 162

MGIINTLQKY	YCRVRGGRC	VLSCLPKEEQ	IGKCSTRGRK	CCRKKKGGSG	SGSGSGSGSG	60
SGGSMVAVPET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120
VIFKEVSSAT	NALRSMQGFP	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSLSQ	LPPLERLTLG	180
SGSGSGSGSG	SGSGSGSMNP	SAPSYPMASL	YVGDLHPDVT	EAMLYEKFSP	AGPILSIRVC	240
RDMITRRSLG	YAYVNFQPPA	DAERALDTMN	FDVIKGPVVR	IMWSQRDPST	RKSGVGNIFI	300
KNLDSIDNK	ALYDTFSAFG	NILSCKVVC	ENGSKGYGFV	HFETQEAER	AIEKMNGMLL	360
NDRKVPVGRF	KSRKERAEEL	GARAKEFTNV	YIKNFGEDMD	DERLKDLPFG	FGPALSVKVM	420
TDESGKSKGF	GFVSFERHED	AQKAVDEMNG	KELNGKQIYV	GRAQKKVERQ	TELKRKFEQM	480
KQDRITRYQG	VNLVVKNLDD	GIDDERLRKE	FSPFGTITSA	KVMMEGGRSK	GFGEVCFSSP	540
EEATKAVTEM	NGRIVATKPL	YVALAQKKEE	ROAHLTNQYM	QRMASVRAVP	NPVINPYQPA	600
PPSGYFMAAI	PQTQNRAAAY	PPSQIAQLRP	SPRWTAQGAR	PHPFQNMPGA	IRPAAPRPPF	660
STMTPASSQV	PRVMSTQVRA	NTSTQTMGPR	PAAAAAATP	AVRTVPQYKY	AAGVRNPQQH	720
LNAQPQVTMQ	QPAVHVQGE	PLTASMLASA	PPQEQKQMLG	ERLFPPLIQAM	HPTLAGKITG	780
MLLEIDNSEL	LHMLESPESL	RSKVDEAVAV	LQAHAQKEAA	QKAVNSATGV	PTVGSYKDD	840
DDK						843

SEQ ID NO: 163 moltype = AA length = 1096
 FEATURE Location/Qualifiers
 source 1..1096
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 163

MGIINTLQKY	YCRVRGGRC	VLSCLPKEEQ	IGKCSTRGRK	CCRKKKGGSG	SGSGSGSGSG	60
SGGSMVAVPET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120
VIFKEVSSAT	NALRSMQGFP	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSLSQ	LPPLERLTLG	180
SGSGSGSGSG	SGSGSGSEPR	KIIATVLMTE	DIKLNKAeka	WKPSKRTAA	DKDRGEEDAD	240
GSKTQDLFR	VRSLNLKLT	QMFQQLMKQV	TQLAIDTEER	LKGVIDLIFE	KAISEPNFSV	300
AYANMCRCLM	ALKVPTTEKP	TVTNNFRKLL	LNRCQKEFEK	DKDDDEVFEK	KQKEMDEAAT	360
AEERGLRLEE	LEEARDIARR	RLGNIKFIG	ELFKLKMTE	AIMHDCVVKL	LKNHDEESLE	420
CLCRLLTTIG	KDLDFEAKAP	RMDQYFNQME	KIIKEKKTSS	RIRFMLQDVL	DLRGSNWVPR	480
RGDQGPKTID	QIHKEAEEMEE	HREHIKVQQL	MAKGS DKRRG	GPPGPPISRG	LPLVDDGGWN	540
TVPIKSGSRP	IDTSRLTKIT	KPGSIDSNNQ	LFAPGGRLSW	GKGSSGGSGA	KPSDAASEAA	600
RPATSTLNR	SALQAVPTE	STDNRVVRQ	SSLSRERGEK	AGDRGDRLER	SERGGDRGDR	660
LDRARTPATK	RSFSKEVEER	SRERPSQPEG	LRKAASLTED	RDRGRDAVKR	EALPPVSP	720
KAALSEEELE	KKSKAIEEY	LHLNDMKEAV	QCVQELASPS	LLFIFVRHGV	ESTLERSAIA	780
REHMQLLHQ	LLCAGHLSTA	QYYQGLYEIL	ELAEDMEIDI	PHVWLYLAEL	VTPILQEGGV	840
PMGELFREIT	KPLRPLGKAA	SLLEILGLL	CKSMGPKKVG	TLWREAGLSW	KEFLPEGQDI	900
GAFVAEQKVE	YTLGESEAP	GQRALPSEEL	NRQLEKLLKE	GSSNQRVFDW	IEANLSEQQI	960
VSNTLVRLAM	TAVCYSAIIF	ETPLRVDVAV	LKARAKLLQK	YLCDEQKELQ	ALYALQALVV	1020
TLEQPPNLLR	MFFDALYDED	VVKEDAFYSW	ESSKDPAEQQ	GKGVALKSVT	APFKWLREAE	1080
EESDHNGSDY	KDDDDK					1096

SEQ ID NO: 164 moltype = AA length = 601
 FEATURE Location/Qualifiers
 source 1..601
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 164

MGIINTLQKY	YCRVRGGRC	VLSCLPKEEQ	IGKCSTRGRK	CCRKKKGGSG	SGSGSGSGSG	60
SGGSMVAVPET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120
VIFKEVSSAT	NALRSMQGFP	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSLSQ	LPPLERLTLG	180
SGSGSGSGSG	SGSGSGSEPR	KIIATVLMTE	DIKLNKAeka	WKPSKRTAA	DKDRGEEDAD	240

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GSKTQDLFR	VRSLNKLTP	QMFQQLMKQV	TQLAIDTEER	LKGVIDLIFE	KAISEPNFSV	300
AYANMCRCLM	ALKVPTTEKP	TVTNNFRKLL	LNRCQKEFEK	DKDDDEVFEK	KQKEMDEAAT	360
AEERGRLLKEE	LEEARDIARR	RSLGNIKFIG	ELFKLKMLTE	AIMHDCVVKL	LKNHDEESLE	420
CLCRLTTTIG	KDLDFEAKAP	RMDQYFNQME	KIIKEKKTSS	RIRFMLQDVL	DLRGSNNWVR	480
RGDQGPKTID	QIHKEAEEMEE	HREHIKVQQL	MAKGS DKRRG	GPPGPPI SRG	LPLVDDGGWN	540
TVPIKSGSRP	IDTSRLTKIT	KPGSIDSNNQ	LFAPGGRLSW	GKGSSGGSGA	KGS DYKDDDD	600
K						601

SEQ ID NO: 165 moltype = AA length = 490
 FEATURE Location/Qualifiers
 source 1..490
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 165

MGIINTLQKY	YCRVRGGRCA	VLSCLPKEEQ	IGKCSTRGRK	CCRKKKGGSG	GSGSGSGSGG	60
SGGSMAPPET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120
VIFKEVSSAT	NALRSMQGF	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSLO	LPPLERLTLG	180
GSGSGSGSGG	SGGSGSGMQT	DSLSPSPNPV	SPVPLNNPTS	APRYGTVIPN	RIFVGGIDFK	240
TNESDLRKFF	SGYGSVKEVK	IVNDRAGVSK	GYGFVTFETQ	EDAQKILQEA	EKLNYKDKKL	300
NIGPAIRKQQ	VGIPRSSIMP	AAGTMYLTTS	TGYPTYHYNG	VAYFHTPEVT	SVPPPWPSPRS	360
VCSSPVMVAQ	PIYQQPAYHY	QATTQYLPQG	WQWSVPQPSA	SSAPFLYLQP	SEVIYQPVFI	420
AQDGGCVPPP	LSLMETSVPE	PYSDHGVQAT	YHQVYAPSAI	TMPAPVMQPE	PIKTVWSIHY	480
GSDYKDDDDK						490

SEQ ID NO: 166 moltype = AA length = 802
 FEATURE Location/Qualifiers
 source 1..802
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 166

MGIINTLQKY	YCRVRGGRCA	VLSCLPKEEQ	IGKCSTRGRK	CCRKKKGGSG	GSGSGSGSGG	60
SGGSMAPPET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120
VIFKEVSSAT	NALRSMQGF	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSLO	LPPLERLTLG	180
GSGSGSGSGG	SGGSGGSMNT	TLHTKASGMA	LLHQIQGNEL	EPLNRPLQKI	PLERPLGEVY	240
LDSSKPAVYN	YPEGAAAYEFN	AAAAANAQVY	GQTGLPYGPG	SEAAAFGSNG	LGGFPPLNSV	300
SPSPMLLLHP	PPQLSPFLQP	HGQQVPYYLE	NEPSGYTVRE	AGPPAFYRPN	SDNRRQGGRE	360
RLASTNDKGS	MAMESAKETR	YCAVCNDYAS	GYHYGVWSCE	GCKAFFKRSI	QGHNDYMCPA	420
TNQTIDKNR	RKSCQACRLR	KCYEVGMMKG	GIRKDRRGR	MLKHQRQDD	GEGRGEVGS	480
GDMRAANLWP	SPLMIKRSK	NSLALSLTAD	QMVSAALLDAE	PPILYSEYDP	TRPFSEASMM	540
GLLTNLADRE	LVHMINWAKR	VPGFVDLTLH	DQVHLLCAW	LEILMIGLVW	RSMEHPGKLL	600
FAPNLLLDNR	QGKCEVGMVE	IFDMLLATSS	RFRMMNLQGE	EFVCLKSIIL	LNSGVYTFLS	660
STLKSLEEKD	HIHRVLDKIT	DTLIHLMAKA	GLTLQQQHQR	LAQLLLILSH	IRHMSNKGME	720
HLYSMKCKNV	VPLYDLLELM	LDARHLHAPT	SRGGASVEET	DQSHLATAGS	TSSHSLQKYY	780
ITGEAEGFPA	TVGSDYKDD	DK				802

SEQ ID NO: 167 moltype = AA length = 538
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 source 1..538
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 167

MGIINTLQKY	YCRVRGGRCA	VLSCLPKEEQ	IGKCSTRGRK	CCRKKKGGSG	GSGSGSGSGG	60
SGGSMAPPET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120
VIFKEVSSAT	NALRSMQGF	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSLO	LPPLERLTLG	180
GSGSGSGSGG	SGGSGGSMNT	GVIIEGLNVT	LTIRLLMHGK	EVGSIIGKKG	ESVKKMREES	240
GARINISEGN	CPERIITLAG	PTNAIFKAPA	MIIDKLEEDI	SSSMTNSTAA	SRPPVTLRLV	300
VPASQCGSLI	GKGGCKIKEI	RESTGAQVQV	AGDMLPNSTE	RAITIAGIPQ	SIIECVKQIC	360
VVMLESPPKG	VTIPYRPKPS	SSPVIFAGGQ	AYTIQQQYAI	PQPDLTKLHQ	LAMQQSHFPM	420
THGNTGFSGI	ESSSPPEVKY	WAGLDASAQT	TSHELTIPND	LIGCIIGRQG	AKINEIROQMS	480
GAQIKIANPV	EGSSTRQVTI	TGSAASISLA	QYLINVRSS	ETGGMGSSGS	DYKDDDDK	538

SEQ ID NO: 168 moltype = AA length = 533
 FEATURE Location/Qualifiers
 source 1..533
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 168

MGIINTLQKY	YCRVRGGRCA	VLSCLPKEEQ	IGKCSTRGRK	CCRKKKGGSG	GSGSGSGSGG	60
SGGSMAPPET	RPNHTIYINN	LNSKIKKDEL	KKSLYAIFSQ	FGQILDILVP	RQRTPRGQAF	120
VIFKEVSSAT	NALRSMQGF	FYDKPMRIQY	AKTDKRIPAK	MKGTFVGSLO	LPPLERLTLG	180
GSGSGSGSGG	SGGSGGSMNS	GYEDHMAEDC	RGDIGRTNLI	VNYLPQNMTQ	DELRSLFSSI	240
GEVESAKLIR	DKVAGHSLGY	GFVNYVTAKD	AERAINTLNG	LRLQSKTIKV	SYARPSSEVI	300
KDANLYISGL	PRMTMQKQVE	DMFSRFGRII	NSRVLVDQTT	GLSRGVAFIR	FDKRSEAEAA	360
ITSFNGHKPP	GSSEPI TVKF	AANPNQNKNV	ALLSOLYHSP	ARRFGGPVHH	QAQRF RFS PM	420
GVDHMSGLSG	VNVPGNASSG	WCIFIYNLGG	DADEGILWQM	FGPFGAVTNV	KVIRDFNTNK	480
CKGFGVVTMT	NYEEAAMAIA	SLNGYRLGDK	ILQVSFKTNK	SHKGS DYKDD	DDK	533

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SEQ ID NO: 169      moltype = AA  length = 512
FEATURE            Location/Qualifiers
source              1..512
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 169
MGIINTLQKY YCRVRGGRC A VLSCLPKEEQ IGKCTRGRK CRRRKKGGSG GSGGSGGSGG 60
SGGSMVAPET RPNHTIYIN LNSKIKKDEL KSLYAIQSQ FGQILDILVP RQRTPRGQAF 120
VIFKEVSSAT NALRSMQGF FDKPMRIQY AKTDKRIPAK MKGTFVGSQ LPLRLTLG 180
GSGGSGGSGG GSGGSGGMDL SGVKKKSLLG VKENNKKSST RAPSPTKRKD RSDEKSKDRS 240
KDKGATKES EKDRGRDKTR KRSASSGSS STRSRSSSTS SSGSSTSTGS SSGSSSSAS 300
SRSGSSSTSR SSSSSSSSGS PPSRRRHDN RRRSRKSKP PKRDEKERKR RSPSPKPTKV 360
HIGRLTRNVT KDHIMEIFST YGKIKMIDMP VERMHPHLSK GYAYVEFENP DEAEKALKHM 420
DGGQIDGQEI TATAVLAPWP RPPRRFSP RMLPPPPMW RSPRRMR RSPRRRSPV 480
RRRSRSPGR RHRSRSSNS SRGSDYKDD DK 512

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1. A polypeptide comprising an effector domain wherein the effector domain comprises the amino acid sequence of one of SEQ ID NOs: 19-39, 60-62, 69-76, 113-121, or 136-143 or comprises an amino acid sequence with at least 80% sequence identity to one of SEQ ID NOs: 19-39, 60-62, 69-76, 113-121, or 136-143; and wherein the polypeptide further comprises one or more of a stabilizer domain, linker, RNA hairpin binding domain, nuclear export signal, and molecular tag.

2. A polypeptide comprising an effector domain, wherein the effector domain comprises a fusion between at least two polypeptides, wherein each polypeptide is independently selected from one of SEQ ID NOs: 22-24, 28-39, 60-62, 69-76, 113-121, or 136-143 or a polypeptide with at least 80% sequence identity to one of SEQ ID NOs: 22-24, 28-39, 60-62, 69-76, 113-121, or 136-143.

3. The polypeptide of claim 2, wherein each polypeptide is independently selected from one of SEQ ID NOs: 32-39 or 69-76 or a polypeptide with at least 80% sequence identity to one of SEQ ID NOs: 32-39 or 69-76.

4. The polypeptide of claim 2 or 3, wherein the polypeptide comprises a linker between the two polypeptides.

5. The polypeptide of claim 4, wherein the linker comprises SEQ ID NO:15.

6. The polypeptide of any one of claims 2-5, wherein the effector domain comprises: a fusion between a polypeptide comprising at least 80% sequence identity to SEQ ID NO:32 and a polypeptide comprising at least 80% sequence identity to SEQ ID NO:33; a fusion between a polypeptide comprising at least 80% sequence identity to SEQ ID NO:34 and a polypeptide comprising at least 80% sequence identity to SEQ ID NO:33; a fusion between a polypeptide comprising at least 80% sequence identity to SEQ ID NO:34 and a polypeptide comprising at least 80% sequence identity to SEQ ID NO:32; a fusion between a polypeptide comprising at least 80% sequence identity to SEQ ID NO:35 and a polypeptide comprising at least 80% sequence identity to SEQ ID NO:36; a fusion between a polypeptide comprising at least 80% sequence identity to SEQ ID NO:36 and a polypeptide comprising at least 80% sequence identity to SEQ ID NO:37; a fusion between a polypeptide comprising at least 80% sequence identity to SEQ ID NO:38 and a polypeptide comprising at least 80% sequence identity to SEQ ID NO:39; a fusion between a polypeptide comprising at least 80% sequence identity to SEQ ID NO:69 and a polypeptide comprising at least 80% sequence identity to SEQ ID NO:70; or a fusion between a polypeptide com-

prising at least 80% sequence identity to SEQ ID NO:69 and a polypeptide comprising at least 80% sequence identity to SEQ ID NO:71.

7. The polypeptide of any one of claims 2-6, wherein the effector domain comprises: a fusion between a polypeptide comprising SEQ ID NO:32 and a polypeptide comprising SEQ ID NO:33; a fusion between a polypeptide comprising SEQ ID NO:34 and a polypeptide comprising SEQ ID NO:33; a fusion between a polypeptide comprising SEQ ID NO:34 and a polypeptide comprising SEQ ID NO:32; a fusion between a polypeptide comprising SEQ ID NO:35 and a polypeptide comprising SEQ ID NO:36; a fusion between a polypeptide comprising SEQ ID NO:36 and a polypeptide comprising SEQ ID NO:37; a fusion between SEQ ID NO:38 and a polypeptide comprising SEQ ID NO:39; a fusion between a polypeptide comprising SEQ ID NO:69 and a polypeptide comprising SEQ ID NO:70; or a fusion between a polypeptide comprising SEQ ID NO:69 and a polypeptide comprising SEQ ID NO:71.

8. The polypeptide of any one of claims 2-7, wherein the effector domain comprises the amino acid sequence of one of SEQ ID NOs: 19-21, 25-27, 72 or 73 or a the effector domain comprises an amino acid sequence with at least 80% sequence identity to one of SEQ ID NOs: 19-21, 25-27, 72 or 73.

9. The polypeptide of any one of claims 2-8, wherein the polypeptide further comprises one or more of a stabilizer domain, linker, RNA hairpin binding domain, nuclear export signal, and molecular tag.

10. The polypeptide of any one of claims 1-9, wherein the polypeptide comprises a RNA hairpin binding domain.

11. The polypeptide of claim 10, wherein the hairpin binding domain comprises SEQ ID NO: 16 or a polypeptide with at least 80% sequence identity to SEQ ID NO:16.

12. The polypeptide of any one of claims 1-11, wherein the polypeptide comprises a stabilizer domain.

13. The polypeptide of claim 12, wherein the stabilizer domain comprises SEQ ID NO:14, 58, or 59 or a polypeptide with at least 80% sequence identity to SEQ ID NO:14, 58, or 59.

14. The polypeptide of any one of claims 1-13, wherein the polypeptide comprises a nuclear export signal.

15. The polypeptide of claim 14, wherein the nuclear export signal comprises SEQ ID NO: 17, or a polypeptide with at least 80% sequence identity to SEQ ID NO:17.

16. The polypeptide of any one of claims 1-15, wherein the polypeptide comprises a molecular tag.

17. The polypeptide of claim 16, wherein the molecular tag comprises SEQ ID NO:18, or a polypeptide with at least 80% sequence identity to SEQ ID NO:18.

18. The polypeptide of any one of claims 1-17, wherein the hairpin binding domain is amino-proximal to the effector domain.

19. The polypeptide of any one of claims 1-18, wherein the stabilizer domain is amino-proximal to the RNA hairpin binding domain and/or effector domain.

20. The polypeptide of any one of claims 1-19, wherein the nuclear export signal is amino-proximal to the effector domain.

21. The polypeptide of any one of claims 1-20, wherein the molecular tag is carboxy-proximal to the effector domain.

22. The polypeptide of any one of claims 1-21, wherein the polypeptide comprises, from N-terminus to the C-terminus, a stabilizer domain, a glycine serine linker, a RNA hairpin binding region, a nuclear export signal, an effector domain, and a molecular tag.

23. The polypeptide of any one of claims 1-22, wherein the polypeptide comprises the amino acid sequence of one of SEQ ID NOs: 1-13, 63-68, 77-82, or 148-169 or an amino acid with at least 80% sequence identity to one of SEQ ID NOs: 1-13, 63-68, 77-82, or 148-169.

24. The polypeptide of claim 23, wherein the polypeptide comprises at least one substitution relative to SEQ ID NOs: 1-13, 63-68, 77-82, or 148-169.

25. A nucleic acid encoding the polypeptide of any one of claims 1-24.

26. A nucleic acid encoding a RNA targeting molecule comprising a RNA targeting region and at least one hairpin structure, wherein the hairpin structure of the RNA targeting molecule specifically binds to the RNA hairpin binding domain of the polypeptide.

27. The nucleic acid of claim 26, wherein the RNA targeting region hybridizes to the 5'UTR of a RNA.

28. The nucleic acid of claim 26, wherein the RNA targeting region hybridizes to the 3'UTR of a RNA.

29. The nucleic acid of claim 26 or 27, wherein the RNA targeting region hybridizes to the 5'UTR or 3'UTR of a mammalian mRNA.

30. The nucleic acid of any one of claims 26-29, wherein the RNA targeting region comprises a nucleic acid sequence of one of SEQ ID NOs: 55, 83-86, 88-112, 122-131, or 144-147 or a nucleic acid sequence having at least 80% sequence identity to one of of SEQ ID NOs: 55, 83-86, 88-112, 122-131, or 144-147.

31. The nucleic acid of any one of claims 26-30, wherein the RNA targeting molecule comprises a TAR hairpin scaffold or a human SLBP hairpin scaffold.

32. The nucleic acid of any one of claims 26-31, wherein the RNA targeting molecule comprises at least one of SEQ ID NO:40-54, 56, 57, or the sequence UCCCGA.

33. The nucleic acid of claim 32, wherein the RNA targeting molecule comprises SEQ ID NO: 40 or 41.

34. An expression vector comprising the nucleic acid of any one of claims 25-33.

35. A host cell comprising the nucleic acid of any one of claims 25-33 or expression vector of claim 34.

36. A delivery vehicle comprising the polypeptide of any one of claims 1-24, nucleic acid of any one of claims 25-33, or expression vector of claim 34.

37. The delivery vehicle of claim 36, wherein the delivery vehicle comprises liposome(s), particle(s), exosome(s), microvesicle(s), a gene-gun or one or more nucleic acid vector(s).

38. The delivery vehicle of claim 36 or 37, wherein the delivery vehicle further comprises a RNA targeting molecule comprising a RNA targeting region and at least one hairpin structure, wherein the hairpin structure of the RNA targeting molecule specifically binds to the RNA hairpin binding domain of the polypeptide.

39. The delivery vehicle of claim 38, wherein the RNA targeting region hybridizes to the 5'UTR of a RNA.

40. The delivery vehicle of claim 38, wherein the RNA targeting region hybridizes to the 3'UTR of a RNA.

41. The delivery vehicle of claim 39 or 40, wherein the RNA targeting region hybridizes to the 5'UTR or 3'UTR of a mammalian mRNA.

42. The delivery vehicle of any one of claims 38-41, wherein the RNA targeting region comprises a nucleic acid sequence of one of SEQ ID NOs: 55, 83-86, 88-112, 122-131, or 144-147 or a nucleic acid sequence having at least 80% sequence identity to one of SEQ ID NOs: 55, 83-86, 88-112, 122-131, or 144-147.

43. The delivery vehicle of any one of claims 38-42, wherein the RNA targeting molecule comprises a TAR hairpin scaffold or a human SLBP hairpin scaffold.

44. The delivery vehicle of any one of claims 38-43, wherein the RNA targeting molecule comprises at least one of SEQ ID NO:40-54, 56, 57, or the sequence UCCCGA.

45. The delivery vehicle of claim 44, wherein the RNA targeting molecule comprises SEQ ID NO: 40 or 41.

46. A composition comprising the polypeptide of any one of claims 1-24, nucleic acid of any one of claims 25-33, expression vector of claim 34, host cell of claim 35, or delivery vehicle of any one of claims 36-45.

47. The composition of claim 46, wherein the composition further comprises a RNA targeting molecule comprising a RNA targeting region and at least one hairpin structure, wherein the hairpin structure of the RNA targeting molecule specifically binds to the RNA hairpin binding domain of the polypeptide.

48. The composition of claim 47, wherein the RNA targeting region hybridizes to the 5'UTR of a RNA.

49. The composition of claim 47, wherein the RNA targeting region hybridizes to the 3'UTR of a RNA.

50. The composition of claim 48 or 49, wherein the RNA targeting region hybridizes to the 5'UTR or 3'UTR of a mammalian mRNA.

51. The composition of any one of claims 47-50, wherein the RNA targeting region comprises a nucleic acid sequence of one of SEQ ID NOs: 55, 83-86, 88-112, 122-131, or 144-147 or a nucleic acid sequence having at least 80% sequence identity to one of SEQ ID NOs: 55, 83-86, 88-112, 122-131, or 144-147.

52. The composition of any one of claims 47-51, wherein the RNA targeting molecule comprises a TAR hairpin scaffold or a human SLBP hairpin scaffold.

53. The composition of any one of claims 47-52, wherein the RNA targeting molecule comprises at least one of SEQ ID NO:40-54, 56, 57, or the sequence UCCCGA.

54. The composition of claim 53, wherein the RNA targeting molecule comprises SEQ ID NO: 40 or 41.

55. A method of making a polypeptide comprising transferring the nucleic acid of claim **25** or expression vector of claim **34** into a cell and isolating the expressed polypeptide.

56. A method of modulating at least one target RNA comprising contacting the target RNA with the polypeptide of any one of claims **1-24**, the delivery vehicle of any one of claims **36-45**, or the composition of any one of claims **46-54**.

57. A method for modulating at least one target RNA in a subject, the method comprising administering the polypeptide of any one of claims **1-24**, the delivery vehicle of any one of claims **36-45**, or the composition of any one of claims **46-54** to the subject.

58. The method of claim **56** or **57**, wherein modulating the at least one target RNA comprises cleaving, demethylating, methylating, activating translation, repressing translation, promoting degradation, and/or binding to the RNA.

59. The method of claim **56** or **58**, wherein the target RNA is in a prokaryotic or eukaryotic cell.

60. The method of claim **59**, wherein the target RNA is in a human cell.

61. The method of claim **59** or **60**, wherein the target RNA is in vitro or in vivo.

62. A cell or progeny thereof comprising modulated target RNA, wherein the target RNA has been modulated according to any one of claims **56-61**.

63. A multicellular organism comprising one or more cells according to claim **62**.

64. A plant or animal comprising one or more cells according to claim **63**.

65. A kit comprising the polypeptide of any one of claims **1-24**, nucleic acid of any one of claims **25-33**, expression vector of claim **34**, host cell of claim **35**, delivery vehicle of any one of claims **36-45**, or the composition of any one of claims **46-54**.

* * * * *